

Single-Variable Calculus

The Curious Family Library · Mathematics · Module Two

Meira Davidson

Single-Variable Calculus — Calculus I. The Curious Family Library · Mathematics ·
Module Two.

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The accompanying source code, figures, and interactive companion notebooks are released as open resources at https://github.com/MeiraDavidson/math_module2.

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*For the Jessica family kids —
be curious, stay open, and be kind.*

The Mathematics Foundation

The Curious Family Library is a family of plain-spoken, build-it-yourself guides across many subjects. This book continues its Mathematics line — **Module Two** of a small, deliberately ordered foundation. Its aim is easy to say and ambitious to do: to give you the backbone of mathematics — quickly, and deeply enough that it lasts — so you can dive into whatever you choose: artificial intelligence, engineering, medicine, science, or a field that doesn't even have a name yet.

The goal is never to memorize. It is to understand well enough to rebuild. Every idea here is earned from a reason, so you walk away with not a list of facts but two durable habits: the critical thinking to ask whether a claim is really true, and the power to derive what you need when you've forgotten the formula — which, in real life, is always. Master the foundation and you needn't carry the whole library in your head; you can regenerate it.

The modules of the foundation:

1. **Numbers and Functions** — the number systems, and the handful of function families that everything else is built from.
2. **Calculus I** (*this book*) — single variable: limits, derivatives, integrals — the mathematics of change and accumulation.
3. **Linear Algebra** — vectors, matrices, and the language of high dimensions. It sits here on purpose: Calculus II leans on it (and it is the language modern AI lives in).
4. **Calculus II** — several variables: the ideas of Calculus I set loose in more than one dimension, standing on the vectors and matrices just learned.
5. **Probability and Statistics** — reasoning carefully about uncertainty, data, and evidence.
6. **The Summary** — a short capstone that ties the whole foundation together and shows how the pieces connect.

They're meant to be read roughly in order — each leans on the ones before — but every module stands on its own feet. Computing and algorithms have a line of their own in the Library; this is the backbone they rest on.

Every chapter of this book has an interactive **companion notebook** — sliders, animations, and drills — and all of its code and figures are open. They live at https://github.com/MeiraDavidson/math_module2, in the `notebooks/` folder.

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Chapter 1

Patterns That Go Somewhere

Or: What does a list of numbers want to become?

Companion notebook. Play with this chapter’s ideas — sliders, animations, and drills — in notebooks/ch1_sequences.ipynb.

By the end of this chapter you will be able to look at a never-ending list of numbers and answer a question that turns out to be the seed of all of calculus: *what number, if any, is this list sneaking up on?*

You will recognize the two great families of sequences — the ones that grow by **adding** the same amount each step, and the ones that grow by **multiplying** by the same amount — and you will have formulas for both. That part is mostly a sharpening of patterns you already met in Book 1.

Then comes the genuinely new idea, the one this whole book is built on. Some lists settle down. The list $1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \dots$ is creeping toward 0 without ever arriving. The list $2, 2.25, 2.37, \dots$ is creeping toward a strange number near 2.718. Pinning down what “creeping toward” *means* — precisely enough that we can prove it rather than just squint at it — is the idea of a **limit**, and once you have it, derivatives and integrals are within reach.

And you will learn the part that separates people who *know* about limits from people who can actually *use* them: what to do when the obvious move fails. By the end you’ll be able to evaluate limits where naive substitution gives nonsense like “ ∞ times 0” — where the answer is decided by a hidden race between two parts of the expression — and you’ll meet a stubborn $\infty \cdot 0$, the n -th root of n , and the whole family of e , each of which looks impossible until you know the trick.

A note on the two layers (please read this once)

Starting in this book, every chapter runs on two layers, and you should know which one you are walking.

Layer 1 is the main road. It is the ideas, the pictures, and how to actually compute things. Everyone walks the main road, start to finish. If you read only Layer 1 you will understand what is going on and be able to do the work.

Layer 2 is the set of marked detours — the boxes labeled with a diamond, $\blacklozenge$. This is where we stop pointing at a picture and *prove* the thing is really true, using exact definitions. On your first pass through a chapter you are welcome to glance at these and keep walking. On your second pass — or right now, if you are hungry — the climbs are the best part, and they are where you stop taking my word for anything.

Why the proofs start *here*, in Chapter 1, when Book 1 mostly let you trust your eyes? Because a limit is the first idea in this series that is genuinely slippery — your intuition about “getting close” is good enough to get started and *not* good enough to keep you safe later. So this is exactly where proof begins to earn its place. Let’s go.

1.1 Sequences: lists that follow a rule

A **sequence** is an ordered, never-ending list of numbers, each one produced by a rule. We write the first term a_1 , the second a_2 , the n -th term a_n , and the whole list as

$$a_1, a_2, a_3, \dots, a_n, \dots$$

The subscript n is just the term’s position in line; a_n is “the number sitting in the n -th seat.” In Book 1 you found patterns like 1, 4, 9, 16, ... and said “those are the squares.” In sequence language that is simply the rule

$a_n = n^2$. Same idea, sharper notation — and the notation matters, because soon we will ask questions about *all* the terms at once, and you cannot do that without a name for the general one.

1.2 The two great families

Almost every sequence you meet at first belongs to one of two families, distinguished by *how you get from one term to the next*.

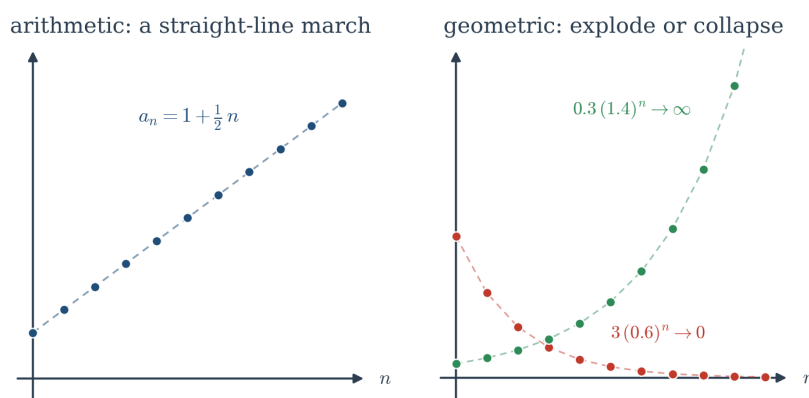


Figure 1.1

An **arithmetic sequence** gets from each term to the next by **adding** a fixed number d , the *common difference*:

$$a_n = a_1 + (n - 1)d.$$

Example: 3, 7, 11, 15, ... has $a_1 = 3$ and $d = 4$, so $a_n = 3 + 4(n - 1)$. The tenth term is $3 + 4(9) = 39$. Arithmetic sequences are the steady ones — they march in a straight line.

A **geometric sequence** gets from each term to the next by **multiply-**ing by a fixed number r , the *common ratio*:

$$a_n = a_1 \cdot r^{n-1}.$$

Example: $1, \frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \dots$ has $a_1 = 1$ and $r = \frac{1}{2}$, so $a_n = \left(\frac{1}{2}\right)^{n-1}$. Geometric sequences are the ones that explode or collapse — and which one they do depends entirely on r , a fact that will matter enormously later (you will see it run the whole show in Chapter 8). Keep your eye on geometric sequences. They are the most important family in this book.

1.3 The new question: what does it approach?

Here is where we leave familiar ground. Look at these four sequences not as lists to extend, but as *journeys* — and ask where each one is headed.

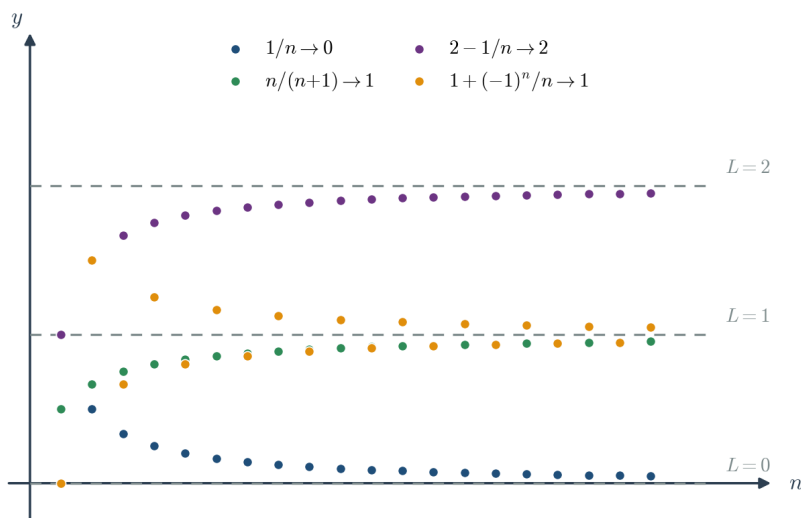


Figure 1.2

n	$\frac{1}{n}$	$\left(\frac{1}{2}\right)^n$	$\frac{n}{n+1}$	$\left(1 + \frac{1}{n}\right)^n$
1	1	0.5	0.5	2
2	0.5	0.25	0.667	2.25
5	0.2	0.031	0.833	2.488
10	0.1	0.000977	0.909	2.594
100	0.01	$\approx 10^{-30}$	0.990	2.705
1000	0.001	≈ 0	0.999	2.717

Read down the columns. The first two are sinking toward 0. The third is climbing toward 1 but will never *reach* 1 (since $\frac{n}{n+1}$ is always a hair below 1). The fourth is climbing too, but toward some odd number near 2.718 that is not 1, not 3, not anything with a name you know yet.

When a sequence's terms get **and stay** arbitrarily close to a single number L , we say the sequence **converges** to L , that L is its **limit**, and we write

$$\lim_{n \rightarrow \infty} a_n = L \quad \text{or simply} \quad a_n \rightarrow L.$$

So $\frac{1}{n} \rightarrow 0$, and $\frac{n}{n+1} \rightarrow 1$. The crucial word is *stay*: it is not enough to brush past L once; the terms must close in and remain close, tighter and tighter, forever.

That fourth sequence is special enough to flag now and chase down later. Its limit is a genuine new constant of mathematics, and we will name it e in Chapter 5 — where the whole chapter is built around it. **For now treat “ $\left(1 + \frac{1}{n}\right)^n$ approaches some number near 2.718” as a promise, not a proven fact.** We will keep returning to it, and we will finally nail it down for good in Chapter 10. Watching a debt like this get tracked and repaid across a book is part of what I want you to learn.

Toolkit: Finding the limit of a sequence

The laws. If $a_n \rightarrow A$ and $b_n \rightarrow B$, then sums, products, and quotients behave as you'd hope: $a_n + b_n \rightarrow A + B$, $a_n b_n \rightarrow AB$, $\frac{a_n}{b_n} \rightarrow \frac{A}{B}$ (provided $B \neq 0$), $c a_n \rightarrow cA$.

The workhorse fact. $\frac{1}{n} \rightarrow 0$, and so does $\frac{1}{n^k}$ for any fixed power $k \geq 1$. Almost every limit below gets reduced to this.

Geometric sequences. For $a_n = r^n$: if $|r| < 1$ then $r^n \rightarrow 0$; if $r = 1$ then $r^n \rightarrow 1$; if $|r| > 1$ then r^n blows up; if $r = -1$ it just flips $1, -1, 1, \dots$ and never settles.

The “divide by the biggest term” trick for ratios of polynomials: divide top and bottom by the highest power of n present, then send each $\frac{1}{n^k} \rightarrow 0$.

Worked example. Find $\lim_{n \rightarrow \infty} \frac{3n^2 + 2n}{n^2 - 5}$. The biggest power is n^2 , so divide top and bottom by n^2 :

$$\frac{3n^2 + 2n}{n^2 - 5} = \frac{3 + \frac{2}{n}}{1 - \frac{5}{n^2}}.$$

Now use the workhorse fact: as $n \rightarrow \infty$, $\frac{2}{n} \rightarrow 0$ and $\frac{5}{n^2} \rightarrow 0$. By the limit laws the whole thing approaches $\frac{3+0}{1-0} = 3$. So the limit is 3. The trick converts a scary-looking ratio into “everything with a $\frac{1}{n}$ in it dies, read off what's left.”

1.4 Three fates

Not every sequence converges. There are exactly three things a sequence can do, and naming them honestly keeps you from forcing a limit where there is none.

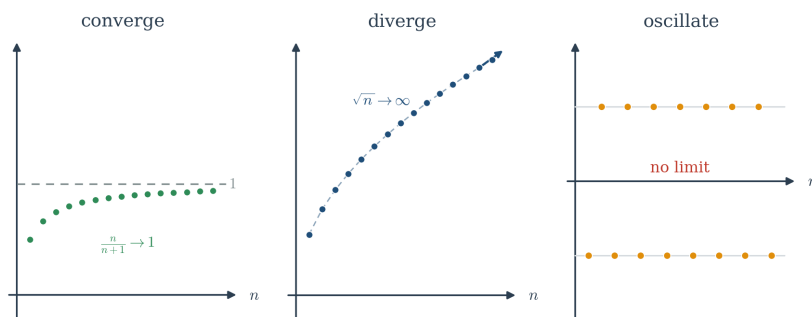


Figure 1.4

It can **converge** — settle to a single number L , like $\frac{1}{n} \rightarrow 0$.

It can **diverge to infinity** — grow without bound, like $a_n = n^2$ or like a geometric sequence with $|r| > 1$. There is no limit; the terms outrun every ceiling.

It can **oscillate** — bounce around and never settle, like $a_n = (-1)^n$, which is $-1, 1, -1, 1, \dots$ forever. It stays bounded but it has no limit, because it never gets *and stays* close to any one number. (It is near 1 half the time and near -1 the other half; “near L half the time” is not convergence.)

That last case is the warning the rest of this section answers. Squinting at a table can fool you — a few terms can look like they’re settling when they’re not. To be *sure*, we need to say exactly what “gets and stays arbitrarily close” means. That is Layer 2.

◆ Layer 2 — What “approaches” really means

Here is the definition that makes calculus rigorous. It looks intimidating for about a day and then becomes the most natural thing in the world. Read it as a *game between two players*.

Definition. $a_n \rightarrow L$ means: for every tolerance $\varepsilon > 0$, there exists a cutoff N such that for all $n > N$, $|a_n - L| < \varepsilon$.

The game. A skeptic hands you a tolerance ε — say, “I bet your terms don’t get within $\varepsilon = 0.001$ of L and stay there.” Your job is to answer with a cutoff N — a seat number — and promise: “every term from seat N onward is within 0.001 of L .” If you can win this game *no matter how tiny a tolerance the skeptic names*, then $a_n \rightarrow L$ is true. The quantity $|a_n - L|$ is just the distance from the term to the target; $|a_n - L| < \varepsilon$ says “this term is inside a window of width ε around L .”

Why this and not “gets closer and closer”? Because “gets closer and closer” is a trap: the sequence 1, 0.9, 0.99, 1.001, 0.5, ... might get closer for a while and then wander off, and the vague phrase can’t tell. The ε - N definition forbids wandering off: past the cutoff, *every* term must stay in the window. That single word “every” is what your intuition was missing.

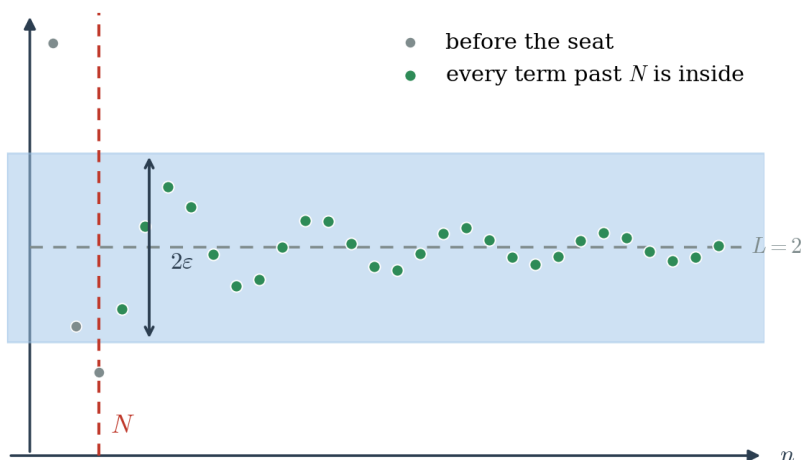


Figure 1.3

◆ **Proof:** $\frac{1}{n} \rightarrow 0$

We play the game. Let a tolerance $\varepsilon > 0$ be handed to us. We must produce a cutoff N so that for all $n > N$, the distance $\left|\frac{1}{n} - 0\right| = \frac{1}{n}$ is less than ε .

We want $\frac{1}{n} < \varepsilon$. Rearranging, that is the same as $n > \frac{1}{\varepsilon}$. So we simply *choose* N to be any whole number with $N \geq \frac{1}{\varepsilon}$. (Such a whole number always exists — the number line never runs out of integers above a given value. That's a basic property of the reals, sometimes called the Archimedean property.)

Now check it works: if $n > N \geq \frac{1}{\varepsilon}$, then $n > \frac{1}{\varepsilon}$, which rearranges back to $\frac{1}{n} < \varepsilon$. So every term past seat N is within ε of 0. Since we won the game for an *arbitrary* ε , the limit is 0. ■

Sit with this one. It is the first real ε - N proof in the book, and almost every other convergence proof is this same three-move dance: *let ε be given* $\rightarrow$ *solve the inequality for n* $\rightarrow$ *choose N accordingly and verify.*

◆ **Proof:** $r^n \rightarrow 0$ **when** $|r| < 1$

This is the fact the whole of Chapter 8 will lean on, so it deserves a real proof. Assume $0 < |r| < 1$ (the case $r = 0$ is trivially 0).

Since $|r| < 1$, its reciprocal $\frac{1}{|r|}$ is bigger than 1, so we can write $\frac{1}{|r|} = 1 + h$ for some $h > 0$. Then $|r|^n = \frac{1}{(1+h)^n}$. Now use

Bernoulli's inequality, $(1 + h)^n \geq 1 + nh$ (proved in problem 30), to push the denominator up and the whole fraction down:

$$|r|^n = \frac{1}{(1 + h)^n} \leq \frac{1}{1 + nh} < \frac{1}{nh}.$$

Now play the game. Given $\varepsilon > 0$, we want $\frac{1}{nh} < \varepsilon$, i.e. $n > \frac{1}{h\varepsilon}$. Choose $N \geq \frac{1}{h\varepsilon}$. Then for every $n > N$ we have $|r^n| = |r|^n < \frac{1}{nh} < \varepsilon$. So $r^n \rightarrow 0$. ■

The engine is the same as before — the only cleverness was rewriting $|r|$ as $\frac{1}{1+h}$ so that Bernoulli's inequality could turn the shrinking power into an honest $\frac{1}{n}$ -style bound we already know how to beat.

◆ Layer 2 — Sequences that *must* arrive (and the birth of e)

Sometimes you can see a sequence climbing steadily but you cannot compute its limit directly — exactly the situation with $\left(1 + \frac{1}{n}\right)^n$. There is a beautiful principle that rescues us, and it comes straight from the property of the real number line you met in Book 1: the line has **no gaps**.

The Monotone Convergence principle. A sequence that is *increasing* and *bounded above* (never climbs past some ceiling) must converge to a limit. (Likewise: decreasing and bounded below must converge.)

The picture is honest and the gaplessness is doing the work: an increasing sequence trapped under a ceiling has nowhere to escape to, and because the real line has no holes for it to pile up against, there is a genuine number it closes in on. (On a number line *with* gaps — like the rationals alone — this can fail, which is precisely the defect Book 1’s irrational numbers fixed.)

Now point it at our mystery sequence. The sequence $\left(1 + \frac{1}{n}\right)^n$ turns out to be **increasing** (each term beats the last — take this on faith here; it’s true and provable) and **bounded above by 3** (it never reaches 3 — you’ll prove this yourself in problem 31, and the proof reuses the geometric sequence $1 + \frac{1}{2} + \frac{1}{4} + \dots$ from this very chapter). Increasing and bounded above: by the principle, it **must converge**. Its limit is a real, specific number, and since it doesn’t equal anything we’ve already named, we give it a new name:

$$e = \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n \approx 2.71828 \dots$$

Notice what just happened: we proved *e exists* without ever computing it exactly — the principle guarantees a limit is there to be found. That is a new kind of power. We will discover in Chapter 5 that this same *e* is the one base whose exponential curve is its own slope, and we’ll prove the two descriptions match in Chapter 10. Born here, fully understood three chapters from now.

1.5 Limits that fight back

Everything so far has been gentle: substitute, simplify, read off the answer. The limits that *teach* you something are the ones where that fails. Look at these three:

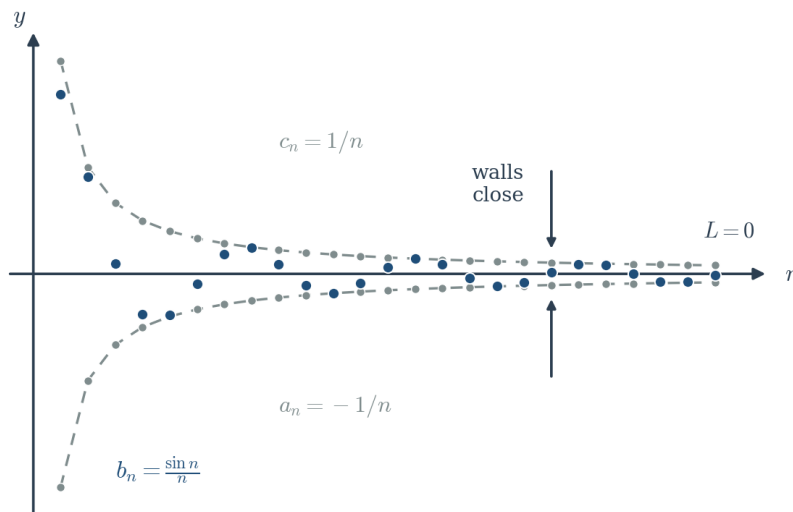


Figure 1.5

$$n \left(\sqrt{1 + \frac{1}{n}} - 1 \right), \quad \sqrt[n]{n} = n^{1/n}, \quad \left(1 + \frac{1}{n} \right)^n.$$

Try the naive move on the first. As $n \rightarrow \infty$, the factor n runs off to infinity while the parenthesis $\sqrt{1 + \frac{1}{n}} - 1$ collapses to 0. So the answer is... infinity times zero? That is not a number. It is a **fight** between a part pulling the product up and a part pulling it down, and “ $\infty \cdot 0$ ” is just our shorthand for “*the winner is not yet decided — look closer.*”

These deadlocks have a name: **indeterminate forms**. The usual suspects are

$$\infty \cdot 0, \quad \frac{\infty}{\infty}, \quad \frac{0}{0}, \quad 1^\infty, \quad \infty - \infty, \quad 0^0.$$

When substitution lands you on one of these, you have learned exactly

one thing: *you cannot stop here*. The limit might be 0, might be ∞ , might be any finite number in between — it depends on **how fast** each part moves, and finding out is the whole skill. Our three sequences are, in order, an $\infty \cdot 0$, an ∞^0 , and a 1^∞ . Let's win all three. First we need one new tool.

Toolkit: The Squeeze Theorem

If a sequence is trapped between two others heading to the same place, it has no choice but to go there too.

Squeeze Theorem. If $a_n \leq b_n \leq c_n$ for all large n , and both $a_n \rightarrow L$ and $c_n \rightarrow L$, then $b_n \rightarrow L$.

Use it when b_n itself is hard but you can *bound* it above and below by things you understand. The art is choosing good walls.

A quick win, to watch it move: what is $\lim_{n \rightarrow \infty} \frac{\sin n}{n}$? The numerator $\sin n$ lurches around forever and has no limit of its own — but it never escapes the corridor $-1 \leq \sin n \leq 1$. Divide through by n :

$$-\frac{1}{n} \leq \frac{\sin n}{n} \leq \frac{1}{n}.$$

Both walls, $\pm \frac{1}{n}$, go to 0. Squeezed between them, $\frac{\sin n}{n} \rightarrow 0$ — and we never had to tame the wild numerator. We just fenced it in.

◆ Proof: the Squeeze Theorem

Play the game. Given $\varepsilon > 0$: since $a_n \rightarrow L$ there's a cutoff N_1 past which $a_n > L - \varepsilon$; since $c_n \rightarrow L$ there's a cutoff N_2 past

which $c_n < L + \varepsilon$. Let N be the larger of the two. Then for every $n > N$, the sandwich $a_n \leq b_n \leq c_n$ gives

$$L - \varepsilon < a_n \leq b_n \leq c_n < L + \varepsilon,$$

so $|b_n - L| < \varepsilon$. Beaten for every ε , so $b_n \rightarrow L$. ■

Fight one: $n \left(\sqrt{1 + \frac{1}{n}} - 1 \right)$ (**an** $\infty \cdot 0$)

The move with an indeterminate form is almost always to *rewrite* it until a shape you can finish. Here the troublemaker is the difference of a square root and 1, and the standard cure is to **rationalize** — multiply by the conjugate $\sqrt{1 + \frac{1}{n}} + 1$ over itself, which turns the difference of roots into a clean difference of the things under them:

$$n \left(\sqrt{1 + \frac{1}{n}} - 1 \right) = n \cdot \frac{\left(1 + \frac{1}{n}\right) - 1}{\sqrt{1 + \frac{1}{n}} + 1} = n \cdot \frac{1/n}{\sqrt{1 + \frac{1}{n}} + 1} = \frac{1}{\sqrt{1 + \frac{1}{n}} + 1}.$$

The n canceled the $\frac{1}{n}$ — the two combatants annihilated, leaving something tame. As $n \rightarrow \infty$, $\frac{1}{n} \rightarrow 0$, so the bottom heads to $\sqrt{1} + 1 = 2$, and

$$n \left(\sqrt{1 + \frac{1}{n}} - 1 \right) \rightarrow \frac{1}{2}.$$

Not 0, not ∞ , but exactly $\frac{1}{2}$: the growing n and the shrinking parenthesis were *perfectly matched*, and they balance at one-half. (Tuck this away: in Chapter 3 you'll recognize $\frac{1}{2}$ as the slope of $\sqrt{x}$ at $x = 1$. You are computing a derivative right now without knowing it — and we'll meet the same $\frac{0}{0}$ -after-rationalizing move again the moment we reach functions in Chapter 2.)

Fight two: $\sqrt[n]{n}$ (an ∞^0 , and a genuine surprise)

This one should bother you. Take the n -th root of n : $\sqrt[2]{2} \approx 1.41$, $\sqrt[3]{3} \approx 1.44$, $\sqrt[10]{10} \approx 1.26$, $\sqrt[100]{100} \approx 1.047$, $\sqrt[1000]{1000} \approx 1.0069$. The thing under the root is racing to infinity, yet the root is *settling toward* 1. The form is ∞^0 — a huge base, a vanishing exponent $\frac{1}{n}$ — and the vanishing exponent wins. (Intuition: roots are ferocious levelers; a giant number rooted hard enough is tamed almost to 1.) The proof is a small gem.

◆ **Proof:** $n^{1/n} \rightarrow 1$

Since $n^{1/n} \geq 1$, write $n^{1/n} = 1 + h_n$ with $h_n \geq 0$; we must show $h_n \rightarrow 0$. Raise to the n -th power: $n = (1 + h_n)^n$. Expand by the binomial theorem and **throw away every term but one** — keep only the $k = 2$ term, legal because all terms are nonnegative:

$$n = (1 + h_n)^n \geq \binom{n}{2} h_n^2 = \frac{n(n-1)}{2} h_n^2.$$

Divide by $\frac{n(n-1)}{2}$: $h_n^2 \leq \frac{2}{n-1}$, so $0 \leq h_n \leq \sqrt{\frac{2}{n-1}}$. The right wall is built from $\frac{1}{n-1} \rightarrow 0$, so it $\rightarrow 0$; the left wall is 0. Squeeze: $h_n \rightarrow 0$, hence $n^{1/n} = 1 + h_n \rightarrow 1$. ■ The trick worth stealing: when a binomial expansion is all-positive, *keeping a single well-chosen term* is a valid lower bound — and one term is often all you need.

Making the theorem earn its living: a sequence that defines itself

The monotone-convergence principle feels abstract until you meet a sequence with **no formula at all**, built by feeding each term back into itself:

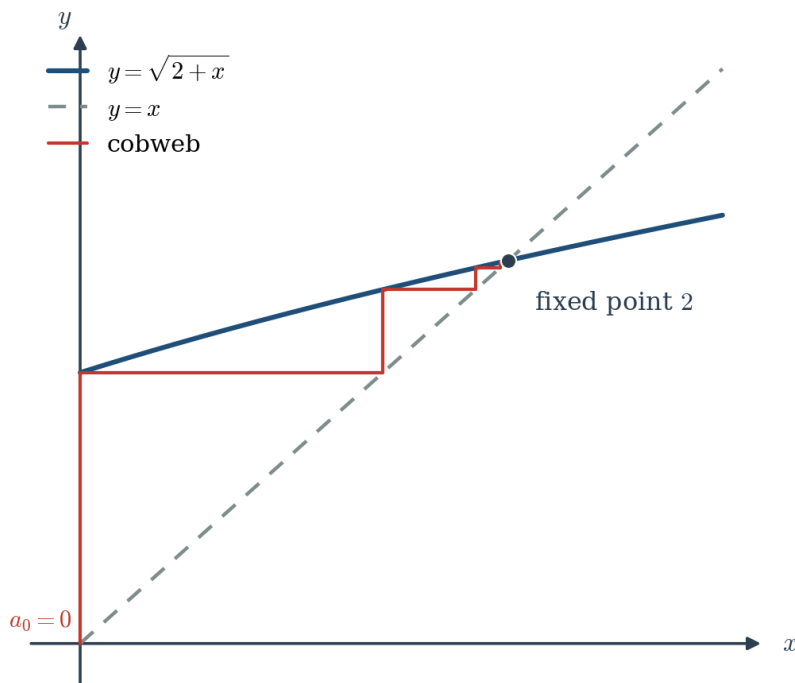


Figure 1.7

$$a_1 = \sqrt{2}, \quad a_{n+1} = \sqrt{2 + a_n}.$$

So $a_2 = \sqrt{2 + \sqrt{2}} \approx 1.848$, $a_3 \approx 1.962$, $a_4 \approx 1.990$, climbing toward... something. We can't write a_n in closed form — but we can still find the limit *exactly*, and this is the principle's finest hour.

◆ Layer 2 — Solving a self-defined sequence

Step 1 — bounded above by 2 (induction): $a_1 = \sqrt{2} < 2$; and if $a_n < 2$, then $a_{n+1} = \sqrt{2 + a_n} < \sqrt{2 + 2} = 2$. So every term stays under 2. **Step 2 — increasing:** $a_{n+1} > a_n$ means $\sqrt{2 + a_n} > a_n$, i.e. (square, both sides positive) $2 + a_n > a_n^2$, i.e. $a_n^2 - a_n - 2 < 0$, i.e. $(a_n - 2)(a_n + 1) < 0$ — true for every a_n with $0 < a_n < 2$, which Step 1 guarantees. **Step 3 — therefore it converges** (increasing and bounded above) to some limit L . **Step 4 — pin down L .** Take the limit of *both sides* of the rule $a_{n+1} = \sqrt{2 + a_n}$. The left side $\rightarrow L$; the right side $\rightarrow \sqrt{2 + L}$. So L must obey its own equation:

$$L = \sqrt{2 + L} \implies L^2 = 2 + L \implies (L - 2)(L + 1) = 0.$$

Hence $L = 2$ (reject -1 : the terms are positive). The sequence we couldn't even write down converges to exactly 2. That move — *let the limit satisfy the rule* — is your first **fixed point**: a value a process leaves unchanged. It returns whenever you study how an iteration settles, including, in spirit, the iterations of Chapter 8.

1.6 The many faces of e

The sequence $\left(1 + \frac{1}{n}\right)^n$ is the most important 1^∞ fight in all of mathematics, and it has a whole family. What if we replace that 1 in the numerator with some other number x ?

$$\boxed{\left(1 + \frac{x}{n}\right)^n \longrightarrow e^x}$$

This single fact is the bridge from this chapter to Chapter 5. It says e is not a one-off curiosity: it spawns an entire *function* e^x , and that function is born from compounding. Set $x = 1$ and recover the original $\rightarrow e$. Set $x = -1$: $\left(1 - \frac{1}{n}\right)^n \rightarrow e^{-1} = \frac{1}{e} \approx 0.368$. Set $x = 2$: $\left(1 + \frac{2}{n}\right)^n \rightarrow e^2$. Try a few on a calculator and watch them land on powers of e .

◆ Layer 2 — Why $\left(1 + \frac{x}{n}\right)^n \rightarrow e^x$ (one honest loan)

Take the logarithm, which turns the frightening exponent into a product (Chapter 5 makes $\ln$ rigorous; here we borrow only that $\ln$ pulls a power down into a multiplier):

$$\ln\left(1 + \frac{x}{n}\right)^n = n \ln\left(1 + \frac{x}{n}\right).$$

Let $t = \frac{x}{n} \rightarrow 0$, so $n = \frac{x}{t}$, and the right side becomes $x \cdot \frac{\ln(1+t)}{t}$. One more famous limit closes it: $\frac{\ln(1+t)}{t} \rightarrow 1$ as $t \rightarrow 0$ (this is the slope of $\ln$ at 1, proven in Chapter 5 — *the loan*). So the logarithm of our sequence $\rightarrow x \cdot 1 = x$, which means the sequence itself $\rightarrow e^x$. ■ Notice it is the same $\frac{\text{small}}{\text{small}} \rightarrow 1$ shape that turned up in Fight one. The limits that fight back are almost always derivatives wearing a costume — you'll see this confirmed in Chapters 2 and 3.

The race decides everything: three sequences, three fates

The form 1^∞ is indeterminate precisely because the answer hinges on a **race** — how fast the base sinks toward 1 versus how fast the exponent

climbs to ∞ . Watch three sequences that all look like “ 1^∞ ” land in three different worlds:

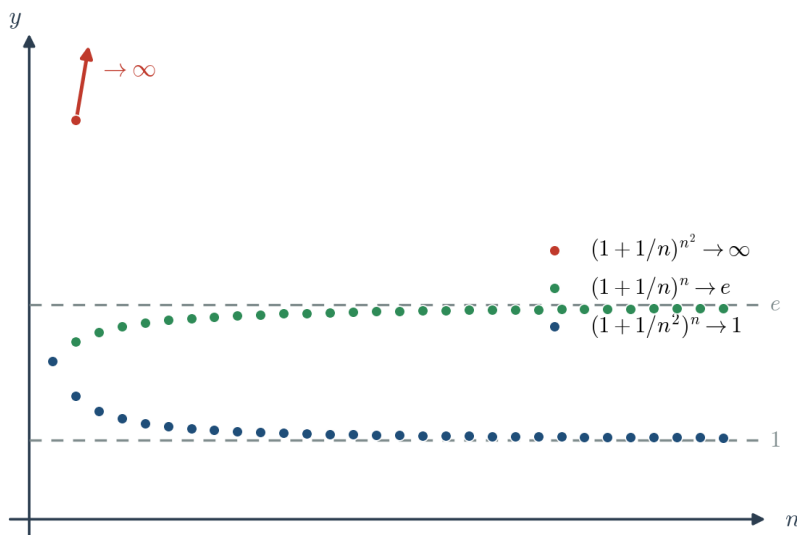


Figure 1.6

$$\left(1 + \frac{1}{n}\right)^n \rightarrow e, \quad \left(1 + \frac{1}{n^2}\right)^n \rightarrow 1, \quad \left(1 + \frac{1}{n}\right)^{n^2} \rightarrow \infty.$$

In the first, base and exponent are matched and balance at e . In the second, the base rushes to 1 *too fast* (like $\frac{1}{n^2}$) for a mere n in the exponent to matter — its logarithm runs like $n \cdot \frac{1}{n^2} = \frac{1}{n} \rightarrow 0$, so the limit is $e^0 = 1$. In the third, the exponent n^2 is *too strong* for the base’s gentle approach — its logarithm runs like $n^2 \cdot \frac{1}{n} = n \rightarrow \infty$, so it explodes.

Same form, three answers, decided entirely by who wins. Here’s the gauge that settles any of them at a glance: for a base near 1, look at (exponent) $\times$ (base $-$ 1); whatever number k *that* approaches, the limit is e^k . Run the three cases through it and out pop $e^1 = e$, then $e^0 = 1$, then $e^\infty = \infty$. That is the entire secret of 1^∞ forms, and you now hold it.

1.7 Where this is heading

The word “approaches” is the hinge of everything that follows. In Chapter 2 we ask what a *function* approaches as its input closes in on a point — the same idea, lifted from lists to curves. In Chapter 3 we use it to do the thing this whole book is named for: take the slope of a straight line between two points on a curve and watch what it *approaches* as the points slide together. That limit is the slope of the curve itself — the derivative. Every limit you compute now is practice for that moment.

And e , which appeared here as nothing more than the destination of a climbing list, will turn out to be one of the two or three most important numbers in all of mathematics. You met it first as a sequence. That’s fitting: in this book, the deep things arrive quietly, as patterns that go somewhere.

1.8 Problems

Six tiers, each with a different job. **Fluency** and **Practice** are doing* — drill until the moves are automatic, then multi-step work where you pick the tool. **Concept & counterexample** and **♦ Diamond** are *thinking* — traps, constructions, and Layer-2 proofs that test whether you understand, not just compute. **Teach it** is *teaching* — explain, write up, or fix reasoning for a peer, which is the surest way to find the gaps in your own. **Looking ahead** hands you Chapter 2 early. Do all of Fluency; do not skip Concept or Teach it — that is where the understanding actually sets.*

Fluency — do these fast, until they’re automatic

1. First four terms of: (a) $a_n = 3n - 1$; (b) $a_n = \left(\frac{1}{2}\right)^n$; (c) $a_n = \frac{n}{n+1}$; (d) $a_n = \frac{(-1)^n}{n}$.

2. Arithmetic, geometric, or neither — and give d or r where it applies: (a) 7, 10, 13, 16, ...; (b) 3, -6, 12, -24, ...; (c) $1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \dots$; (d) 2, 5, 10, 17,
3. Use a formula, not a list: (a) the 15th term of the arithmetic sequence with $a_1 = 4$, $d = 5$; (b) the 6th term of the geometric sequence with $a_1 = 2$, $r = 3$.
4. Guess the limit (or say “no limit”): (a) $\frac{1}{n}$; (b) $\frac{2n-1}{n}$; (c) $(-1)^n$; (d) $(0.8)^n$; (e) n^2 ; (f) $\frac{(-1)^n}{n}$.
5. Divide by the highest power: (a) $\lim_{n \rightarrow \infty} \frac{3n+1}{n+2}$; (b) $\lim_{n \rightarrow \infty} \frac{n^2-1}{2n^2+n}$; (c) $\lim_{n \rightarrow \infty} \frac{5n}{n^2+1}$; (d) $\lim_{n \rightarrow \infty} \frac{2n^3}{n^2+1}$.
6. For each r , does $r^n \rightarrow 0$, $\rightarrow 1$, blow up, or oscillate? $r = \frac{1}{3}$; $r = -\frac{1}{2}$; $r = 1$; $r = -1$; $r = 2$.

Practice — *choose the right tool*

7. $\lim_{n \rightarrow \infty} \frac{n^2 - 3n}{2n^2 + 1}$.
8. An $\infty - \infty$ that hides a finite answer: $\lim_{n \rightarrow \infty} (\sqrt{n^2 + n} - n)$. (Rationalize, then divide by the top power.)
9. An $\infty \cdot 0$: $\lim_{n \rightarrow \infty} n \left(\sqrt{1 + \frac{1}{n}} - 1 \right)$. (Mirror Fight one from the chapter exactly.)
10. The e -family: (a) $\lim_{n \rightarrow \infty} \left(1 + \frac{3}{n} \right)^n$; (b) $\lim_{n \rightarrow \infty} \left(1 - \frac{2}{n} \right)^n$; (c) $\lim_{n \rightarrow \infty} \left(1 + \frac{1}{n} \right)^{3n}$.
11. Use the gauge “(exponent) $\times$ (base - 1) $\rightarrow k \Rightarrow$ limit = e^k ”: evaluate or classify $\left(1 + \frac{1}{n} \right)^{n^2}$, $\left(1 + \frac{1}{n^2} \right)^n$, and $\left(1 + \frac{5}{n} \right)^n$. State k first each time.
12. A self-defined sequence: $a_1 = 1$, $a_{n+1} = \sqrt{12 + a_n}$. Compute a_2, a_3 , then find the limit by letting L satisfy the rule. (You’ll prove it converges in the Diamond tier.)
13. A geometric sequence has $a_1 = 1$, $r = 0.9$. About how many terms until it drops below 0.001? (Use the idea behind the $r^n \rightarrow 0$ proof, or trial and error.)

14. Decompose, then take the limit: $\lim_{n \rightarrow \infty} \frac{n + (-1)^n}{n}$. (Split into two pieces, one obvious, one you've already handled.)

Concept & counterexample — *think; almost no arithmetic. State true/false and justify, or build the example asked for.*

15. (a) If $a_n \rightarrow L$, then *every* term is within 0.01 of L . (b) If a sequence is bounded, it converges. (c) If $a_n \rightarrow 0$, the sequence is eventually decreasing. (d) A sequence can converge to two different limits.
16. **Construct** a bounded sequence that does not converge.
17. **Construct** an increasing sequence that does not converge — and explain in one line which hypothesis of the Monotone Convergence principle your example is missing.
18. **Construct** a sequence that converges to 5 but whose terms are *sometimes above and sometimes below* 5.
19. **Find the error.** A student writes: “ $(-1)^n$ gets arbitrarily close to 1 infinitely often and to -1 infinitely often, so it has two limits, 1 and -1 .” Identify precisely what's wrong, and state what *is* true.
20. **Is it possible?** A sequence with $a_n \rightarrow 0$ whose running totals $a_1 + a_2 + \cdots + a_n$ nonetheless grow without bound. (Name an example if yes — this is a quiet preview of infinite series.)
21. The Monotone Convergence principle needs *two* hypotheses, “increasing” and “bounded above.” Give one sequence showing the conclusion fails if you drop “bounded above,” and one showing it fails if you drop “increasing.” (This proves both hypotheses are load-bearing.)

◆ **Diamond — Layer 2: prove it**

22. ◆ Prove $\frac{1}{n} \rightarrow 0$ from the ε - N definition.

23. ♦ Prove $\frac{1}{n^2} \rightarrow 0$ from the definition.
24. ♦ Prove $\frac{n}{n+1} \rightarrow 1$ from the definition. (First show $\left| \frac{n}{n+1} - 1 \right| = \frac{1}{n+1}$.)
25. ♦ Prove $\frac{2n+1}{n} \rightarrow 2$ from the definition.
26. ♦ Suppose $a_n \rightarrow 5$. Using the limit laws, find $\lim \left(3a_n - \frac{1}{n} \right)$, naming the law at each step.
27. ♦ Prove $\frac{\sin n}{n} \rightarrow 0$ via the Squeeze Theorem: name the two walls, cite the theorem, and explain in one sentence why $\sin n$ never needed a limit of its own.
28. ♦ Prove the **uniqueness of limits**: a sequence cannot converge to two different values $L \neq M$. (Hint: if it did, take $\varepsilon = \frac{|L-M|}{3}$ and derive a contradiction — a term can't be within ε of both.) Connect this to your answer in 15(d).
29. ♦ For the sequence in problem 12, prove it is increasing and bounded above by 4, completing the argument that it converges.
30. ♦ Prove **Bernoulli's inequality** $(1+h)^n \geq 1+nh$ for $h > -1$, $n \geq 1$, by induction.
31. ♦♦ Prove $\left(1 + \frac{1}{n}\right)^n < 3$ for every n . Steps: expand by the binomial theorem; bound the k -th term by $\frac{1}{k!}$; bound $\frac{1}{k!} \leq \frac{1}{2^{k-1}}$; sum the geometric bound to get below $1 + \left(1 + \frac{1}{2} + \frac{1}{4} + \dots\right) = 3$.
32. ♦♦ Adapt the $\sqrt[n]{n} \rightarrow 1$ proof to show $\sqrt[n]{2^n + 3^n} \rightarrow 3$. (Factor out the dominant 3^n , then squeeze.)

Teach it — *the surest test of mastery is explaining it*

33. A friend has only ever heard “a limit is what a sequence gets closer and closer to.” In **three or four sentences written to them**, explain why that phrasing is unsafe, using a single concrete sequence that “gets closer” for a while but does not converge.
34. Write a complete, textbook-quality **worked solution** to $\lim_{n \rightarrow \infty} \frac{3n^2 + 2n}{n^2 - 5}$, narrating each step in words a classmate could follow without you in the room.

35. **Debug your classmate.** They argue: “ $\left(1 + \frac{1}{n}\right)^n \rightarrow 1$, because $1 + \frac{1}{n} \rightarrow 1$ and 1 to any power is 1.” Pinpoint the exact flawed step and write the one-sentence correction. (Name the indeterminate form they walked into.)
36. **Design a problem.** Invent a self-defined sequence $a_{n+1} = \sqrt{c + a_n}$ whose limit is exactly 5. State the c you need, then write the four-step solution key (bounded, increasing, converges, solve for L) as if for a teacher’s answer book.
37. In two or three sentences, **teach the difference between “bounded” and “convergent.”** Give one sequence that is bounded but not convergent, and say in plain words why one property does not imply the other.

Looking ahead — Chapter 2 in your hands early

38. In Chapter 2 the input switches from a whole number n to a real number x sliding toward a point. Predict the value of $\lim_{x \rightarrow 2} \frac{x^2 - 4}{x - 2}$ by canceling the common factor, exactly as you’d simplify a sequence ratio. (Next chapter makes this rigorous — and explains why the cancel is legal.)
39. You met e as $\lim \left(1 + \frac{1}{n}\right)^n$. Locate it from the *other* direction: with a calculator, compute $\frac{2^h - 1}{h}$ and $\frac{3^h - 1}{h}$ for small h (say $h = 0.001$). One is below 1, one above. Between which two whole-number bases must the “slope = 1” base — which Chapter 5 will reveal *is* e — lie?
40. Your recursive sequences found their limit by demanding $L =$ (the rule applied to L) — a *fixed point*. In Chapter 8 you’ll iterate $x \leftarrow x - \frac{f(x)}{f'(x)}$ to hunt a root. Predict: if that iteration settles to a value L , what equation must L satisfy?
-

Chapter 2

Getting Arbitrarily Close

Or: What a function is doing right next to a point it might never reach

Companion notebook. Play with this chapter’s ideas — sliders, animations, and drills — in `notebooks/ch2_limits.ipynb`.

By the end of this chapter you will be able to find what a *function* approaches as its input slides toward a target — which is the same idea as Chapter 1, with one new and crucial wrinkle. You will compute these limits three ways: by simply plugging in (when the function is well-behaved), by **factoring and canceling** (when plugging in gives the nonsense $\frac{0}{0}$), and out at infinity. You will know what it means for a function to be **continuous** — the “draw it without lifting your pen” idea — and why, underneath, continuity is really the promise that *a small change in input makes only a small change in output*. You’ll also meet the two great theorems that continuity *guarantees* — the **Intermediate Value Theorem** (a continuous function can’t skip a value, which is how we’ll prove that solutions exist) and the **Extreme Value Theorem** (a continuous function on a closed interval actually reaches its highest and lowest points) — both of which the back half of the book quietly runs on.

And you will meet one specific limit, $\frac{\sin x}{x} \rightarrow 1$, that looks like it should be undefined and isn’t. It is the single most important limit in the book, because in the very next chapter it turns out to *be* a slope — your first derivative, computed before we even have the word.

Here is the one new idea, and it is worth saying before anything else. In Chapter 1 the input was a whole number n that marched off to infinity.

Now the input is a real number x that you slide toward a finite point a — and it can approach a **from either side**. The limit asks what $f(x)$ is doing *in the neighborhood* of a . It does **not** ask, and does not care, what f does *at* a itself. That gap between “near a ” and “at a ” looks like a technicality. It is actually the loophole that makes all of calculus possible.

2.1 From a marching n to a sliding x

We write

$$\lim_{x \rightarrow a} f(x) = L$$

to mean: as x gets arbitrarily close to a — closing in from the left *and* from the right — the value $f(x)$ gets and stays arbitrarily close to L . Everything you earned in Chapter 1 carries straight over: the limit laws (sums, products, quotients of limits behave as you’d hope), the squeeze idea, and the challenge–response game behind the word “arbitrarily.” We are lifting the same machinery from a discrete staircase to a continuous slide.

For most functions, at most points, this is gloriously boring. The function $f(x) = x^2 + 2$ heads toward 11 as $x \rightarrow 3$, for the unsurprising reason that $f(3) = 11$. When a function is well-behaved, the limit is just the value, and you find it by **substitution**. The interesting cases — the ones we built this whole subject to handle — are exactly the cases where substitution *fails*.

2.2 The point you’re approaching might not be on the graph

Look at

$$f(x) = \frac{x^2 - 1}{x - 1}.$$

Try to evaluate it at $x = 1$ by substitution: you get $\frac{0}{0}$, which is not a number. The function is *undefined* at $x = 1$ — there is a literal hole in its graph there. A careless reader concludes “no value, no limit, move on.” Watch what actually happens.

For every $x \neq 1$, we are allowed to cancel, because the thing we’re canceling isn’t zero:

$$\frac{x^2 - 1}{x - 1} = \frac{(x - 1)(x + 1)}{x - 1} = x + 1.$$

So *everywhere except the single point $x = 1$* , this function is identical to the simple line $x + 1$. And $x + 1$ obviously heads toward 2 as $x \rightarrow 1$. Therefore

$$\lim_{x \rightarrow 1} \frac{x^2 - 1}{x - 1} = 2,$$

even though $f(1)$ flat-out does not exist. Read that again. **The limit is 2 at a point where the function has no value at all.** This is the loophole promised above, stated plainly: the limit listens to what the function does on the *approach*, and the destination point itself is allowed to be missing. (Figure 2.1: the line $y = x + 1$ with a small open circle punched out at $(1, 2)$ — the hole the function has, that the limit sees right past.)

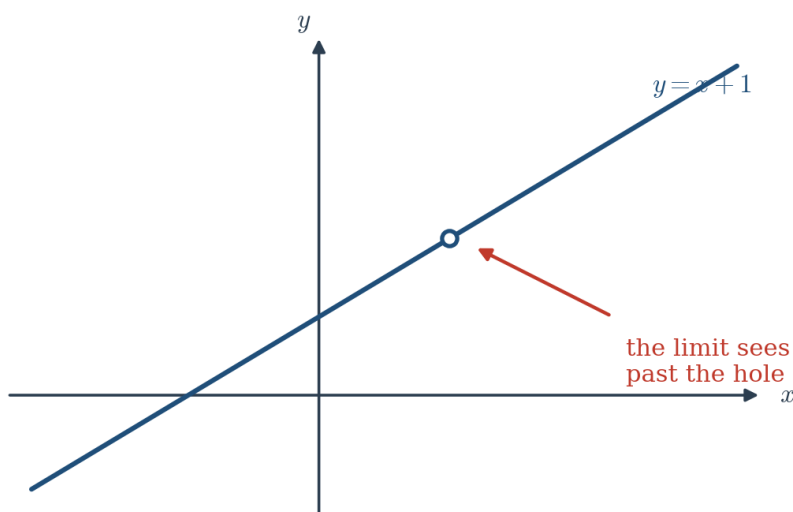


Figure 2.1

Hold onto this example with both hands, because it is the entire shape of the next chapter. In Chapter 3, the slope of a curve will be defined as a fraction that is *exactly* $\frac{0}{0}$ at the point of interest — and we will get a real answer out of it by canceling, in precisely this way. Every derivative you ever compute is a $\frac{0}{0}$ limit tamed by cancellation. You just did your first one.

2.3 Approaching from two sides

Because x slides along a line, it can reach a from the left or from the right, and sometimes the two stories disagree. We write $x \rightarrow a^-$ for the approach from below (the **left-hand limit**) and $x \rightarrow a^+$ for the approach from above (the **right-hand limit**).

$$\lim_{x \rightarrow a} f(x) = L \quad \text{exactly when} \quad \lim_{x \rightarrow a^-} f(x) = \lim_{x \rightarrow a^+} f(x) = L.$$

The two-sided limit exists only if both one-sided limits exist *and agree*. The classic disagreement is

$$f(x) = \frac{|x|}{x},$$

which equals $+1$ for every positive x and -1 for every negative x . As $x \rightarrow 0^+$ it sits at $+1$; as $x \rightarrow 0^-$ it sits at -1 . The two sides refuse to meet, so $\lim_{x \rightarrow 0} \frac{|x|}{x}$ **does not exist** — not because of a hole, but because of a *jump*. The function can't make up its mind, so the limit declines to exist. (Figure 2.2: the two flat arms at $+1$ and -1 with the cliff between them at 0 .)

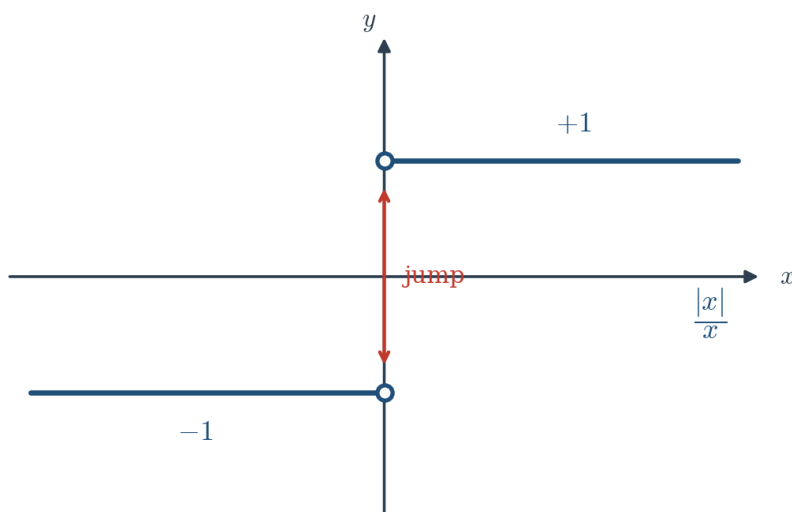


Figure 2.2

Toolkit: Evaluating the limit of a function

Step 1 — try substitution. If f is well-behaved (continuous) at a , then $\lim_{x \rightarrow a} f(x) = f(a)$. Done. Always try this first.

Step 2 — if you hit $\frac{0}{0}$, don't stop — rewrite. The $\frac{0}{0}$ means a common factor is hiding. Two moves flush it out: - **Factor and cancel:** $\frac{x^2 - 4}{x - 2} = \frac{(x - 2)(x + 2)}{x - 2} = x + 2 \rightarrow 4$ as $x \rightarrow 2$. - **Rationalize** (when roots appear): multiply top and bottom by the conjugate to clear the square root.

Step 3 — limits at infinity: divide top and bottom by the highest power of x , exactly as you divided by the highest power of n in Chapter 1.

Worked example (rationalizing a $\frac{0}{0}$). Find $\lim_{x \rightarrow 0} \frac{\sqrt{x+4} - 2}{x}$. Substituting gives $\frac{0}{0}$. Multiply top and bottom by the conjugate $\sqrt{x+4} + 2$:

$$\frac{\sqrt{x+4} - 2}{x} \cdot \frac{\sqrt{x+4} + 2}{\sqrt{x+4} + 2} = \frac{(x+4) - 4}{x(\sqrt{x+4} + 2)} = \frac{x}{x(\sqrt{x+4} + 2)} = \frac{1}{\sqrt{x+4} + 2}.$$

The troublesome x canceled. Now substitution is safe: as $x \rightarrow 0$ this heads to $\frac{1}{\sqrt{4} + 2} = \frac{1}{4}$. (And yes — this too is secretly a derivative, the slope of $\sqrt{x+4}$ at $x = 0$. They're everywhere once you know the shape.)

2.4 Continuity: the graph you can draw without lifting your pen

A function is **continuous at a** when the limit and the value finally agree:

$$\lim_{x \rightarrow a} f(x) = f(a).$$

Three things have to line up: $f(a)$ has to exist, the limit has to exist, and they have to be equal. When this holds at every point of an interval, you can draw the whole graph there without lifting your pen off the paper

— no holes, no jumps, no flying off to infinity. Polynomials are continuous everywhere; so are $\sin x$, $\cos x$, and e^x . The failures come in three flavors, and naming them is useful:

A **removable** discontinuity is a single missing point — our $\frac{x^2-1}{x-1}$ hole. The limit exists; the function just forgot to take the value. You can “repair” it by defining the missing point to be the limit.

A **jump** is when the two one-sided limits exist but disagree — our $\frac{|x|}{x}$ cliff. No repair possible; the gap is real.

An **infinite** discontinuity is when the function blows up, like $\frac{1}{x}$ at $x = 0$. There’s no finite limit to repair to.

(Figure 2.3: a continuous curve beside the three failures — the punched hole, the cliff, the blow-up.)

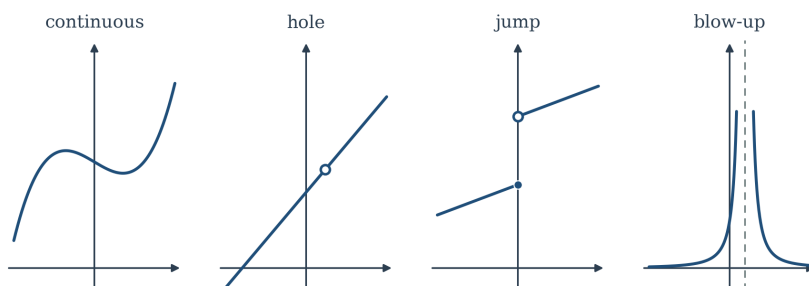


Figure 2.3

Why an engineer cares

Continuity sounds like a tidiness rule for graphs. It is really a statement about **robustness**, and that is the first reason this idea reaches far beyond math class. Continuity says: *nudge the input a little, and the output moves only a little*. Nothing snaps. A continuous control system doesn’t lurch when a sensor reading wobbles; a continuous model doesn’t return a wildly different answer because a measurement was off in the third decimal. Where a system is discontinuous — a circuit breaker tripping, water flashing to steam, a market gapping — is exactly where it behaves unpredictably and where engineers pay closest attention. We will sharpen “a little” into

something measurable in the diamond box below, and that sharpening is the same one that makes the whole subject rigorous.

2.5 What continuity guarantees: two theorems you'll use everywhere

Continuity does more than keep graphs tidy — it *forces* a continuous function to behave in two powerful ways, and the entire back half of this book leans on both. Neither is obvious, and both turn out to rest on the same deep fact about the real numbers.

The Intermediate Value Theorem: you can't cross without touching

Intermediate Value Theorem (IVT). If f is continuous on $[a, b]$, then for *every* value N between $f(a)$ and $f(b)$, there is at least one point c in $[a, b]$ with $f(c) = N$.

The picture is exactly the “no lifting the pen” idea. To get from height $f(a)$ to height $f(b)$, a continuous graph must sweep through *every* height in between — it cannot teleport over a value. So every intermediate height N is hit somewhere. (*Figure 2.4: a continuous curve from $f(a)$ to $f(b)$, a horizontal line at height N , and the crossing point c where the theorem promises a solution.*)

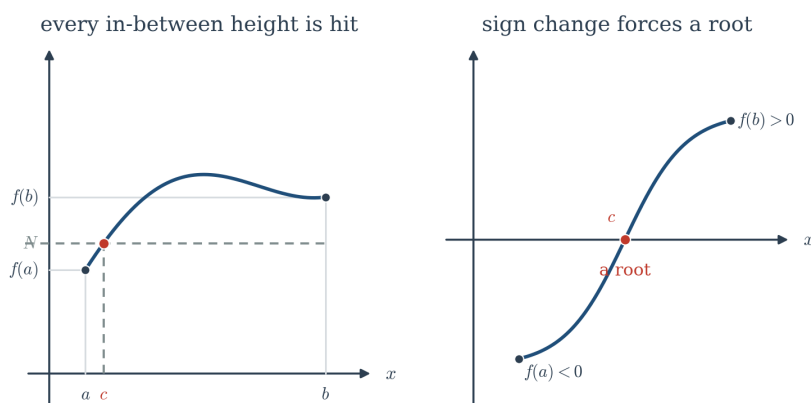


Figure 2.4

Its most useful form is **root-finding**. Suppose f is continuous and *changes sign* on $[a, b]$ — say $f(a) < 0$ and $f(b) > 0$. Then 0 is a value between $f(a)$ and $f(b)$, so the IVT guarantees a c with $f(c) = 0$: a **root**. This lets you prove a solution *exists* without ever solving the equation. For instance, $f(x) = x^5 + x - 1$ has $f(0) = -1 < 0$ and $f(1) = 1 > 0$ — so it has a root between 0 and 1, even though no formula will ever produce it.

This also pays back a debt. In Chapter 1, and again in Chapter 5, we claimed the number e *exists* because $M(a)$ — the slope of a^x as it crosses the axis — slides from below 1 (at $a = 2$) to above 1 (at $a = 3$), so it “must equal 1 somewhere in between.” That “must equal 1 somewhere” was an Intermediate Value Theorem argument all along, taken on faith at the time. Now it is a theorem, and the loan is repaid.

The IVT doesn’t say *where* the root is — but it suggests how to hunt it. Check the midpoint m of $[a, b]$: the sign change must survive in one of the two halves $[a, m]$ or $[m, b]$, so keep that half and repeat, halving the interval around the root every step. This is the **bisection method** — slow, but utterly reliable: it cannot fail or fly off. In Chapter 9 it becomes the safety net beneath Newton’s method, which is fast but fragile; real root-finders combine the two.

The Extreme Value Theorem: a peak that is actually reached

Extreme Value Theorem (EVT). If f is continuous on a *closed* interval $[a, b]$, then f attains a maximum and a minimum on $[a, b]$ — there are points where it reaches its highest and lowest values.

Intuition: a continuous function on a closed interval cannot run off to infinity (that would require a jump), and it cannot merely *approach* its highest value without reaching it — there is no gap for the peak to slip into. So the high point and low point are genuinely *attained*, not just approached. (Figure 2.5: a continuous curve on $[a, b]$ with its actual maximum and minimum points marked.)

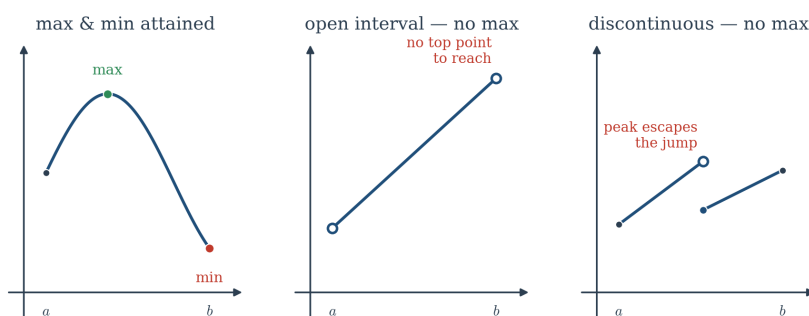


Figure 2.5

Both words in the hypothesis are load-bearing, and watching them fail is the fastest way to understand them: - Drop **closed**: on the *open* interval $(0, 1)$, $f(x) = x$ has no maximum and no minimum — the values creep toward 0 and 1 but never reach them, because those endpoints aren't included. - Drop **continuity**: on $[0, 1]$, define $f(x) = x$ for $x < 1$ and $f(1) = 0$. The values climb toward 1 but the function never takes the value 1 (it drops to 0 at the end), so there is no maximum — the discontinuity let the peak escape.

This is exactly the theorem Chapter 6 will open with: Rolle’s proof begins “by the Extreme Value Theorem, f attains a maximum and a minimum,” and the Mean Value Theorem — the engine of the whole back half — follows from there. It is also what guarantees, in Chapter 8, that a global maximum on a closed interval *exists* to be found before you go looking for it.

The hidden foundation: no gaps

Why are these true? Both fail for the *rational* numbers and hold only for the *reals* — and the reason is the one you met in Chapter 1: the real number line has **no gaps** (completeness). On a number line full of holes, a continuous-looking function could slip *between* two values without ever equaling the one in the middle (no IVT), or climb toward a peak that simply isn’t there (no EVT). The no-gaps property is doing the real work — the very same property that guaranteed e exists as the limit of a climbing sequence. The fully rigorous proofs belong to a more advanced course, but the engine driving them is completeness, which you already own.

◆ Layer 2 — Proving the IVT by bisection (and why completeness is the secret)

Here is the IVT’s proof for the root-finding case ($f(a) < 0 < f(b)$), and it *is* the bisection algorithm made rigorous. Look at the midpoint $m = \frac{a+b}{2}$. If $f(m) = 0$ we’re done. Otherwise $f(m)$ is either positive or negative, so the sign change persists across exactly one of $[a, m]$, $[m, b]$ — keep that half and call it $[a_1, b_1]$. Repeat forever, producing nested intervals $[a_1, b_1] \supset [a_2, b_2] \supset \dots$, each half the width of the last, each still straddling a sign change. The left endpoints $a_1 \leq a_2 \leq \dots$ climb

but stay bounded above (by b), so by the **Monotone Convergence principle of Chapter 1** they converge to a limit c — *and this is the only step that uses completeness; on the rationals the a_n might pile up against a hole and there'd be no c .* The interval widths shrink to 0, so the right endpoints converge to the same c . Since f is continuous and each $f(a_n) \leq 0$ while each $f(b_n) \geq 0$, taking limits forces $f(c) \leq 0$ and $f(c) \geq 0$ — so $f(c) = 0$. The root exists, and bisection is exactly the process that finds it. ■ (The EVT's proof runs on the same completeness fuel but is more delicate — first showing a continuous function on $[a, b]$ is *bounded*, then that it *attains* its bound — and is the natural next step in a real analysis course.)

◆ Layer 2 — What “arbitrarily close” really means: the ε - δ definition

This is the grown-up version of Chapter 1's ε - N game, and it is the definition the entire subject rests on. Same two players, one change: the skeptic now challenges the *output*, and you answer by controlling the *input*.

Definition. $\lim_{x \rightarrow a} f(x) = L$ means: for every output tolerance $\varepsilon > 0$, there exists an input closeness $\delta > 0$ such that whenever $0 < |x - a| < \delta$, we have $|f(x) - L| < \varepsilon$.

The skeptic says “keep the output within ε of L .” You reply “then keep the input within δ of a , and I guarantee it.” If you can win for *every* ε the skeptic throws, the limit is L .

Stare at the “ $0 < |x-a|$ ” on the left. That “ $0 <$ ” says x is allowed to be near a but **never equal to a** — we deliberately exclude the point itself. *That single symbol is the loophole made formal:* it is exactly why the function can have a hole at a (as $\frac{x^2-1}{x-1}$ does) and still have a perfectly good limit there. The definition was built from the start to look only at the neighborhood, never the point.

And notice continuity in this language: f is continuous at a precisely when “input within δ forces output within ε ” — which is the robustness promise from the last section, now exact. Small input change, small output change, with the skeptic allowed to define “small.”

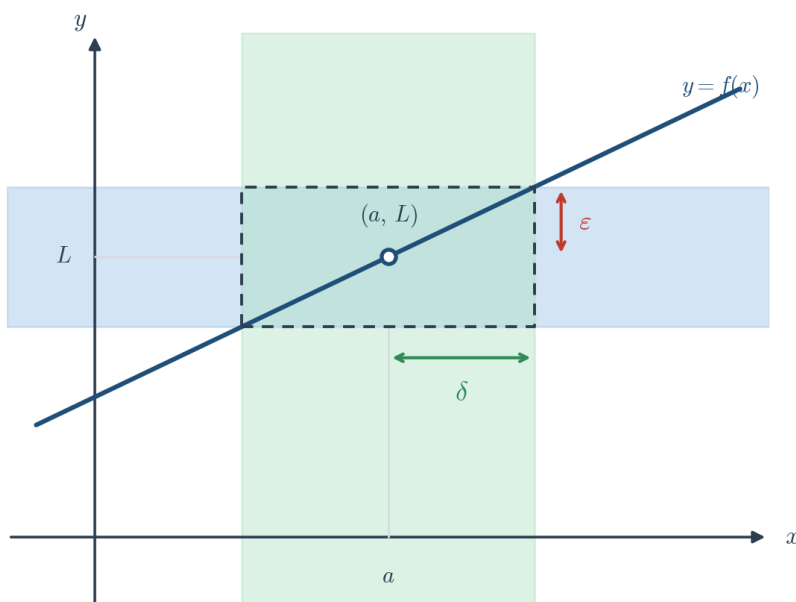


Figure 2.6

◆ **Proof:** $\lim_{x \rightarrow 2} (3x - 1) = 5$

Let $\varepsilon > 0$ be given. We need a δ so that $0 < |x - 2| < \delta$ forces $|(3x - 1) - 5| < \varepsilon$. Work backward from the goal:

$$|(3x - 1) - 5| = |3x - 6| = 3|x - 2|.$$

We want this below ε , i.e. $|x - 2| < \frac{\varepsilon}{3}$. So **choose** $\delta = \frac{\varepsilon}{3}$. Then whenever $0 < |x - 2| < \delta$, we get $|(3x - 1) - 5| = 3|x - 2| < 3\delta = \varepsilon$. Won for every ε . ■

The pattern is the same three-move dance as the sequence proofs: *let ε be given* $\rightarrow$ *solve the output inequality back into an input inequality* $\rightarrow$ *name δ and verify*. Linear functions give a clean $\delta = \varepsilon/(\text{slope})$; curved ones (problem 26) take one extra step.

◆ The squeeze theorem, and the limit that becomes a slope

The squeeze theorem from Chapter 1 lifts to functions word for word:

Squeeze Theorem. If $g(x) \leq f(x) \leq h(x)$ for all x near a , and $g(x) \rightarrow L$ and $h(x) \rightarrow L$, then $f(x) \rightarrow L$.

Its headline application is the limit this whole chapter has been pointing at. (*This box uses sin, cos, tan, the unit circle, and angles in **radians** — the precalculus trigonometry you bring with*

you. We'll prove the derivatives of $\sin$ and $\cos$ in Chapter 4, and this limit is exactly the tool that makes those proofs work.)

Claim: $\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$. Substituting gives $\frac{0}{0}$, so there is real work to do.

Take $0 < x < \frac{\pi}{2}$ and compare three nested regions in the unit circle — a thin triangle inside a circular wedge inside a larger triangle. Their areas give the classic chain

$$\sin x < x < \tan x.$$

This is the step that requires radians: the wedge's arc length equals x only in radian measure. (In degrees, an ugly factor of $\frac{\pi}{180}$ appears and the limit is no longer 1 — which is the honest reason mathematicians abandon degrees for calculus.) Divide all three by $\sin x$ (positive), then flip the fractions:

$$1 < \frac{x}{\sin x} < \frac{1}{\cos x} \quad \implies \quad \cos x < \frac{\sin x}{x} < 1.$$

As $x \rightarrow 0$, $\cos x \rightarrow 1$, so $\frac{\sin x}{x}$ is pinned between something heading to 1 and 1 itself. Squeeze, and $\frac{\sin x}{x} \rightarrow 1$. (For $x < 0$ nothing changes — swapping $x \rightarrow -x$ leaves both $\frac{\sin x}{x}$ and the bounds alone.) ■

Why this is the most important limit in the book: notice it is a $\frac{0}{0}$ limit, and its answer, 1, is the slope of the sine curve as it leaves the origin. In Chapter 3 we will define “slope of a curve” as exactly this kind of $\frac{0}{0}$ limit — so you have, right now and ahead of schedule, computed that the graph of $\sin$ sets off at a perfect 45° . (Figure 2.7: $y = \frac{\sin x}{x}$ near 0, a smooth curve sailing through height 1 with a single open hole exactly at $x = 0$.)

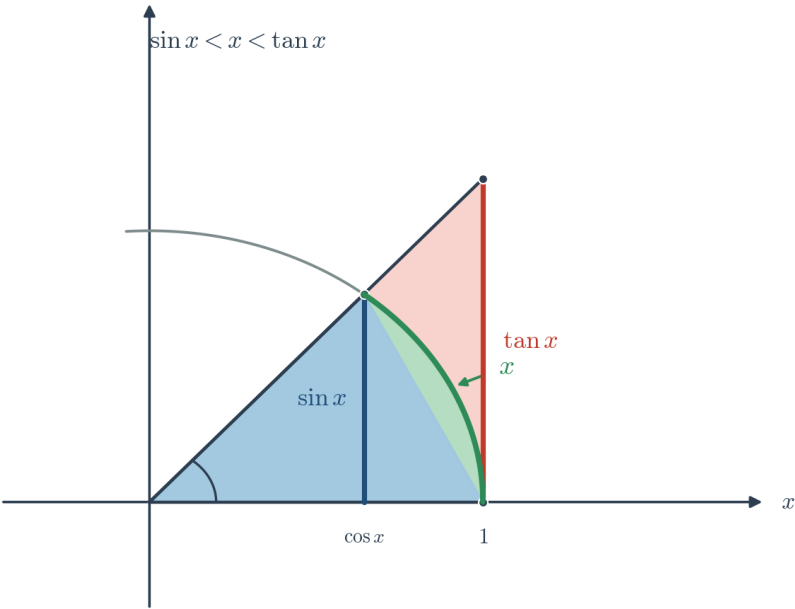


Figure 2.8

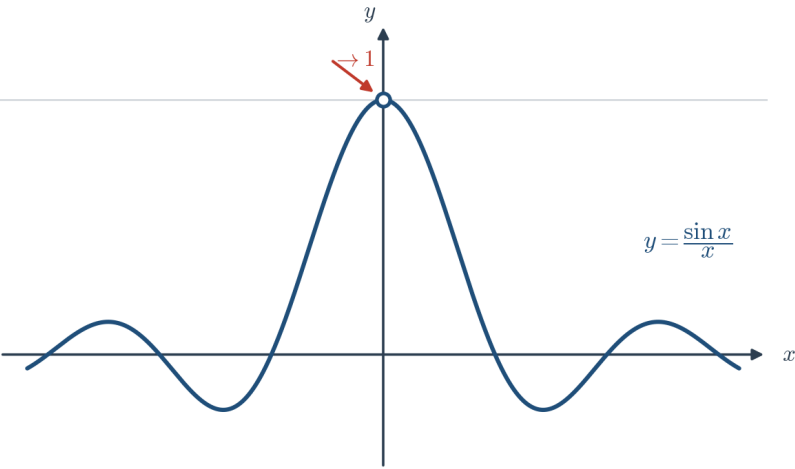


Figure 2.7

2.6 Where this is heading

You now have the one tool Chapter 3 is built from. The slope of a curve at a point is the slope of the line joining two nearby points, $\frac{f(a+h) - f(a)}{h}$, in the limit as the two points slide together — that is, as $h \rightarrow 0$. Substitute $h = 0$ directly and you get $\frac{0}{0}$, every single time. But you are no longer afraid of $\frac{0}{0}$. You know it is not a wall; it is an invitation to cancel. Take that fear away and the derivative — the whole engine of the book — is just one more limit of exactly the kind you spent this chapter learning to read.

The slope of a curve is a limit. Turn the page and we'll take it.

2.7 Problems

Six tiers, and they do different jobs. **Fluency** builds speed until the techniques are automatic — do all of them, fast. **Practice** is multi-step work where you must choose the right tool. **Concept & counterexample** has almost no computation; it tests whether you actually understand limits, and it is where most of the real learning hides — do not skip it. ♦ **Diamond** is Layer 2: proofs and stretch. **Teach it** asks you to explain, write up, or debug. **Looking ahead** quietly hands you Chapter 3. Several problems now exercise the chapter's two theorems — the Intermediate and Extreme Value Theorems — alongside the limit techniques.

Fluency

- By substitution: (a) $\lim_{x \rightarrow 1} (x^3 - 2x + 1)$; (b) $\lim_{x \rightarrow -2} (3x + 1)$; (c) $\lim_{x \rightarrow 0} \cos x$; (d) $\lim_{x \rightarrow 9} \sqrt{x}$.
- Factor and cancel: (a) $\lim_{x \rightarrow 2} \frac{x^2 - 5x + 6}{x - 2}$; (b) $\lim_{x \rightarrow 1} \frac{x^3 - 1}{x - 1}$; (c) $\lim_{x \rightarrow 3} \frac{x^2 - x - 6}{x^2 - 9}$.

3. Rationalize: (a) $\lim_{x \rightarrow 4} \frac{\sqrt{x} - 2}{x - 4}$; (b) $\lim_{x \rightarrow 0} \frac{\sqrt{x+1} - 1}{x}$.
4. At infinity: (a) $\lim_{x \rightarrow \infty} \frac{3x^3 - x}{2x^3 + 5}$; (b) $\lim_{x \rightarrow \infty} \frac{5x + 1}{x^2 + 2}$; (c) $\lim_{x \rightarrow \infty} \frac{4x^2 - 3}{2x^2 + x}$; (d) $\lim_{x \rightarrow \infty} \frac{2x^2 + 1}{x + 1}$.
5. One-sided: for $f(x) = x$ if $x < 1$ and $f(x) = x + 2$ if $x \geq 1$, find $\lim_{x \rightarrow 1^-} f$, $\lim_{x \rightarrow 1^+} f$, and say whether $\lim_{x \rightarrow 1} f$ exists.
6. Continuity check (one line each): is $\frac{1}{x}$ continuous at $x = 2$? at $x = 0$? Is $\sqrt{x}$ continuous at $x = -1$?
7. **Intermediate Value Theorem.** Use the IVT to show each function has a root in the given interval (check the sign at each endpoint): (a) $x^3 + x - 1$ on $[0, 1]$; (b) $\cos x - x$ on $[0, 1]$; (c) $x^5 - 3x + 1$ on $[0, 1]$.
8. **Extreme Value Theorem.** Does the EVT *guarantee* a maximum and a minimum? Answer yes or no for each, with the reason: (a) x^2 on $[-1, 3]$; (b) $\frac{1}{x}$ on $(0, 1]$; (c) x on $[0, 5]$; (d) any continuous function on $[2, 7]$.

Practice

9. $\lim_{x \rightarrow 3} \frac{x^3 - 27}{x^2 - 9}$. (Difference of cubes on top; common factor with the bottom.)
10. Complex fraction: $\lim_{x \rightarrow 3} \frac{\frac{1}{x} - \frac{1}{3}}{x - 3}$. (Combine the small fractions first, then cancel.)
11. An $\infty - \infty$ that hides a finite answer: $\lim_{x \rightarrow \infty} (\sqrt{x^2 + 3x} - x)$. (Rationalize, then divide by the highest power.)
12. $\lim_{x \rightarrow \infty} (\sqrt{4x^2 + x} - 2x)$.
13. Find constants a and b that make f continuous everywhere, where $f(x) = x^2$ for $x \leq 1$, $f(x) = ax + b$ for $1 < x < 3$, and $f(x) = 4$ for $x \geq 3$. (Two matching conditions, two unknowns.)
14. Is f continuous at $x = 1$, where $f(x) = \frac{x^2 - 1}{x - 1}$ for $x < 1$ and $f(x) = 2x$ for $x \geq 1$? Justify with the three-part continuity test. (A repaired

hole meeting a line — do they actually agree?)

15. **Bisection by hand.** You showed in problem 7 that $x^3 + x - 1$ has a root in $[0, 1]$. Do two steps of bisection: evaluate f at the midpoint 0.5, decide which half holds the root, then bisect that half. Report the interval of width 0.25 that traps the root.
16. **The IVT in the world.** At 6 a.m. the temperature was 40°F; at 2 p.m. it was 75°F. Explain why, assuming temperature varies continuously, there must have been a moment when it read exactly 60°F — and explain why this argument cannot tell you *when*.

Concept & counterexample

State true or false and **justify** — a one-line reason or, for a false claim, an explicit counterexample.

17. (a) If $\lim_{x \rightarrow a} f(x) = L$, then $f(a) = L$. (b) If $f(a) = 5$, then $\lim_{x \rightarrow a} f(x) = 5$. (c) If $\lim_{x \rightarrow a^-} f$ and $\lim_{x \rightarrow a^+} f$ both exist, then $\lim_{x \rightarrow a} f$ exists. (d) If f and g each fail to have a limit at a , then $f + g$ fails to have a limit at a .
18. **Construct** a function f with $\lim_{x \rightarrow 2} f(x) = 4$ but $f(2) = 10$. (Then say which type of discontinuity you built.)
19. **Construct** a single function that is continuous everywhere except for a *jump* at $x = 0$ and a *removable hole* at $x = 1$. (Write it piecewise; sketch it.)
20. **Find the error.** A student writes: “To find $\lim_{x \rightarrow 3} \frac{x^2 - 9}{x - 3}$, plug in $x = 3$ to get $\frac{0}{0}$. Since $\frac{0}{0}$ is undefined, the limit does not exist.” Identify the exact mistake and give the correct answer.
21. **Construct** two functions f and g , *each* with no limit as $x \rightarrow 0$, whose product fg *does* have a limit at 0. (This is the multiplicative version of 17(d).)
22. Your friend says: “A function has a hole at $x = 2$. Just define $f(2)$ to be any number you like and now it’s continuous.” When are they right, and when are they wrong? State the precise condition.

23. True or false, and **justify**: (a) If f is continuous on $[a, b]$ and $f(a)$ and $f(b)$ have the same sign, then f has no root in $[a, b]$. (b) A continuous function on an *open* interval (a, b) always attains a maximum. (c) If a continuous f takes both a positive and a negative value on $[a, b]$, it must have a root there. (d) The IVT tells you exactly how many roots a function has.
24. **Construct** a function on the *closed* interval $[0, 1]$ that attains **no maximum**, and state precisely which hypothesis of the Extreme Value Theorem your example breaks. (Why must any such example be discontinuous?)

◆ Diamond — Layer 2 reach problems.

25. ◆ Prove $\lim_{x \rightarrow 2} (3x - 1) = 5$ from the ε - δ definition, finding δ in terms of ε .
26. ◆ Prove $\lim_{x \rightarrow 3} x^2 = 9$ from ε - δ . (Hint: $|x^2 - 9| = |x - 3||x + 3|$; first restrict $\delta \leq 1$ so $|x + 3| \leq 7$, then take $\delta = \min(1, \frac{\varepsilon}{7})$.)
27. ◆ Prove $\lim_{x \rightarrow 2} \frac{1}{x} = \frac{1}{2}$ from ε - δ . (The new wrinkle: the denominator. Show $|\frac{1}{x} - \frac{1}{2}| = \frac{|x-2|}{2|x|}$, then restrict $\delta \leq 1$ to keep $|x| \geq 1$ so the denominator can't sabotage you; conclude $\delta = \min(1, 2\varepsilon)$.)
28. ◆ Prove the **sum law**: if $\lim_{x \rightarrow a} f = L$ and $\lim_{x \rightarrow a} g = M$, then $\lim_{x \rightarrow a} (f + g) = L + M$. (Given ε , secure each piece within $\frac{\varepsilon}{2}$, then use the triangle inequality $|(f + g) - (L + M)| \leq |f - L| + |g - M|$.)
29. ◆ Prove the **squeeze-by-zero lemma**: if $|f(x)| \leq M$ for all x near a and $g(x) \rightarrow 0$, then $f(x)g(x) \rightarrow 0$. Then use it to re-derive $\lim_{x \rightarrow 0} x^2 \sin \frac{1}{x} = 0$ in one line.
30. ◆ Prove the Squeeze Theorem for functions from the ε - δ definition.
31. ◆ Prove $\lim_{x \rightarrow 0} \frac{1 - \cos x}{x} = 0$ using $\frac{\sin x}{x} \rightarrow 1$. (Multiply by $\frac{1 + \cos x}{1 + \cos x}$, turn $1 - \cos^2 x$ into $\sin^2 x$, then split.)
32. ◆ Using $\frac{\sin x}{x} \rightarrow 1$, evaluate: (a) $\lim_{x \rightarrow 0} \frac{\sin 5x}{x}$; (b) $\lim_{x \rightarrow 0} \frac{\sin 2x}{\sin 3x}$; (c) $\lim_{x \rightarrow 0} \frac{\tan x}{x}$. (For each, manufacture the $\frac{\sin(\text{stuff})}{\text{stuff}}$ shape by hand.)

33. **♦ A fixed point.** Prove that every continuous function $f : [0, 1] \rightarrow [0, 1]$ has a *fixed point* — a c with $f(c) = c$. (Apply the IVT to $g(x) = f(x) - x$: what are the signs of $g(0)$ and $g(1)$, given that f 's outputs lie in $[0, 1]$?) This is a one-dimensional cousin of a celebrated theorem of Brouwer.

Teach it — *explain it, write it up, or fix it for someone else*

34. A classmate has absorbed the rule “if substitution gives $\frac{0}{0}$, the limit doesn't exist.” In **three or four sentences written to them**, explain why that's wrong and give them the correct rule, using $\frac{x^2-9}{x-3}$ as your worked counterexample.
35. Write a complete, textbook-quality **worked solution** to $\lim_{x \rightarrow 0} \frac{\sqrt{x+4} - 2}{x}$, narrating *why* you multiply by the conjugate and what it accomplishes, so a peer could reproduce the move on a new problem.
36. **Debug your classmate.** They claim $\lim_{x \rightarrow 0} \frac{|x|}{x} = 0$ “because the left side gives -1 and the right side gives $+1$, and those average to 0 .” Pinpoint the precise error and state what is actually true about this limit.
37. **Design a problem.** Invent a single piecewise function that has a *removable hole* at $x = -1$ and a *jump* at $x = 2$. Write it out, then write the solution key that identifies each discontinuity and the one-sided limits at $x = 2$.
38. In two or three sentences, **teach the single most important idea of this chapter** — that $\lim_{x \rightarrow a} f(x)$ and $f(a)$ are different questions — to someone who has never seen a limit, using one concrete example where the two answers disagree.
39. In **three or four sentences with a real-world picture** (a hiker's altitude, a warming room, a car's speedometer), teach a friend the Intermediate Value Theorem — why a continuously changing quantity cannot skip over a value on its way from one number to another

— and name the single hypothesis that makes it true.

Looking ahead — *you are about to compute derivatives with nothing but this chapter*

40. For $f(x) = x^2$, expand and simplify $\frac{f(2+h) - f(2)}{h}$, then take $\lim_{h \rightarrow 0}$. (You should get a clean number. That number is the slope of x^2 at $x = 2$.)
 41. Repeat 40 at a *general* point: find $\lim_{h \rightarrow 0} \frac{(a+h)^2 - a^2}{h}$ in terms of a . (You have just found the slope of x^2 at every point at once — keep the answer; Chapter 3 will call it $\frac{d}{dx}x^2$.)
 42. For $f(x) = \frac{1}{x}$, find $\lim_{h \rightarrow 0} \frac{\frac{1}{a+h} - \frac{1}{a}}{h}$. (This is problem 10's complex-fraction move again — notice you already practiced exactly this.)
 43. For $f(x) = \sqrt{x}$, find $\lim_{h \rightarrow 0} \frac{\sqrt{a+h} - \sqrt{a}}{h}$. (Rationalize, as in problem 3.) When you finish 40–43, look back: every technique in this chapter existed to make these four limits computable. That is the whole point, and it is the first page of Chapter 3.
 44. In problem 15 you trapped a root by bisection — reliable, but slow, halving the interval each step. Chapter 9 builds a far *faster* root-finder, Newton's method, which uses the **slope** of f (the derivative Chapter 3 is about) to leap toward the root. Predict one thing that could go wrong with a method that blindly follows the tangent line — a failure the plodding bisection method never suffers.
-

Chapter 3

The Slope of a Single Point

Or: how fast is something changing right now, when “rate” seems to need two moments?

Companion notebook. Play with this chapter’s ideas — sliders, animations, and drills — in notebooks/`ch3_derivative.ipynb`.

This is the chapter the whole book is named for. By the end of it you will be able to do the central act of calculus: find the **exact rate at which a function is changing at a single instant** — the steepness of a curve at one point, the speed of a moving thing at one moment — and you will do it *from scratch*, with nothing but the limits you learned in Chapter 2. No formulas handed to you, no rules to memorize yet. You will compute derivatives the way they were first computed: by hand, from the definition, watching the answer fall out.

You’ll find the slope of x^2 , of x^3 , of $\frac{1}{x}$, of $\sqrt{x}$ — each one earned, not received. You’ll see that the result is itself a **function**, a machine that returns the slope at every point at once. You’ll connect it to **velocity**, the most physical version of the idea. And in the Layer-2 climb you’ll discover the one place this can fail — a function so sharp it has no slope at all — which turns out to be deeply connected to the corners and jumps of Chapter 2.

Here is the paradox we have to resolve, because it’s what makes the whole thing interesting. “Rate of change” normally compares two moments: you went 120 miles in 2 hours, so your *average* speed was 60. But

your speedometer shows a speed at a single instant, with no interval at all. How can there be a rate of change *at one point*, when measuring a rate seems to require two? The answer is the limit, and it is the most important idea you will learn this year.

3.1 From two points to one: the difference quotient

Start where it's easy: the rate of change over an **interval**. Take a function f and look at it between $x = a$ and a nearby point $x = a+h$ (here h is a small step, the gap between the two points). Over that interval, the **average rate of change** is the change in output divided by the change in input:

$$\frac{f(a+h) - f(a)}{h}.$$

Geometrically, this is the **slope of the secant line** — the straight line through the two points $(a, f(a))$ and $(a+h, f(a+h))$ on the graph. (You'll meet this same quantity again in Chapter 6, where the Mean Value Theorem calls it the “average slope.”) There's nothing mysterious here yet; it's rise over run between two points. (*Figure 3.1: a curve with two points marked and the secant line through them, the rise and run labeled.*)

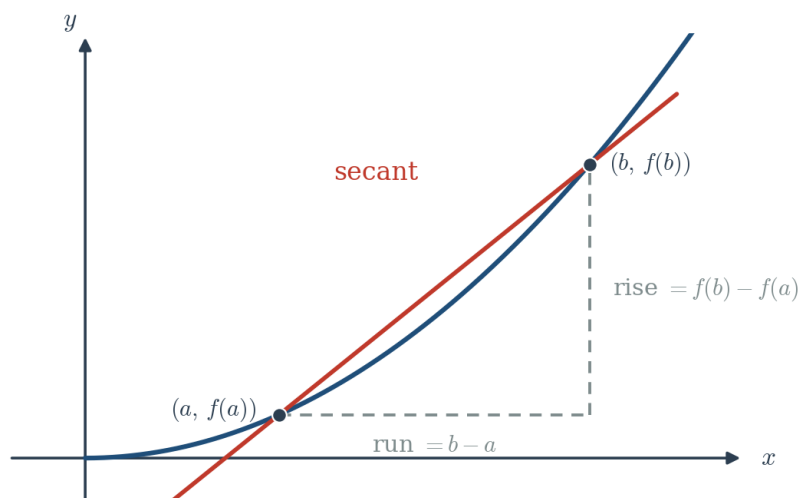


Figure 3.1

Now the move that creates calculus. We don't want the average rate over an interval — we want the rate *at the single point a* . So we slide the second point toward the first: let h shrink toward 0. The two points merge, the secant line pivots, and — if all goes well — it settles onto a single line just **touching** the curve at a : the **tangent line**. Its slope is the rate of change at that one instant. (Figure 3.2: the same curve with a sequence of secant lines as the second point slides in, converging onto the tangent line at a .)

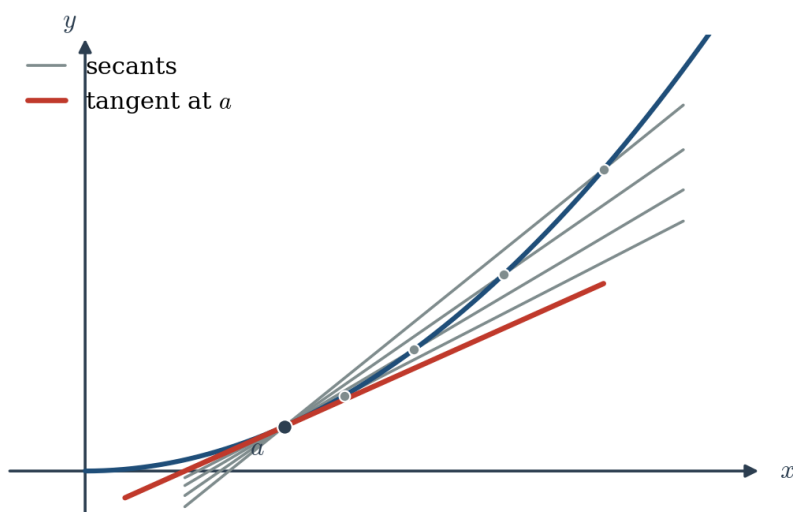


Figure 3.2

But look at what happens to the formula as $h \rightarrow 0$. Both the top, $f(a + h) - f(a)$, and the bottom, h , go to 0. We are staring straight at $\frac{0}{0}$ — the indeterminate form from Chapter 2. And that is exactly why we spent a whole chapter learning that $\frac{0}{0}$ is **not a wall**. It's an invitation to simplify and cancel. The rate of change at a point is a $\frac{0}{0}$ limit, and we already know how to handle those.

3.2 The derivative

Here is the definition the entire back half of this book rests on.

The derivative of f at the point a , written $f'(a)$, is

$$f'(a) = \lim_{h \rightarrow 0} \frac{f(a + h) - f(a)}{h},$$

when this limit exists. It is the instantaneous rate of change of f at a , and the slope of the tangent line to the graph at a .

That's it. The derivative is the limit of the average rate as the interval shrinks to nothing. Three names for one number: *instantaneous rate of change*, *slope of the tangent line*, *the derivative*. Now let's actually compute some — and notice that every single computation is the same dance: **write the difference quotient, simplify until the h in the denominator cancels, then let $h \rightarrow 0$.**

3.3 Computing derivatives by hand

The slope of x^2 . Let $f(x) = x^2$. Build the difference quotient at a general point a and simplify:

$$\frac{(a+h)^2 - a^2}{h} = \frac{a^2 + 2ah + h^2 - a^2}{h} = \frac{2ah + h^2}{h} = \frac{h(2a + h)}{h} = 2a + h.$$

The h canceled — the $\frac{0}{0}$ dissolved, exactly as in Chapter 2. Now let $h \rightarrow 0$:

$$f'(a) = \lim_{h \rightarrow 0} (2a + h) = 2a.$$

So the slope of $y = x^2$ at any point a is $2a$. At $x = 3$ the slope is 6; at $x = 0$ it's 0 (the bottom of the parabola is flat); at $x = -1$ it's -2 (sloping down). You computed a derivative — and if you did the Chapter 2 “Looking ahead” problems, you've seen this exact calculation before; now it has a name.

The slope of x^3 . Same dance, one more term to expand:

$$\frac{(a+h)^3 - a^3}{h} = \frac{3a^2h + 3ah^2 + h^3}{h} = 3a^2 + 3ah + h^2 \xrightarrow{h \rightarrow 0} 3a^2.$$

So the derivative of x^3 is $3x^2$.

The slope of $\frac{1}{x}$. Now a fraction. Let $f(x) = \frac{1}{x}$. The difference quotient is a “fraction of fractions” — combine the top over a common denominator first:

$$\frac{\frac{1}{a+h} - \frac{1}{a}}{h} = \frac{\frac{a-(a+h)}{a(a+h)}}{h} = \frac{-h}{h a(a+h)} = \frac{-1}{a(a+h)} \xrightarrow{h \rightarrow 0} \frac{-1}{a^2}.$$

So the derivative of $\frac{1}{x}$ is $-\frac{1}{x^2}$ — always negative, because $\frac{1}{x}$ always slopes downward.

The slope of $\sqrt{x}$. A square root: rationalize, the trick from Chapter 2. Let $f(x) = \sqrt{x}$ and multiply top and bottom by the conjugate $\sqrt{a+h} + \sqrt{a}$:

$$\frac{\sqrt{a+h} - \sqrt{a}}{h} = \frac{(a+h) - a}{h(\sqrt{a+h} + \sqrt{a})} = \frac{1}{\sqrt{a+h} + \sqrt{a}} \xrightarrow{h \rightarrow 0} \frac{1}{2\sqrt{a}}.$$

So the derivative of $\sqrt{x}$ is $\frac{1}{2\sqrt{x}}$.

The two simplest of all. A constant function $f(x) = c$ never changes, so its difference quotient is $\frac{c-c}{h} = 0$ — derivative 0, as it must be (a flat line has slope 0). A line $f(x) = mx + b$ has difference quotient $\frac{[m(a+h)+b] - [ma+b]}{h} = \frac{mh}{h} = m$ — derivative m , the slope it always had. A line is its own slope; that’s the whole point of a line.

Toolkit: Computing a derivative from the definition

1. Write the difference quotient $\frac{f(a+h) - f(a)}{h}$.
2. **Simplify until the h in the denominator cancels** — expand and combine (x^2, x^3), find a common denominator (fractions), or rationalize (roots).
3. Let $h \rightarrow 0$ in what’s left.

Every derivative in this chapter is these three steps. The algebra changes; the dance never does.

3.4 The derivative is a function

Look back at what we found: $f'(a) = 2a$ held for *every* value of a . So the derivative isn't really a single number attached to one point — it's a **rule** that takes any point and returns the slope there. That rule is itself a function, and we write it with the variable x :

$$\frac{d}{dx} x^2 = 2x, \quad \frac{d}{dx} x^3 = 3x^2, \quad \frac{d}{dx} \frac{1}{x} = -\frac{1}{x^2}, \quad \frac{d}{dx} \sqrt{x} = \frac{1}{2\sqrt{x}}.$$

This reframe is bigger than it looks, and the rest of the book depends on it. The derivative of f is a new function f' , which you can then graph, evaluate, or **differentiate again** (giving f'' , the second derivative — the star of Chapters 7 and 8). The derivative is not just an answer; it's an object you can keep working with.

A word on **notation**, because you'll see two styles everywhere. Lagrange's $f'(x)$ is compact — “ f prime of x .” Leibniz's $\frac{df}{dx}$ or $\frac{dy}{dx}$ keeps the idea of the difference quotient visible: it's the limit of $\frac{\Delta y}{\Delta x}$ (change in output over change in input) as the changes shrink to nothing. The symbol $\frac{d}{dx}$ all by itself is read as an *operator* — “take the derivative with respect to x of whatever comes next.” Both names mean the identical thing; we'll use whichever is clearer, and you should grow comfortable with both. (Leibniz's notation will earn its keep especially in Chapter 4's chain rule, where it almost does the thinking for you.)

3.5 The most physical version: velocity

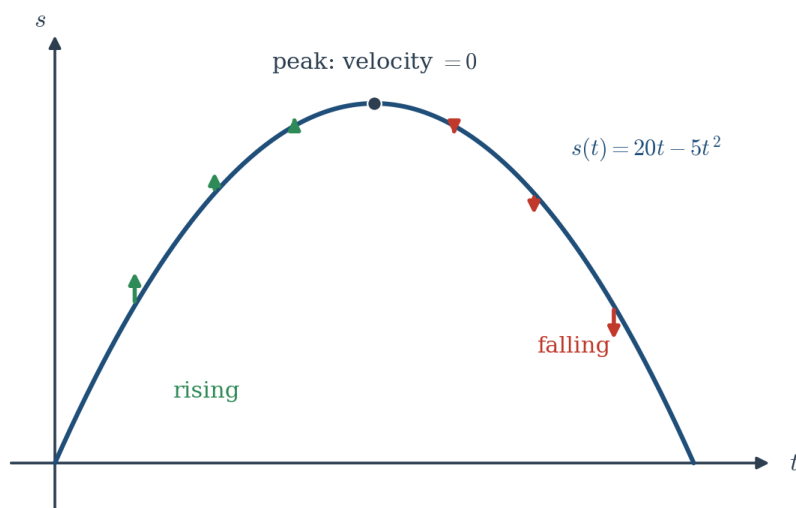
The reason this idea felt urgent to the people who invented it was **motion**. Let $s(t)$ be the position of a moving object at time t . Over a time interval, its average speed is (change in position)/(change in time) — a difference quotient. Shrink the interval to an instant, and you get the **instantaneous velocity**: the number on the speedometer, the rate of change of position *right now*.

$$v(t) = s'(t) = \lim_{h \rightarrow 0} \frac{s(t+h) - s(t)}{h}.$$

Worked example — a thrown ball. Throw a ball straight up, and (ignoring air, with gravity rounded to 10 m/s^2) its height after t seconds is $s(t) = -5t^2 + 20t$ meters. Its velocity is the derivative — which we compute the only way we know, from the definition:

$$\frac{s(t+h) - s(t)}{h} = \frac{-5(t+h)^2 + 20(t+h) - (-5t^2 + 20t)}{h} = \frac{-10th - 5h^2 + 20h}{h} = -10t - 5h$$

so $v(t) = s'(t) = -10t + 20$. At launch ($t = 0$) the ball rises at 20 m/s ; the velocity drops steadily; and at $t = 2$ it hits $v = 0$ — the ball is momentarily *not moving*, which is exactly the **top** of its flight, the instant before it falls back. Hold onto that: the highest point is where the velocity (the derivative) is zero. That is your first glimpse of Chapter 8, where every maximum and minimum is found by setting the derivative to zero. (Figure 3.3: the height curve $s(t)$ as an arch, with the velocity arrows shrinking to zero at the peak and reversing after.)

*Figure 3.3*

3.6 The tangent line: a function's best straight-line stand-in

The derivative hands you the slope of the tangent at a , and with a slope and a point you can write the tangent line:

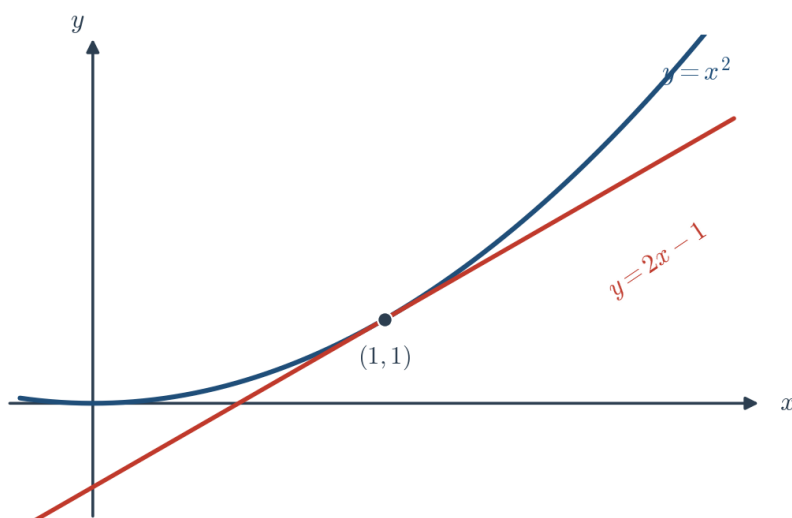


Figure 3.4

$$y = f(a) + f'(a)(x - a)$$

Example. The tangent to $y = x^2$ at $x = 1$: here $f(1) = 1$ and $f'(1) = 2(1) = 2$, so the line is $y = 1 + 2(x - 1) = 2x - 1$. Near $x = 1$, the curve x^2 and the line $2x - 1$ are almost indistinguishable.

That “almost indistinguishable near a ” is a profound idea in disguise, and we’ll name it fully in Chapter 7: **the tangent line is the best straight-line approximation of a curve near a point**. Zoom in on any smooth curve far enough and it looks straight — and the straight line it looks like is the tangent, whose slope is the derivative. To know a function’s derivative at a is to know the simplest possible model of it right there.

◆ Layer 2 — When a function has no slope, and what that has to do with Chapter 2

The derivative is a *limit*, and limits don't always exist — so neither do derivatives. The cleanest example is one you already know the shape of: $f(x) = |x|$ at $x = 0$. It's a perfectly continuous function (you can draw it without lifting your pen), but it has a sharp **corner** at the origin, and a corner has no single slope. Watch the difference quotient confirm it. At $a = 0$, the quotient is $\frac{|0 + h| - 0}{h} = \frac{|h|}{h}$, which is the sign function from Chapter 2: - from the **right** ($h > 0$): $\frac{|h|}{h} = +1$; - from the **left** ($h < 0$): $\frac{|h|}{h} = -1$.

The two one-sided limits disagree, so the two-sided limit — the derivative — **does not exist**. This is *exactly* the $\frac{|x|}{x}$ jump from Chapter 2, reappearing as the slope of $|x|$. The corner of one graph is the jump of its derivative.

So differentiability is genuinely *stronger* than continuity — every differentiable function is continuous (proved below), but not every continuous function is differentiable. Corners (like $|x|$), cusps, and vertical tangents are all continuous-but-not-differentiable. This is the third floor of the “smoothness has consequences” story from Chapters 2 and 6, seen from below: differentiability asks for more than continuity, and a function can pass the first test and fail the second.

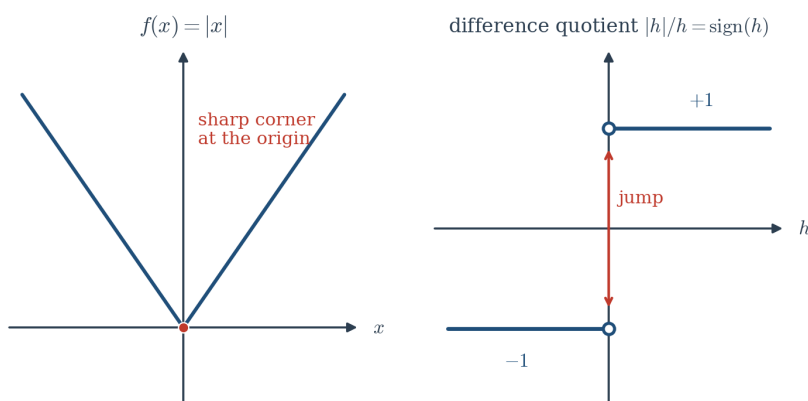


Figure 3.5

◆ Layer 2 – Proof that differentiable $\Rightarrow$ continuous

Suppose $f'(a)$ exists. We must show f is continuous at a , i.e. $f(a+h) \rightarrow f(a)$ as $h \rightarrow 0$. The trick is to write the change in f as the difference quotient times h :

$$f(a+h) - f(a) = \underbrace{\frac{f(a+h) - f(a)}{h}}_{\rightarrow f'(a)} \cdot h.$$

As $h \rightarrow 0$, the first factor approaches the finite number $f'(a)$ and the second approaches 0, so the product approaches $f'(a) \cdot 0 = 0$. Therefore $f(a+h) \rightarrow f(a)$: f is continuous at a . ■ (The converse fails — $|x|$ is the counterexample — so “differentiable” sits *strictly* above “continuous.”)

3.7 Where this is heading

You now have the engine of all of calculus — but computing each derivative from the definition is laborious, and you may already be tired of expanding $(a + h)^n$ and rationalizing roots. **Chapter 4** fixes that: it proves a handful of **rules** (power, product, quotient, chain) — each derived once from the very definition you just learned — so that you can differentiate almost anything in seconds, never touching a difference quotient again. (In the Diamond problems below you'll even derive the power rule $\frac{d}{dx}x^n = nx^{n-1}$ for yourself, straight from the binomial theorem.) The same definition powers everything after: **Chapter 5** computes the derivative of e^x from first principles; **Chapter 6**'s Fermat's lemma is built from the one-sided difference quotients you just met at the corner of $|x|$; **Chapter 8** finds every peak and valley by setting the derivative to zero, exactly as the thrown ball went still at the top of its arc.

One limit — the slope of a single point — and the rest of the book is its consequences.

3.8 Problems

*Six tiers. **Fluency and Practice** are doing* — compute derivatives from the definition until the three-step dance is automatic. **Concept & counterexample** and ♦ **Diamond** are *thinking* — where derivatives fail, and the proofs. **Teach it** is *teaching*. **Looking ahead** opens Chapter 4. Do all of Fluency; the Concept tier is where the continuity/differentiability distinction sets. (Every derivative here is computed **from the definition** — you have no rules yet. That's the point.)**

Fluency — *do these from the definition, fast*

- Using the difference quotient, find the derivative of: (a) $f(x) = x^2$; (b) $f(x) = 3x^2$; (c) $f(x) = x^2 + 5$.
- From the definition, find the derivative of: (a) $f(x) = 5$ (a constant); (b) $f(x) = 7x - 2$ (a line); (c) $f(x) = -x$.
- From the definition, find the derivative of: (a) $f(x) = x^3$; (b) $f(x) = x^4$. (Expand $(a + h)^n$.)
- From the definition, find the derivative of $f(x) = 3x^2 - 4x + 1$, then evaluate it at $x = 2$.
- From the definition, find the derivative of: (a) $f(x) = \frac{1}{x}$; (b) $f(x) = \frac{1}{x^2}$. (Combine over a common denominator.)
- From the definition, find the derivative of $f(x) = \sqrt{x}$, then its value at $x = 9$. (Rationalize.)
- For $f(x) = x^2$, find the slope at $x = 0$, $x = 1$, and $x = -2$. Where is the slope zero?
- Find the equation of the tangent line to $y = x^2$ at: (a) $x = 1$; (b) $x = 3$.
- A ball's height is $s(t) = -5t^2 + 30t$ meters. From the definition, find its velocity $v(t)$, then $v(0)$ and the time when $v = 0$.

Practice — *harder from-scratch, tangents, and motion*

- From the definition, find the derivative of $f(x) = \frac{1}{x+1}$.
- From the definition, find the derivative of $f(x) = \sqrt{x+3}$.
- From the definition, find the derivative of $f(x) = x^2 + \frac{1}{x}$. (Handle the two pieces in one difference quotient.)
- Find the tangent line to $y = \frac{1}{x}$ at $x = 2$, and the tangent line to $y = \sqrt{x}$ at $x = 4$.
- For $f(x) = x^3 - 3x$, find $f'(x)$ from the definition, then find **where the tangent line is horizontal** (where $f'(x) = 0$). (You're previewing how Chapter 8 finds peaks and valleys.)
- A particle's position is $s(t) = t^2 - 6t + 5$. Find $v(t)$ from the definition, the time the particle is momentarily at rest, and whether it's moving

left or right at $t = 1$.

16. Find the point on $y = x^2$ where the tangent line has slope 10. (Set the derivative equal to 10 and solve.)
17. The derivative of $\sqrt{x}$ is $\frac{1}{2\sqrt{x}}$. Use the tangent line at $x = 100$ to *estimate* $\sqrt{101}$ without a calculator. (This is the “best linear approximation” idea at work.)

Concept & counterexample — *think; true/false with justification, or build the example*

18. (a) The derivative at a point is a number, but the derivative is also a function. (b) Every continuous function is differentiable. (c) If $f'(a) = 0$, the graph has a horizontal tangent at a . (d) A function can be differentiable at a point where it is not continuous.
19. **Find the error.** A student computes the derivative of x^2 as: “ $\frac{(a+h)^2 - a^2}{h} = \frac{2ah + h^2}{h}$, and setting $h = 0$ gives $\frac{0}{0}$, so the derivative doesn’t exist.” What did they do wrong?
20. **Construct** a function that is continuous everywhere but **not differentiable** at $x = 0$. Explain, using one-sided difference quotients, why the derivative fails to exist there.
21. The function $|x|$ is continuous at 0 but not differentiable there. Is the reverse possible — a function **differentiable** at 0 but **not continuous** there? Justify with the theorem from the chapter.
22. **Is it possible?** A function whose derivative is 0 at every point. What must such a function look like, and why? (You’ll prove this rigorously in Chapter 6 — guess the answer now.)
23. Sketch (or describe) the graph of f' for a function f that increases, levels off at a peak, then decreases. What is the sign of f' in each region, and what is f' at the peak?

◆ **Diamond — the proofs, and the rule you’ll meet next chapter**

24. ◆ Prove that **differentiable implies continuous**: if $f'(a)$ exists, then f is continuous at a . (Write $f(a+h) - f(a)$ as the difference quotient times h , and take $h \rightarrow 0$.)
25. ◆ Prove that $|x|$ is **not** differentiable at 0 by computing the left- and right-hand limits of the difference quotient and showing they disagree.
26. ◆◆ **Derive the power rule**. Using the binomial theorem $(x+h)^n = x^n + nx^{n-1}h + \binom{n}{2}x^{n-2}h^2 + \dots$, show from the definition that $\frac{d}{dx}x^n = nx^{n-1}$ for every positive integer n . (Every term after the first carries an h and dies in the limit — the same “everything but one term survives” pattern as Chapter 1’s $n^{1/n}$ and Chapter 7’s Taylor coefficients.)
27. ◆◆ A function with a **vertical** tangent. Show that $f(x) = x^{1/3}$ has difference quotient at 0 equal to $\frac{h^{1/3}}{h} = h^{-2/3}$, and that this $\rightarrow +\infty$ as $h \rightarrow 0$. Conclude the derivative doesn’t exist at 0 — but for a *different* reason than $|x|$. (One has a corner; this has a vertical tangent. Describe the difference geometrically.)
28. ◆◆ **One-sided derivatives**. Define the right-hand derivative as $\lim_{h \rightarrow 0^+} \frac{f(a+h) - f(a)}{h}$ and the left-hand as the same with $h \rightarrow 0^-$. Show f is differentiable at a exactly when both exist and are equal, and use this to re-explain the $|x|$ corner. (These one-sided difference quotients are precisely the tool behind Chapter 6’s Fermat’s lemma.)

Teach it — explain it, write it up, or fix it

29. In **three or four sentences with the secant-to-tangent picture**, teach a friend what a derivative is and how shrinking h turns an average rate into an instantaneous one.
30. **Debug your classmate**. They write: “The derivative of x^2 is

$\frac{(x+h)^2 - x^2}{h}$.” Explain what they’ve left out, and write the correct two final steps.

31. Write a complete **worked solution** finding the derivative of $\frac{1}{x}$ from the definition, narrating *why* you combine fractions before letting $h \rightarrow 0$, so a peer could redo it for $\frac{1}{x+2}$.
32. **Design a problem.** Invent a function whose derivative-from-the-definition gives a clean answer, and write the full solution key showing all three steps of the dance.
33. In a few sentences, **teach the difference between continuous and differentiable**, using $|x|$ at 0 as the example that separates them.

Looking ahead — Chapter 4 in your hands early

34. You’ve now computed $\frac{d}{dx}x^2 = 2x$, $\frac{d}{dx}x^3 = 3x^2$, $\frac{d}{dx}x^4 = 4x^3$. **Predict** $\frac{d}{dx}x^5$ and $\frac{d}{dx}x^{10}$ from the pattern — then notice that doing each from the definition would be miserable, which is exactly why Chapter 4 proves the rule once and for all.
 35. You found $\frac{d}{dx}(3x^2 - 4x + 1) = 6x - 4$ — the derivative of a sum, term by term. **Predict** the rule: how does the derivative of $f(x) + g(x)$ relate to $f'(x)$ and $g'(x)$?
 36. Chapter 5 will compute $\frac{d}{dx}e^x$ from this same definition — and the answer will be e^x itself. Using the definition, write down the limit you’d have to evaluate, and identify the one piece of it ($\frac{e^h - 1}{h}$) whose value will decide the whole thing.
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Chapter 4

A Toolkit for Slopes

Or: how to differentiate almost anything without ever touching a difference quotient again

Companion notebook. Play with this chapter’s ideas — sliders, animations, and drills — in `notebooks/ch4_rules.ipynb`.

In Chapter 3 you computed derivatives the hard way — every one from the definition, expanding $(a + h)^n$ and rationalizing roots until the h canceled. That was the right way to *learn* what a derivative is. It is a terrible way to *use* them. This chapter fixes that, permanently.

By the end you will own a handful of **rules** — power, sum, product, quotient, chain — that let you differentiate polynomials, roots, fractions, compositions, trigonometric functions, and exponentials in seconds, never again returning to a limit. But here is the rule of *this* book: you will not be handed these rules to memorize. **Every one of them is derived from the definition you learned last chapter.** You’ll see exactly why each is true, which is the only way to know when you’re using it correctly.

One of them — the **chain rule** — stands above the rest. It is the rule for differentiating a function *inside* a function, and it is the single most important computation in modern machine learning: training a neural network is the chain rule applied, backward, across millions of nested functions. We’ll get there. First, the easy wins.

4.1 The power rule

In Chapter 3 you computed, by hand, that $\frac{d}{dx}x^2 = 2x$, $\frac{d}{dx}x^3 = 3x^2$, $\frac{d}{dx}x^4 = 4x^3$ — and in the Diamond problems you proved the pattern for *every* positive whole number using the binomial theorem:

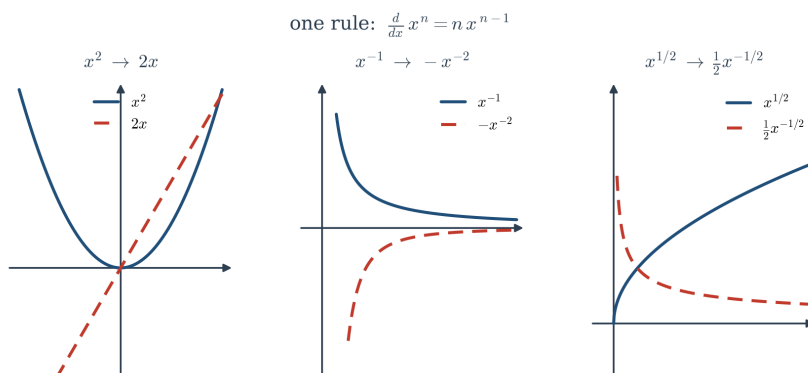


Figure 4.1

The Power Rule. $\frac{d}{dx}x^n = nx^{n-1}$.

What's remarkable is that this one formula also absorbs the *other* derivatives you computed from scratch. Recall $\frac{1}{x} = x^{-1}$ and $\sqrt{x} = x^{1/2}$. Plug those exponents into the power rule:

$$\frac{d}{dx}x^{-1} = -1 \cdot x^{-2} = -\frac{1}{x^2}, \quad \frac{d}{dx}x^{1/2} = \frac{1}{2}x^{-1/2} = \frac{1}{2\sqrt{x}}.$$

Both match exactly what you ground out by hand in Chapter 3 — the reciprocal and the square root were special cases of the power rule all along. In fact the rule nx^{n-1} holds for **every real exponent** n — negative, fractional, even irrational. (The whole-number case you've proven; the negative case follows from the quotient rule below; the general case needs $x^n = e^{n \ln x}$ and waits for Chapter 5. We'll use the full rule now and settle the last piece there — an honest loan.) From now on, no more expanding $(a + h)^n$. Just bring the exponent down and subtract one.

4.2 Sums, differences, and constant multiples

In Chapter 3 you found $\frac{d}{dx}(3x^2 - 4x + 1) = 6x - 4$ — you differentiated term by term, and you predicted the rule in the Looking-ahead problems. Here it is, and it's immediate from the definition (the difference quotient of a sum is the sum of the difference quotients):

$$\text{Sum / Constant-Multiple Rules. } \frac{d}{dx}[f(x) + g(x)] = f'(x) + g'(x), \quad \frac{d}{dx}[c f(x)] = c f'(x).$$

Together with the power rule, these differentiate **any polynomial instantly**. For example, $\frac{d}{dx}(5x^4 - 2x^3 + 7x - 9) = 20x^3 - 6x^2 + 7$ — bring each exponent down, subtract one, drop the constant. What took a page in Chapter 3 now takes a glance.

4.3 The product rule (and why it is *not* what you'd guess)

Here intuition fails, so we derive carefully. You might guess that the derivative of a product is the product of the derivatives — that $(fg)' = f'g'$. **It is not**, and the derivation shows why. Start from the definition and use one clever move: *add and subtract* the same term, $f(a+h)g(a)$, to split the quotient into two recognizable pieces.

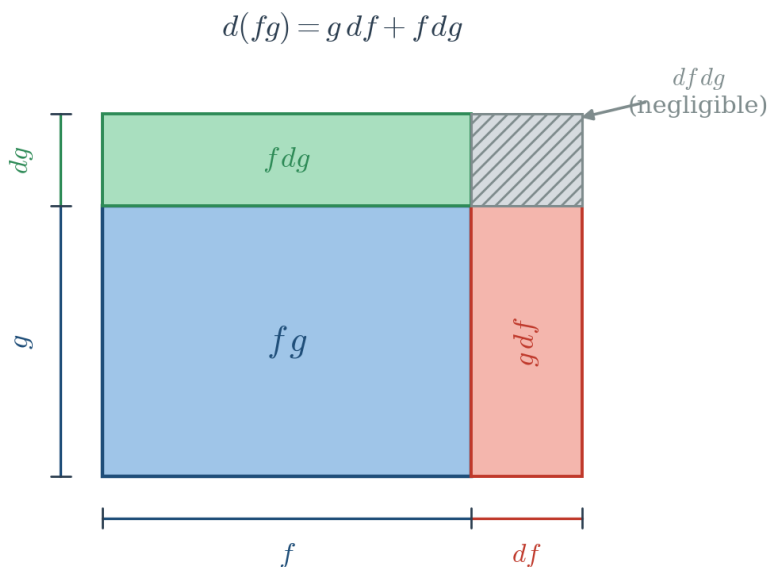


Figure 4.2

$$\begin{aligned} \frac{f(a+h)g(a+h) - f(a)g(a)}{h} &= \frac{f(a+h)g(a+h) - f(a+h)g(a)}{h} + \frac{f(a+h)g(a) - f(a)g(a)}{h} \\ &= f(a+h) \cdot \frac{g(a+h) - g(a)}{h} + g(a) \cdot \frac{f(a+h) - f(a)}{h}. \end{aligned}$$

Now let $h \rightarrow 0$. The two difference quotients become $g'(a)$ and $f'(a)$; and $f(a+h) \rightarrow f(a)$ because f is continuous (which, by Chapter 3, follows from f being differentiable). So:

The Product Rule. $\frac{d}{dx}[f(x)g(x)] = f'(x)g(x) + f(x)g'(x).$

In words: *derivative of the first times the second, plus the first times the derivative of the second.* The extra term — the one the naive guess misses — is exactly the cross-term that the add-and-subtract trick exposed. **Example:** $\frac{d}{dx}[x^2(x^3+1)] = (2x)(x^3+1) + (x^2)(3x^2) = 2x^4 + 2x + 3x^4 = 5x^4 + 2x.$

(Check: expanding first gives $x^5 + x^2$, whose derivative is $5x^4 + 2x$. They agree.)

4.4 The chain rule: differentiating a function inside a function

This is the one that matters most, because most real functions are **compositions** — a function applied to the output of another function, like $(x^2 + 1)^5$ (raise-to-the-fifth applied to $x^2 + 1$) or $\sqrt{x^2 + 1}$ or e^{x^2} . The chain rule tells you how to differentiate them, and Leibniz's notation makes it almost think for you. Write $y = f(u)$ where $u = g(x)$ — an *outer* function f and an *inner* function g . Then:

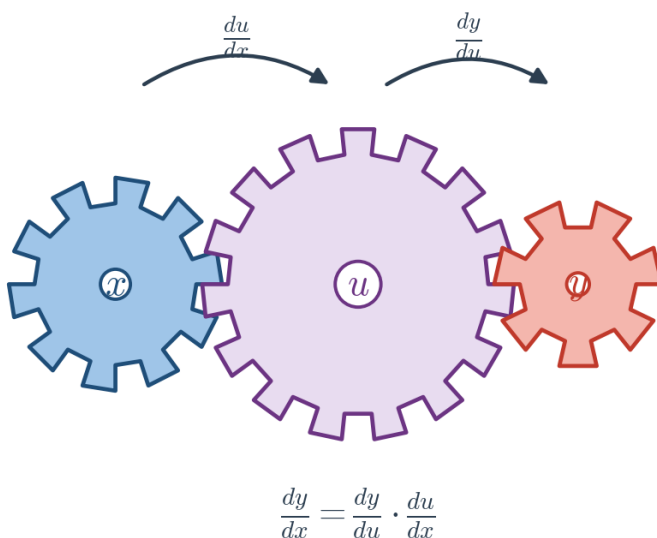


Figure 4.3

The Chain Rule. $\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}$, or $\frac{d}{dx} f(g(x)) = f'(g(x)) \cdot g'(x)$.

The idea behind it is beautifully simple: rates of change *multiply* through a chain. If y changes $\frac{dy}{du}$ times as fast as u , and u changes $\frac{du}{dx}$ times as fast as x , then y changes $\frac{dy}{du} \cdot \frac{du}{dx}$ times as fast as x — the same way gear ratios multiply. (The honest derivation writes $\frac{\Delta y}{\Delta x} = \frac{\Delta y}{\Delta u} \cdot \frac{\Delta u}{\Delta x}$ and takes limits; a careful proof handles the case $\Delta u = 0$ separately, which is the Diamond problem.) In practice: **differentiate the outer function, leaving the inside alone, then multiply by the derivative of the inside.**

- $\frac{d}{dx}(x^2 + 1)^5 = 5(x^2 + 1)^4 \cdot \underbrace{2x}_{\text{inside}'} = 10x(x^2 + 1)^4.$
- $\frac{d}{dx}\sqrt{x^2 + 1} = \frac{1}{2\sqrt{x^2 + 1}} \cdot 2x = \frac{x}{\sqrt{x^2 + 1}}.$
- $\frac{d}{dx}e^{x^2} = e^{x^2} \cdot 2x$ (using the exponential's derivative, below).

A glimpse of why machine learning runs on this rule

A neural network is a tower of functions — the input passes through layer after layer, each one a function of the layer below: $f(g(h(\dots)))$, sometimes hundreds deep. To *train* it, you need the derivative of the final error with respect to every knob in every layer — and you get it by the chain rule, multiplying the rates of change layer by layer, working from the output back toward the input. That backward sweep has a name — **backpropagation** — and it is nothing more exotic than the chain rule, applied across an enormous composition, a few billion times per second. The rule you're learning on $(x^2 + 1)^5$ is the literal engine of modern AI.

4.5 The quotient rule (which the chain rule hands us)

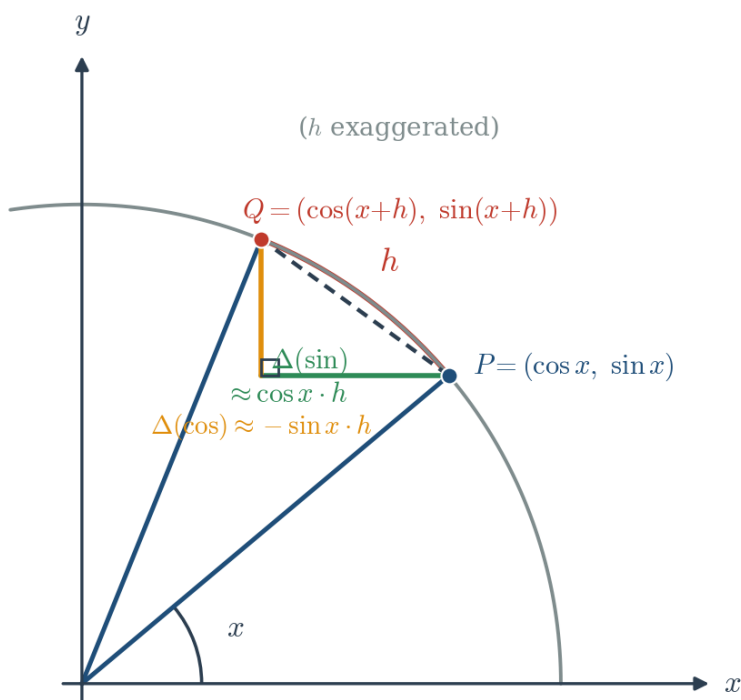
Now that we have the product and chain rules, the quotient rule is almost free. Write a quotient as a product with a negative power: $\frac{f}{g} = f \cdot g^{-1}$. The derivative of g^{-1} is, by power-rule-plus-chain, $-g^{-2}g'$. So the product rule gives $f'g^{-1} + f(-g^{-2}g') = \frac{f'}{g} - \frac{fg'}{g^2}$, and combining over g^2 :

$$\text{The Quotient Rule. } \frac{d}{dx} \left[\frac{f(x)}{g(x)} \right] = \frac{f'(x)g(x) - f(x)g'(x)}{(g(x))^2}.$$

In words: *bottom times derivative of top, minus top times derivative of bottom, all over the bottom squared.* (The order matters — the minus sign makes it non-symmetric.) **Example:** $\frac{d}{dx} \frac{x}{x^2+1} = \frac{(1)(x^2+1) - (x)(2x)}{(x^2+1)^2} = \frac{1-x^2}{(x^2+1)^2}.$

4.6 The trigonometric derivatives (where Chapter 2 finally pays off)

We promised, back in Chapter 2, that the two strange limits we proved there — $\frac{\sin h}{h} \rightarrow 1$ and $\frac{\cos h - 1}{h} \rightarrow 0$ — were the tools that would unlock the derivatives of sine and cosine. Here is that promise kept, and it's worth savoring because it's the whole reason those limits earned a place in the book.



$$\text{as } h \rightarrow 0: \quad \frac{\sin h}{h} \rightarrow 1, \quad \frac{1 - \cos h}{h} \rightarrow 0$$

Figure 4.4

Differentiate $\sin x$ from the definition. The key is the angle-addition formula $\sin(x + h) = \sin x \cos h + \cos x \sin h$:

$$\frac{\sin(x + h) - \sin x}{h} = \frac{\sin x \cos h + \cos x \sin h - \sin x}{h} = \sin x \cdot \frac{\cos h - 1}{h} + \cos x \cdot \frac{\sin h}{h}.$$

Now let $h \rightarrow 0$. The first fraction goes to 0 and the second goes to 1 — *exactly* the two Chapter 2 limits — so the whole thing becomes $\sin x \cdot 0 + \cos x \cdot 1$:

$$\frac{d}{dx} \sin x = \cos x$$

The identical computation with $\cos(x + h) = \cos x \cos h - \sin x \sin h$ gives

$$\frac{d}{dx} \cos x = -\sin x$$

(the minus sign because cosine is *falling* as it leaves 0). And $\tan x = \frac{\sin x}{\cos x}$ now yields to the quotient rule: $\frac{d}{dx} \tan x = \frac{\cos x \cos x - \sin x(-\sin x)}{\cos^2 x} = \frac{\cos^2 x + \sin^2 x}{\cos^2 x} = \frac{1}{\cos^2 x} = \sec^2 x$. These are the derivatives that Chapter 7 used to build the Taylor series of sin and cos (and, through them, Euler's formula) — now derived from the ground up.

4.7 The exponential, and the derivatives of inverse functions

The exponential cameo. One function deserves a special mention here and a full chapter (Chapter 5) of its own: e^x , the function that is *its own derivative*,

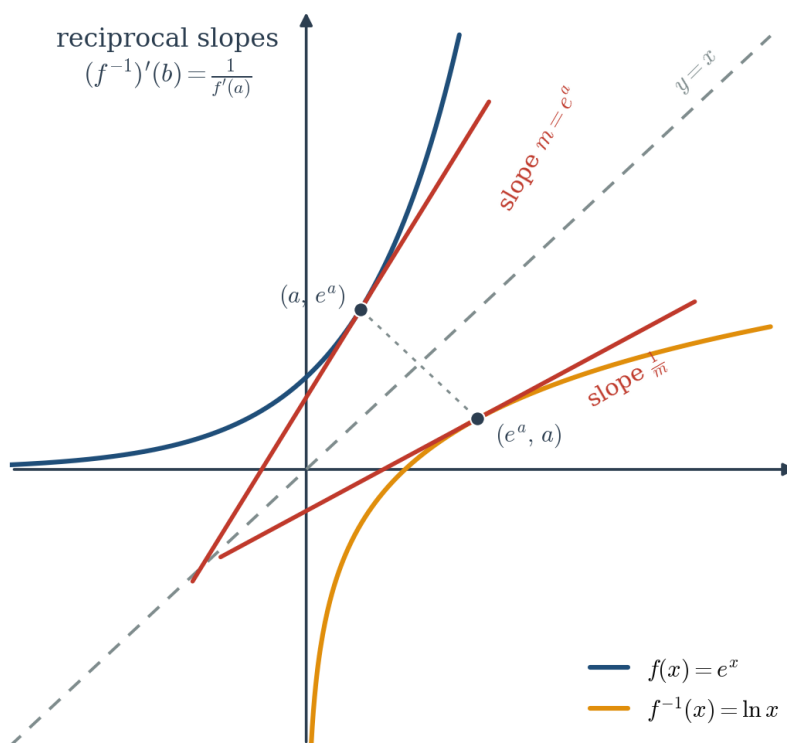


Figure 4.5

$$\frac{d}{dx}e^x = e^x.$$

(From the definition, the difference quotient is $e^x \cdot \frac{e^h - 1}{h}$, and the fact that $\frac{e^h - 1}{h} \rightarrow 1$ is precisely what singles out the number e — the loan you flagged in Chapter 3's Looking-ahead, repaid in Chapter 5.) Combined with the chain rule, this differentiates every exponential you'll meet: $\frac{d}{dx}e^{kx} = k e^{kx}$, $\frac{d}{dx}e^{x^2} = 2x e^{x^2}$.

Inverse functions. The last tool in the kit, and the one Chapter 5 needs to differentiate the logarithm. Suppose g is the inverse of f — meaning $f(g(x)) = x$ for all x . Differentiate *both sides* with the chain rule: the left side gives $f'(g(x)) \cdot g'(x)$, the right side gives 1. Solve:

Inverse-Function Rule. If $g = f^{-1}$, then $g'(x) = \frac{1}{f'(g(x))}$.

The picture is clean: an inverse reflects a graph across the line $y = x$, which swaps “rise” and “run” — so the slope of the inverse at a point is the **reciprocal** of the slope of the original at the matching point. And it passes an honesty check against Chapter 3: $\sqrt{x}$ is the inverse of $f(x) = x^2$ (for $x \geq 0$), and since $f'(x) = 2x$, the rule gives $\frac{d}{dx}\sqrt{x} = \frac{1}{f'(\sqrt{x})} = \frac{1}{2\sqrt{x}}$ — the same answer you rationalized by hand, now obtained without a single difference quotient. This is exactly the machinery Chapter 5 will turn on $\ln x$ (the inverse of e^x) to discover that its derivative is $\frac{1}{x}$.

Toolkit: The rules of differentiation

Rule	Formula
Power	$\frac{d}{dx}x^n = nx^{n-1}$
Sum	$(f + g)' = f' + g'$
Constant multiple	$(cf)' = cf'$
Product	$(fg)' = f'g + fg'$
Quotient	$\left(\frac{f}{g}\right)' = \frac{f'g - fg'}{g^2}$
Chain	$\frac{d}{dx}f(g(x)) = f'(g(x))g'(x)$
Sine / Cosine	$(\sin x)' = \cos x, \quad (\cos x)' = -\sin x$
Exponential	$(e^x)' = e^x$
Inverse	$(f^{-1})'(x) = \frac{1}{f'(f^{-1}(x))}$

With these, you can differentiate essentially every function in this book — no difference quotients required. The art is recognizing *which* rule (or combination) a given function calls for.

4.8 Where this is heading

You now hold the complete differentiation toolkit, and the rest of the book simply *uses* it. **Chapter 5** turns the inverse-function rule on $\ln x$ and the exponential on e^x to build the whole theory of growth and decay. **Chapter 6** (the Mean Value Theorem) and everything after assume you can differentiate anything — and now you can. **Chapter 7**'s Taylor series demanded the trig and exponential derivatives you just proved. **Chapter 8** finds maxima and minima by setting derivatives to zero, which means differentiating first.

And one quiet thing worth noticing: every rule in this chapter takes a function and returns its derivative *as a new function* — which you can feed right back in. Differentiate x^4 to get $4x^3$, differentiate again to get $12x^2$: that's the **second derivative**, the curvature that runs Chapters 7 and 8. The toolkit doesn't just compute slopes; it lets you climb the whole tower of derivatives, as high as you need to go.

4.9 Problems

*Six tiers. **Fluency** and **Practice** are doing** — apply the rules until choosing the right one is automatic. **Concept & counterexample** and **◆ Diamond** are *thinking* — the rule-traps and the proofs (each rule derived from the definition). **Teach it** is *teaching*. **Looking ahead** opens Chapter 5. Do all of Fluency — and there's a lot of it, more than any other chapter, on purpose: differentiation is a mechanical skill, and the only way to own it is volume. These rules should become as automatic as multiplication.*

Fluency — *differentiate, fast; this tier is the whole point*

1. Power rule, whole-number powers: (a) x^9 ; (b) $4x^6$; (c) x^{100} ; (d) 7; (e) x .

2. Power rule, negative and fractional powers: (a) x^{-3} ; (b) $\frac{1}{x^5}$; (c) $x^{1/3}$; (d) $\sqrt[4]{x}$; (e) $x^{-1/2}$; (f) $\frac{1}{\sqrt{x}}$.
3. Rewrite first, then power rule: (a) $\frac{1}{x^2} + \sqrt{x}$; (b) $\frac{x^3-1}{x}$ (split the fraction); (c) $x^2\sqrt{x}$ (combine into one power first).
4. Polynomials, term by term: (a) $5x^3 - 2x^2 + 7x - 4$; (b) $\frac{1}{3}x^6 + 4x$; (c) $9x^5 - x^4 + 2x - 11$; (d) $(2x - 1)(x + 3)$ (expand first).
5. Product rule: (a) x^2e^x ; (b) $x \sin x$; (c) $(x^2 + 1)(x^3 - 2)$; (d) $x^3 \cos x$; (e) $e^x \cos x$.
6. Product rule, then evaluate at $x = 0$: (a) $(x + 1)e^x$; (b) $x^2 \sin x$.
7. Quotient rule: (a) $\frac{x}{x^2+1}$; (b) $\frac{x^2}{x-1}$; (c) $\frac{\sin x}{x}$; (d) $\frac{e^x}{x}$; (e) $\frac{x-1}{x+1}$.
8. Quotient rule with trig: (a) $\tan x = \frac{\sin x}{\cos x}$; (b) $\frac{x \cos x}{\sin x}$; (c) $\frac{1}{\sin x}$.
9. Chain rule, powers: (a) $(x^2 + 1)^5$; (b) $(1 - x^2)^{10}$; (c) $(3x + 2)^7$; (d) $(x^3 - x)^4$.
10. Chain rule, roots: (a) $\sqrt{3x + 2}$; (b) $\sqrt{x^2 + 1}$; (c) $\sqrt{1 - x^2}$; (d) $\frac{1}{\sqrt{x^2+4}}$.
11. Chain rule, exponentials: (a) e^{4x} ; (b) e^{-x} ; (c) e^{x^2} ; (d) $e^{-x^2/2}$; (e) e^{3x+1} .
12. Chain rule, trig: (a) $\sin(3x)$; (b) $\cos(x^2)$; (c) $\sin(x^2 + 1)$; (d) $\cos(5x)$; (e) $\tan(2x)$.
13. Trig, basic: (a) $\cos x$; (b) $\tan x$; (c) $3 \sin x - 2 \cos x$; (d) $x + \cos x$; (e) $x^2 + \sin x$.
14. Powers of trig (chain): (a) $\sin^2 x$; (b) $\cos^3 x$; (c) $\tan^2 x$.
15. Mixed — *pick the right rule first*: (a) x^2e^x ; (b) $\frac{x}{\cos x}$; (c) $\sin(x^3)$; (d) $(x^2 + x)^6$; (e) $e^x + e^{-x}$.
16. Evaluate the derivative at the given point: (a) $f(x) = x^3 - x$ at $x = 2$; (b) $f(x) = e^x \sin x$ at $x = 0$; (c) $f(x) = \frac{1}{x}$ at $x = \frac{1}{2}$; (d) $f(x) = \sqrt{x}$ at $x = 4$.
17. Find the equation of the tangent line: (a) $y = (x^2 - 1)^3$ at $x = 1$; (b) $y = e^x$ at $x = 0$; (c) $y = \sin x$ at $x = 0$; (d) $y = \frac{1}{x}$ at $x = 1$.
18. Differentiate, then simplify: (a) $\frac{e^x}{x}$; (b) $x^2 \cos x$; (c) $\sqrt{x} e^x$; (d) $\frac{\sin x}{1+\cos x}$.
19. Second derivatives — find f'' : (a) x^4 ; (b) $\sin x$; (c) e^{2x} ; (d) $x^3 - 2x$.
20. Higher-order: (a) the third derivative of x^5 ; (b) the fourth derivative of $\sin x$; (c) the second derivative of $\frac{1}{x}$.

Practice — combine the rules

21. A chain inside a product: $f(x) = x^2(x^3 + 1)^4$.
22. A chain inside a chain: $f(x) = \sqrt{1 + \sqrt{x}}$. (Outer root, then inner root.)
23. Order of composition matters: (a) $e^{\sin x}$; (b) $\sin(e^x)$; (c) $(\sin x)^2$. (Note how (a) and (b) differ.)
24. A product of *three*: $f(x) = x e^x \sin x$. (Apply the product rule twice, or group as $x \cdot (e^x \sin x)$.)
25. A quotient with a chain on top: $f(x) = \frac{(x^2 + 1)^3}{x}$.
26. Differentiate $f(x) = \frac{e^x}{x^2 + 1}$ and find where its tangent is horizontal.
27. For $f(x) = x^3 - 6x^2 + 9x$, find $f'(x)$, then all points where the tangent is horizontal (preview of Chapter 8's critical points).
28. Find where $f(x) = x e^{-x}$ has a horizontal tangent, and compute f'' there.
29. Inverse-function rule: find the derivative of $g(x) = x^{1/3}$ as the inverse of $f(x) = x^3$, and check it against the power rule. Then do the same for $g(x) = \sqrt{x}$ as the inverse of x^2 .
30. **The hyperbolic functions** $\cosh x = \frac{e^x + e^{-x}}{2}$ and $\sinh x = \frac{e^x - e^{-x}}{2}$. Differentiate each from the rules, and show $\cosh' x = \sinh x$ and $\sinh' x = \cosh x$ — note the *absence* of a minus sign, unlike $\cos$ and $\sin$. (A Chapter 5 preview, built entirely from e^x .)

Concept & counterexample — think; true/false with justification, or build the example

31. (a) $(fg)' = f'g'$. (b) $\left(\frac{f}{g}\right)' = \frac{f'}{g'}$. (c) $(f \circ g)' = f' \cdot g'$ (composition).
(d) The derivative of e^{kx} is e^{kx} .
32. **Find the error.** A student writes " $\frac{d}{dx}(x^2 \sin x) = 2x \cos x$." Identify the rule they misused and give the correct derivative.
33. **Find the error.** Another writes " $\frac{d}{dx}(x^2 + 1)^5 = 5(x^2 + 1)^4$." What

did they forget, and what's the correct answer?

34. **Construct** two functions f and g for which $(fg)'$ happens to equal $f'g'$ at some point, illustrating that the naive product rule is occasionally *accidentally* right (and explain why you can't rely on it).
35. Explain, using the chain rule, why $\frac{d}{dx}e^{\sin x}$ and $\frac{d}{dx}\sin(e^x)$ are different — what does swapping the inner and outer function do?
36. **Is it possible?** A nonzero function that equals its own derivative. (You differentiated one in this chapter — which, and is it the *only* such function up to a constant multiple?)
37. Without computing it fully, use the chain-rule factor structure to predict the **sign** of $\frac{d}{dx}e^{-x^2}$ for $x > 0$ and for $x < 0$ (look at the $-2x$ that the chain rule produces). What does this tell you about the shape of the bell curve — where it rises, where it falls, and where it peaks?

◆ Diamond — *derive the rules you've been using*

38. ◆ Prove the **product rule** from the definition, using the add-and-subtract trick. State clearly where you use the continuity of f (and why Chapter 3 guarantees it).
39. ◆ Prove the **quotient rule** by writing $\frac{f}{g} = f \cdot g^{-1}$ and applying the product and chain rules. (You'll need $\frac{d}{dx}g^{-1} = -g^{-2}g'$ first.)
40. ◆ Derive $\frac{d}{dx}\sin x = \cos x$ from the definition, using the angle-addition formula and the two Chapter 2 limits $\frac{\sin h}{h} \rightarrow 1$, $\frac{\cos h - 1}{h} \rightarrow 0$. Then do the same for $\cos x$.
41. ◆ Prove the **inverse-function rule** $g'(x) = \frac{1}{f'(g(x))}$ by differentiating $f(g(x)) = x$ with the chain rule. Then use it to derive $\frac{d}{dx}\ln x = \frac{1}{x}$ (taking $\ln$ as the inverse of e^x , and $\frac{d}{dx}e^x = e^x$ as given) — the result Chapter 5 will lean on.
42. ◆◆ Prove the **power rule for negative integers**: using the quotient rule on $\frac{1}{x^n}$, show $\frac{d}{dx}x^{-n} = -n x^{-n-1}$, confirming the power rule extends below zero.
43. ◆◆ Prove the **power rule for rational exponents**, closing the loan

from this chapter. For $y = x^{p/q}$, raise both sides to the q to get $y^q = x^p$, differentiate both sides (chain rule on the left, power rule on the right), and solve for y' — you should land on exactly $\frac{p}{q} x^{p/q-1}$. (This trick — differentiating an equation neither side of which is solved for y — is *implicit differentiation*, and you’ve just invented it.)

44. ♦♦ **The chain rule, carefully.** The heuristic $\frac{\Delta y}{\Delta x} = \frac{\Delta y}{\Delta u} \cdot \frac{\Delta u}{\Delta x}$ breaks when $\Delta u = 0$ for some small Δx (you’d be dividing by zero). Explain why this case can arise, and sketch how a rigorous proof gets around it. (You don’t need the full proof — just identify the gap the heuristic hides.)

Teach it — *explain it, write it up, or fix it*

45. In **three or four sentences with the gear-ratio picture**, teach a friend the chain rule — why rates of change *multiply* through a composition.
46. **Debug your classmate.** They insist “ $(fg)' = f'g'$ — derivatives just distribute over multiplication.” Give them a single concrete counterexample (try $f = g = x$) that settles it in one line, then state the correct rule.
47. Write a complete **worked solution** differentiating $\sqrt{x^2 + 1}$, naming the outer and inner functions and narrating each chain-rule step, so a peer could redo it for $\sqrt{x^3 + 1}$.
48. **Design a problem.** Invent a function that requires *two* rules together (say, a chain inside a product), and write the full solution key.
49. In a few sentences, **teach why the trigonometric derivatives depend on radians** — trace it back through the $\frac{\sin h}{h} \rightarrow 1$ limit to the reason mathematicians measure angles in radians at all.

Looking ahead — *Chapter 5 in your hands early*

50. You proved (Diamond 41) that $\frac{d}{dx} \ln x = \frac{1}{x}$ using the inverse rule. **Predict** how you’d differentiate a^x for a general base a , given the

hint that $a^x = e^{(\ln a)x}$. (Chapter 5 confirms it.)

51. The number e is the one base for which $\frac{e^h - 1}{h} \rightarrow 1$, making e^x its own derivative. **Predict** what the derivative of 2^x would be — specifically, what constant multiple of 2^x — and why that constant is *not* 1. (Chapter 5 will name the constant.)
52. You differentiated $\sin x$ twice in problem 19 and found you return to $\pm \sin x$ after four derivatives. **Predict** the 100th derivative of $\sin x$. (Hint: the cycle has length four — what is 100 divided by 4?)
53. You found (problem 30) that $\cosh' = \sinh$ and $\sinh' = \cosh$. **Predict** $\cosh''$ and $\sinh''$, and notice each hyperbolic function *is its own second derivative*: $y'' = y$. Which functions from this chapter instead satisfy $y'' = -y$? (That contrast — $y'' = y$ for the hyperbolics, $y'' = -y$ for sine and cosine — is the seed of the exponential–trig duality behind Euler’s formula in Chapter 7.)
-

Chapter 5

The Function That Is Its Own Slope

Or: Why one strange number between 2 and 3 refuses to change

Companion notebook. Play with this chapter’s ideas — sliders, animations, and drills — in `notebooks/ch5_exponentials.ipynb`.

By the end of this chapter you will be able to do four things.

You will know what e^x is, and — more importantly — *why* the number $e \approx 2.718$ is the natural base for an exponential, and not some friendlier-looking number like 2 or 10.

You will know that e^x is its own derivative, and you will know this is not a magic trick or a coincidence to memorize. It is the *reason we picked e in the first place*. We are going to go looking for a function that is its own slope, and e is what we find.

You will know how logarithms undo exponentials, and you will be able to prove — not assert, prove — that the slope of $\ln x$ is $1/x$.

And you will be able to point all of this at the real world: money that grows, populations that explode, atoms that decay, and a drug leaving the bloodstream. By the end, “half-life” will be something you can *derive*, not something you look up.

One honest warning before we start. This chapter takes a small loan. We are going to use a fact about e before we have fully earned it, and I am going to tell you exactly where the loan is and exactly when we pay

it back (Chapter 10, once we have integrals). Borrowing a result you are confident in, using it, and settling the debt later is not cheating. It is how working mathematicians actually move. I want you to see it done in the open instead of pretended away.

5.1 The question nobody asks first

Most books hand you e^x on a plate. Here is the function, here is its graph, here is the rule, memorize it. We are going to do the opposite. We are going to ask a question, and the number e is going to be the *answer*.

Here is the question. **Is there a function whose rate of change is equal to itself?**

That sounds abstract, so look at where it comes from. Three completely different situations from the real world:

- **Money in an account** earning interest. The more money you have, the more interest you earn each year — so the rate at which the money grows is proportional to how much money is there.
- **A population** of rabbits, or bacteria, with plenty of food. The more rabbits there are, the more baby rabbits appear each month — the growth rate is proportional to the current number.
- **A radioactive lump**, or a drug in your blood. The more atoms (or drug molecules) are present, the more of them break down or get cleared each second — the rate of *decrease* is proportional to how much is left.

Strip the stories away and they are all the same sentence:

$$\text{rate of change of the amount} = (\text{a constant}) \times (\text{the amount}).$$

In the language we have been building, if $N(t)$ is the amount at time t , this says

$$\frac{dN}{dt} = k N.$$

This little equation is one of the most important sentences in all of science, and we will meet it again. For now, notice what it is *asking* of us. We want a function $N(t)$ whose derivative is a constant times itself. So before we can solve any of those three problems, we have to answer the bare question: what kind of function is proportional to its own slope?

That is the whole chapter. Let us go find the function.

5.2 Every exponential is already proportional to itself

Start with a plain exponential function, $f(x) = a^x$, where a is some fixed positive number — maybe 2, maybe 10. You met these in Book 1. Let us just *take its derivative from scratch*, the way we have taken every derivative so far: write down the difference quotient and see what happens.

$$\frac{f(x+h) - f(x)}{h} = \frac{a^{x+h} - a^x}{h}.$$

Now use the one exponent rule that does all the work here: $a^{x+h} = a^x \cdot a^h$. Factor the a^x out:

$$\frac{a^x a^h - a^x}{h} = a^x \cdot \frac{a^h - 1}{h}.$$

Look hard at what just happened, because it is the heart of the chapter. The a^x came out front and *did not change as h shrank*. The only part that depends on h is the fraction $\frac{a^h - 1}{h}$. So when we take the limit as $h \rightarrow 0$:

$$\frac{d}{dx}a^x = a^x \cdot \underbrace{\left(\lim_{h \rightarrow 0} \frac{a^h - 1}{h} \right)}_{\text{a number that depends only on } a}.$$

That underbraced limit is just a *number*. It does not depend on x at all. Let us give it a name and call it $M(a)$, for the moment, because it is the **mystery number** attached to the base a :

$$\boxed{\frac{d}{dx}a^x = M(a) \cdot a^x} \quad \text{where} \quad M(a) = \lim_{h \rightarrow 0} \frac{a^h - 1}{h}.$$

Read that boxed line out loud, because it is astonishing and most people never notice it. **Every exponential function is already proportional to its own slope.** The derivative of a^x is just a^x scaled by some fixed number $M(a)$. We have *almost* answered our question for free. The only thing standing between us and a function that equals its own derivative is that pesky constant $M(a)$ out front. If only we could find a base a that made $M(a)$ exactly equal to 1.

One more thing to notice before we hunt for that base. What is $M(a)$, geometrically? Plug $x = 0$ into the boxed rule. Since $a^0 = 1$, the slope of a^x at $x = 0$ is exactly $M(a) \cdot 1 = M(a)$. So:

$M(a)$ is simply **the slope of the curve $y = a^x$ as it crosses the y -axis.**

That gives us something we can actually look at. (*Figure 5.1: the curves $y = 2^x$, $y = 3^x$, and a couple of others, each with its tangent line drawn where it crosses the y -axis. The steeper the curve leaves the axis, the bigger $M(a)$.*)

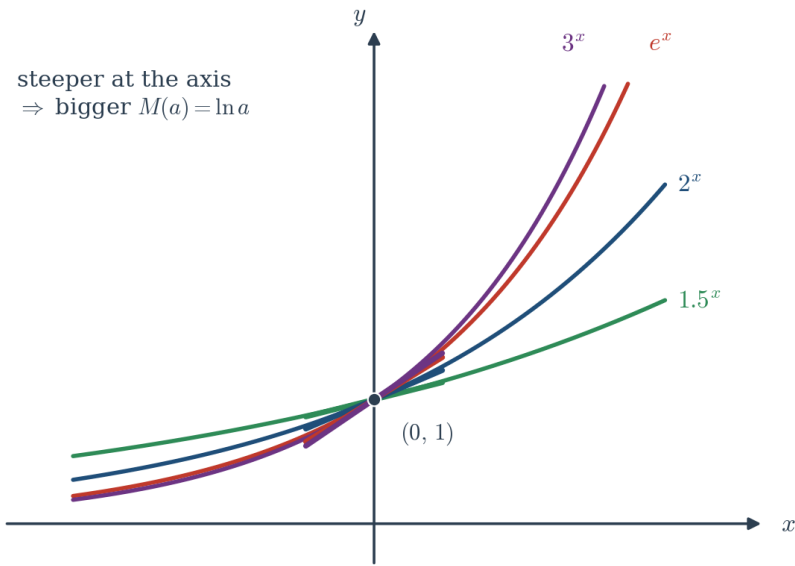


Figure 5.1

5.3 Hunting for the base that leaves the axis at slope 1

We want the base a where the curve crosses the y-axis at slope exactly 1 — where $M(a) = 1$. Let us just measure $M(a)$ for a few bases by taking very small h in $\frac{a^h - 1}{h}$. (This is the kind of thing a calculator, or the notebook for this chapter, does in a heartbeat. Try $h = 0.0001$.)

base a	slope at the axis, $M(a)$
2	≈ 0.693
2.5	≈ 0.916
2.718	≈ 1.000
3	≈ 1.099
10	≈ 2.303

For $a = 2$ the curve leaves the axis too gently — slope only 0.693. For $a = 3$ it leaves a little too steeply — slope 1.099. So *somewhere between 2 and 3* there is a base where the slope is exactly 1. (If you want this argument made airtight: $M(a)$ slides smoothly upward as a grows, it is below 1 at $a = 2$ and above 1 at $a = 3$, so it must hit 1 somewhere in between. We will sharpen “slides smoothly” later, but the picture is honest.)

That special base gets its own letter. We call it e :

$$e \approx 2.71828 \dots$$

and it is *defined* to be the base for which $M(e) = 1$. The instant we say that, the boxed rule gives us our prize:

$$\frac{d}{dx} e^x = M(e) \cdot e^x = 1 \cdot e^x = e^x.$$

$$\boxed{\frac{d}{dx} e^x = e^x}$$

This is the function that is its own slope. We did not pull it out of a hat. We went looking for a function equal to its own derivative, discovered that every exponential is *almost* one (off only by a constant $M(a)$), and then picked the one base that makes the constant vanish. The number e is exactly the price of that wish.

(Figure 5.2: $y = e^x$ with tangent lines drawn at several points. At every point, the steepness of the tangent equals the height of the curve. Where the curve is at height 5, the slope is 5. That is what “its own derivative” looks like.)

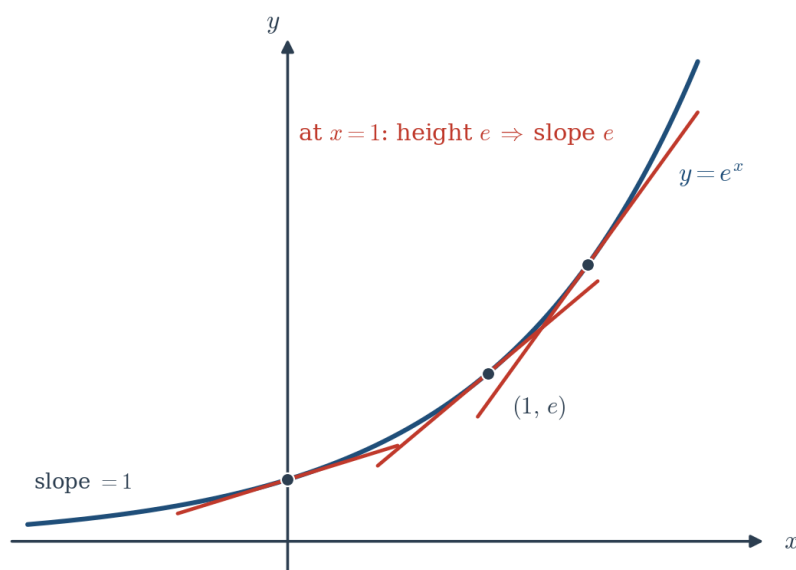


Figure 5.2

A promise about e , to be kept in Chapter 1's other form

Back in Chapter 1, when we played with sequences, we watched the quantity $\left(1 + \frac{1}{n}\right)^n$ creep upward as n grew — 2, then 2.25, then 2.4883 ..., settling toward 2.718 That was the same number e . It is not obvious that “the base whose curve leaves the axis at slope 1” and “the number that $\left(1 + \frac{1}{n}\right)^n$ sneaks up on” are the *same* number. They are. **This is the loan I warned you about.** For now, take it as a promise backed by the matching decimals; in Chapter 10, with integrals in hand, we will prove the two descriptions of e are genuinely one number.

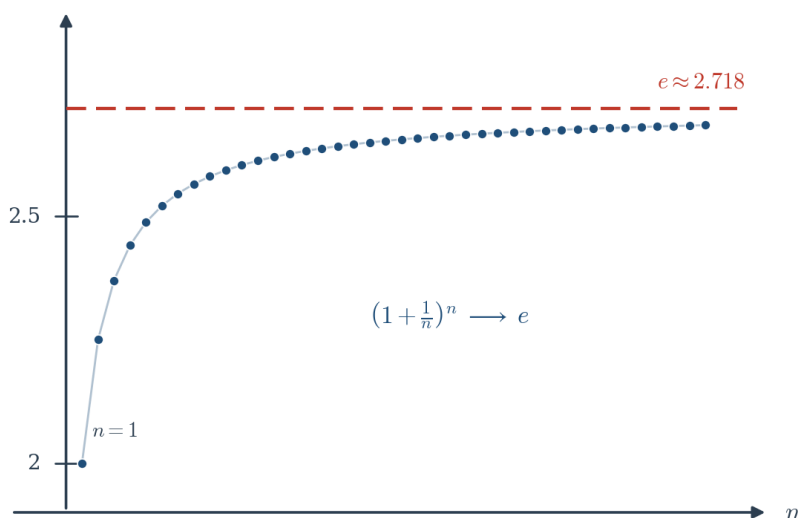


Figure 5.3

Toolkit: Differentiating exponentials

- $\frac{d}{dx} e^x = e^x$ — the headline. The function is its own derivative.
- $\frac{d}{dx} e^{kx} = k e^{kx}$ — by the **chain rule** from Chapter 4 (outer function $e^{(\cdot)}$, inner function kx , inner derivative k).
- $\frac{d}{dx} a^x = (\ln a) a^x$ — the mystery constant $M(a)$ turns out to be $\ln a$. We prove this two pages from now.

The middle line is the one you will use most. The constant k being positive means growth; k negative means decay. *That single sign is the difference between rabbits and radioactivity.*

5.4 Logarithms: undoing the exponential

We have a function that grows. We will constantly need to run it backward: “the account reached \$5000 — *how long did that take?*” Running e^x backward is exactly what the **natural logarithm** $\ln$ does. In the language of Chapter 4, $\ln$ is the *inverse function* of e^x :

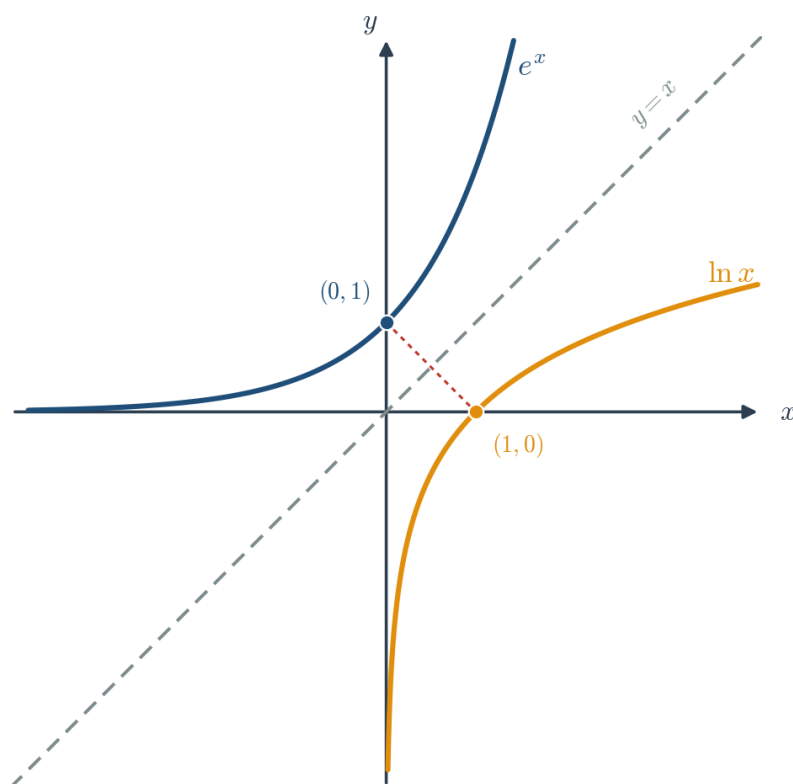


Figure 5.5

$$y = e^x \iff x = \ln y.$$

$\ln$ takes in “what the exponential grew to” and gives back “the input that got us there.” Two facts fall straight out of being an inverse: $\ln(e^x) = x$ and $e^{\ln y} = y$. And the exponent rules become logarithm rules — for

instance, since $e^x e^y = e^{x+y}$, taking $\ln$ of both sides turns multiplication into addition:

$$\ln(AB) = \ln A + \ln B.$$

(That single property — *turns multiplication into addition* — is the entire reason logarithms were invented, centuries before anyone cared about calculus. It is also why log scales work, which we will get to.)

Now the part worth the price of admission: **what is the derivative of $\ln x$?** We do not need a new limit. We have a tool for differentiating inverse functions from Chapter 4, and this is its finest hour. Let $y = \ln x$. By the definition of $\ln$, that means

$$e^y = x.$$

Differentiate *both sides* with respect to x . The right side is easy: $\frac{d}{dx}x = 1$. The left side needs the chain rule, because y is itself a function of x :

$$e^y \cdot \frac{dy}{dx} = 1.$$

Solve for $\frac{dy}{dx}$, then use $e^y = x$:

$$\frac{dy}{dx} = \frac{1}{e^y} = \frac{1}{x}.$$

$\frac{d}{dx} \ln x = \frac{1}{x}$

Take a moment with how clean that was. We never computed a single difference quotient for $\ln$. We leaned entirely on “ $\ln$ is the inverse of e^x ” plus the chain rule, and the answer dropped out in three lines. This is what it feels like when the earlier tools start paying for themselves.

Paying off one small debt right now: the mystery constant was $\ln a$

I promised that the mysterious $M(a)$ — the slope of a^x at the axis — is really $\ln a$. We can settle that one immediately, because we now have the machine to do it. The trick is to write *every* exponential in terms of the one base we understand. Since $a = e^{\ln a}$,

$$a^x = (e^{\ln a})^x = e^{(\ln a)x}.$$

Now differentiate using the chain-rule line from the toolkit, with $k = \ln a$:

$$\frac{d}{dx}a^x = \frac{d}{dx}e^{(\ln a)x} = (\ln a)e^{(\ln a)x} = (\ln a)a^x.$$

So $M(a) = \ln a$, exactly as advertised. The “mystery number” attached to every base was never mysterious; it was the natural log of the base all along. And notice the consistency check: $M(e) = \ln e = 1$, which is precisely the condition we used to *define* e . Everything closes up.

5.5 What all of this is for

Time to collect on the three real-world situations that started the chapter. They were all the same equation, $\frac{dN}{dt} = kN$ — “the rate of change is proportional to the amount.” We can now write down its solution and *check it ourselves*:

$$N(t) = N_0 e^{kt}.$$

Here N_0 is the amount at the start (set $t = 0$: $e^0 = 1$, so $N(0) = N_0$ — good). Does it actually solve the equation? Differentiate, using the chain rule:

$$\frac{dN}{dt} = N_0 \cdot k e^{kt} = k (N_0 e^{kt}) = k N.$$

Yes. The derivative came back equal to k times the function itself, which is exactly the equation we needed to satisfy. We have not just been handed the answer — we have verified it.

Growth: money and rabbits

When $k > 0$, $N(t) = N_0 e^{kt}$ grows, and it grows *faster the bigger it gets*. This is **continuous compounding**: a bank account growing at rate r , compounded not yearly or monthly but every instant, reaches $N_0 e^{rt}$ after time t . (This is also where $\left(1 + \frac{1}{n}\right)^n$ secretly lives — compounding n times a year and letting $n \rightarrow \infty$ is what produces e in the first place. The promise from earlier, peeking through again.)

Decay and half-life: atoms and medicine

When $k < 0$, write it as $k = -\lambda$ with $\lambda > 0$, and the same function now *decays*:

$$N(t) = N_0 e^{-\lambda t}.$$

This describes a radioactive sample losing atoms, and — the example to keep in your pocket if you care about drug design — a medicine being cleared from the bloodstream. Most drugs leave the body by **first-order kinetics**: the liver and kidneys remove an amount proportional to how much is currently circulating, which is exactly $\frac{dN}{dt} = -\lambda N$.

The number everyone quotes for such a process is its **half-life**: the time $t_{1/2}$ for the amount to fall to half of whatever it was. We can *derive* it instead of memorizing it. We want the time when N has dropped to $N_0/2$:

$$N_0 e^{-\lambda t_{1/2}} = \frac{1}{2} N_0.$$

Cancel N_0 , then take $\ln$ of both sides to pull the exponent down (this is the moment logarithms were built for — undoing an exponential):

$$e^{-\lambda t_{1/2}} = \frac{1}{2} \implies -\lambda t_{1/2} = \ln \frac{1}{2} = -\ln 2 \implies t_{1/2} = \frac{\ln 2}{\lambda}.$$

Worked example. A painkiller has an elimination rate constant $\lambda = 0.231$ per hour. What is its half-life, and how much of a 400 mg dose remains after 9 hours?

Half-life: $t_{1/2} = \frac{\ln 2}{0.231} = \frac{0.693}{0.231} = 3.0$ hours. So the amount halves every 3 hours.

After 9 hours: that is exactly three half-lives, so we expect three halvings: $400 \rightarrow 200 \rightarrow 100 \rightarrow 50$ mg. Check it the precise way too: $N(9) = 400 e^{-0.231 \times 9} = 400 e^{-2.079} = 400 \times 0.125 = 50$ mg. The two methods agree, which is how you know you did it right. (Figure 5.4: the decay curve with the dose halving at $t = 3, 6, 9$ hours marked.)

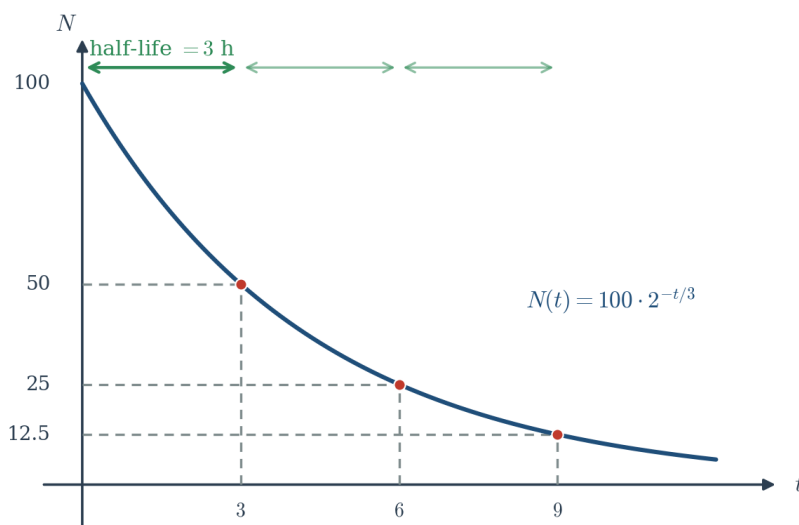


Figure 5.4

Why log scales exist

Because $\ln$ turns multiplication into addition, it squashes enormous multiplicative ranges into walkable additive ones. That is why the loud, scientific scales of the world are logarithmic: earthquake energy (Richter), sound (decibels), acidity (pH). A jump of “one unit” on these scales means *times ten*, not *plus ten*. When someone tells you a magnitude-7 quake is “a bit worse” than a magnitude-5, the logarithm is the reason they are off by a factor of a hundred.

5.6 The honest part: what we borrowed, and when we pay

This is the section most textbooks leave out, and I think leaving it out is a quiet form of lying. So, plainly: where in this chapter did we lean on something we had not proven?

1. We assumed the limit $M(a) = \lim_{h \rightarrow 0} \frac{a^h - 1}{h}$ actually *exists* — that the curve a^x has a definite slope at the axis. The numbers behaved, and the picture is convincing, but “the numbers behaved” is not a proof.
2. We assumed $M(a)$ slides smoothly from below 1 (at $a = 2$) to above 1 (at $a = 3$), so that *some* base hits slope exactly 1. Again: true, pictured honestly, not yet proven.
3. We took on faith that this e — “the slope-1 base” — is the *same* number as Chapter 1’s $\left(1 + \frac{1}{n}\right)^n$.

None of these is wrong. All three are repairable. But the clean, no-loans way to build e runs in the *opposite* direction from how we did it: you first **define** $\ln x$ as the area under the curve $1/t$ from 1 to x — an integral — prove from that definition that it turns multiplication into addition and has slope $1/x$, and then define e^x as its inverse and e as the input where $\ln$

equals 1. Built that way, every fact above becomes a theorem with no gaps. We cannot do it yet because we have not built integrals. We build them in Chapters 9 and 10, and the very last thing Chapter 10 does is come back here and discharge all three loans at once.

Why borrow at all, then? Because waiting would have meant not touching exponentials — the engine of half the science you care about — until the end of the book. Mathematicians borrow like this constantly: you convince yourself a tool is sound, you *use* it to get real work done, and you return to nail down the foundations once you have the right machinery. Watching where the loan sits, and never forgetting to repay it, is itself one of the skills this series is trying to teach you.

5.7 Where this is heading

Two threads to carry forward.

The fact that e^x *is its own derivative* is not just convenient — it is about to make e^x the single cleanest example in the next chapter. When we learn to **rebuild any function out of its own derivatives** (Taylor’s idea, Chapter 6), e^x produces the most beautiful pattern of all, precisely because differentiating it changes nothing.

And the same self-similarity is why e^x shows up everywhere once you reach machine learning: it is inside the “softmax” that turns scores into probabilities, and inside the bell curve that describes noise and uncertainty. You are not learning e^x because it is on a test. You are learning the one function that nature and computation both reach for first.

5.8 Problems

Six tiers. **Fluency** and **Practice** are doing* — drill the derivatives and log rules until automatic, then apply them. **Concept & counterexample** and **◆ Diamond** are *thinking* — traps, constructions, and the proofs (including the honest treatment of where e comes from). **Teach it** is *teaching*. **Looking ahead** reaches into Chapter 6. Do all of Fluency; don't skip Concept or Teach it.*

Fluency — *do these fast, until they're automatic*

1. Differentiate: (a) e^{5x} ; (b) e^{-x} ; (c) $3e^{2x}$; (d) $e^x + x^3$; (e) $e^{x/2}$; (f) $7e^{-0.1x}$.
2. Differentiate: (a) $\ln x$; (b) $\ln(3x)$; (c) $5 \ln x$; (d) $\ln(x) + x^2$.
3. Use the log rules to rewrite as sums/differences/multiples: (a) $\ln(6)$; (b) $\ln \frac{x}{5}$; (c) $\ln(x^3)$; (d) $\ln(2x^2)$.
4. Evaluate without a calculator: (a) $\ln 1$; (b) $\ln e$; (c) $\ln e^4$; (d) $e^{\ln 7}$; (e) $\ln e^{-2}$.
5. A colony follows $N(t) = 200 e^{0.4t}$ (t in hours). Find $N(0)$ and $N(5)$.
6. A drug has half-life 4 hours; you start with 80 mg. How much remains after 4, 8, and 12 hours? (In your head.)

Practice — *choose the rule and apply it*

7. Differentiate, naming outer and inner functions first: (a) e^{3x^2} ; (b) $\ln(x^2 + 1)$; (c) e^{-x^2} ; (d) $\ln(\sqrt{x})$.
8. Show by differentiating that $N(t) = N_0 e^{kt}$ satisfies $\frac{dN}{dt} = kN$, and state in one sentence what N_0 and k mean.
9. A radioactive isotope decays with $\lambda = 0.05$ per year. Find its half-life, then the fraction remaining after 30 years.
10. \$1000 grows at a continuously compounded 5%, so $A(t) = 1000 e^{0.05t}$. How long until it doubles? (Then explain why $\ln 2 \approx 0.69$ makes “ $70 \div \text{rate}\%$ ” a good mental shortcut.)

11. Two drugs: A has half-life 2 h, B has half-life 6 h. Write each as $N_0 e^{-\lambda t}$ (find each λ), and find how long until each falls to 10% of its dose.
12. A pH is defined by $\text{pH} = -\log_{10}[\text{H}^+]$. If one solution has $[\text{H}^+]$ one hundred times another's, how far apart are their pH values? (One line — this is the log-scale point.)
13. Differentiate $h(x) = e^{2x} + \ln(5x)$, then find the slope of h at $x = 1$.

Concept & counterexample — *think; state true/false and justify, or build the example*

14. (a) $\frac{d}{dx}e^x = xe^{x-1}$ (treating e^x “like a power”). (b) $\ln(A + B) = \ln A + \ln B$. (c) The graph of e^x has the same slope as height at every point. (d) 2^x grows but its slope-at-zero is less than 1.
15. Explain in one or two sentences why e , not 10, is called the *natural* base — what specifically is natural about it?
16. **Construct / identify:** among $f(x) = 2^x$, e^x , 3^x , which one has the property that its derivative equals itself, and which two do not? For the two that don't, what does each derivative equal?
17. **Find the error.** A student writes: “ $\frac{d}{dx} \ln(x^2) = \frac{1}{x^2}$, by the rule $\frac{d}{dx} \ln x = \frac{1}{x}$.” Identify the missing step and give the correct derivative two different ways (chain rule, and first using a log rule).
18. **Is it possible?** A quantity decaying by first-order kinetics that reaches exactly zero in finite time. Argue from the form $N_0 e^{-\lambda t}$ why or why not. (What does this say about “when is the drug *completely* gone?”)
19. The chapter took out three “loans” about e . State them in your own words, and say which chapter repays each.

◆ **Diamond — Layer 2: prove it**

20. ◆ Prove $\frac{d}{dx} \ln x = \frac{1}{x}$ using only that $\ln$ is the inverse of e^x and the chain rule (reproduce the text’s three-line argument from scratch).
21. ◆ Prove $M(a) = \ln a$ by writing $a^x = e^{(\ln a)x}$ and differentiating. Then deduce $\frac{d}{dx} a^x = (\ln a)a^x$ and check it gives the right answer at $a = e$.
22. ◆ Derive the half-life formula $t_{1/2} = \frac{\ln 2}{\lambda}$ from $N_0 e^{-\lambda t}$, justifying each logarithm step. Then prove the “memoryless” property: the time to fall from *any* level to half of it is always $t_{1/2}$, independent of where you start.
23. ◆ Using the slope-at-the-axis idea, explain why $M(4) > 1$ and $M(2) < 1$, and why this forces a base with $M(a) = 1$ to exist between them. Name the property of M you’re assuming (it’s one of the chapter’s loans).
24. ◆ Prove $\frac{d}{dx} \ln(kx) = \frac{1}{x}$ for any constant $k > 0$ — and explain in one sentence why the constant vanishes (what does multiplying the input by k do to the *graph* of $\ln$?).
25. ◆◆ Uniqueness. Suppose $f'(x) = k f(x)$ for all x , with $f(0) = N_0$. Show $g(x) = f(x)e^{-kx}$ has $g'(x) = 0$ everywhere, conclude g is constant, and deduce $f(x) = N_0 e^{kx}$. (This proves the exponential is *the* solution to the growth/decay equation, not merely *a* solution.)
26. ◆◆ Logarithmic differentiation preview: by writing $a^x = e^{(\ln a)x}$, and more generally any $u(x)^{v(x)} = e^{v(x) \ln u(x)}$, differentiate x^x for $x > 0$. (You’ll get a clean answer; notice the trick is to *never* differentiate a “variable-to-a-variable” directly — always route through e and $\ln$ first.)

Teach it — explain it, write it up, or fix it for someone else

27. In **three or four sentences to a friend**, explain why e^x is “the function that is its own slope,” starting from the question “what function is proportional to its own rate of change?” — not from the formula.

28. Write a **worked solution** to “a drug with half-life 3 hours: how long until 5% remains?” narrating each step (set up the model, find λ , take the log) so a peer could redo it for any half-life.
29. **Debug your classmate.** They claim “ $\frac{d}{dx}e^{2x} = e^{2x}$, because e to anything is its own derivative.” Pinpoint the error, name the rule they skipped, and give the correct derivative.
30. **Design a problem.** Invent a continuously-compounded investment problem whose answer is “it doubles in exactly 10 years,” stating the rate you need and writing the solution key.
31. In two or three sentences, **teach why mathematicians prefer $\ln$ (base e) over $\log_{10}$** , even though $\log_{10}$ seems friendlier — what does base e buy you that base 10 doesn’t? (Hint: look at the derivatives.)

Looking ahead — Chapter 6 in your hands early

32. e^x is its own derivative, so its first, second, third... derivatives are *all* e^x , which all equal 1 at $x = 0$. In Chapter 6 you’ll build a function out of its derivatives at a point. Guess the first four terms of the polynomial that best matches e^x near 0. (Try matching value, slope, and curvature at $x = 0$ — you may recognize the pattern $1 + x + \frac{x^2}{2} + \dots$)
 33. Compute $e^{0.1}$ on a calculator, then compare it to $1 + 0.1$ and to $1 + 0.1 + \frac{0.1^2}{2}$. Which is closer? (You’re watching a Taylor approximation get better — Chapter 6.)
 34. The clean, no-loans construction of e defines $\ln x = \int_1^x \frac{1}{t} dt$ — an *area*. You don’t have integrals yet. Predict: if $\ln x$ is “the area under $\frac{1}{t}$ from 1 to x ,” what should $\ln 1$ be, and why should $\ln x$ increase as x grows? (Chapter 10 makes this airtight and repays the chapter’s loans.)
-

Chapter 6

The Mean Value Theorem

Or: somewhere, your speedometer read exactly your average speed

Companion notebook. Play with this chapter's ideas — sliders, animations, and drills — in `notebooks/ch6_mvt.ipynb`.

By the end of this chapter you will have the single most useful theorem in all of Calculus 1 — the one that quietly justifies almost everything in the second half of this book.

You will be able to state and use **Rolle's theorem** and the **Mean Value Theorem**, and picture what they say: if you average 60 miles an hour over a trip, then at some instant your speedometer read *exactly* 60. Average rate of change, achieved as an instantaneous rate of change, somewhere in between.

But the theorem itself is not the point. Its **three consequences** are, and they are the load-bearing beams of the rest of the book. You will prove that a function with zero slope everywhere is constant — finally repaying a debt we took out back in Chapter 5. You will prove that the *sign* of the derivative controls whether a function rises or falls — the engine that will let you find maxima and minima in Chapter 8. And you will prove a bound that says a function can't change faster than its steepest slope allows — the idea that makes every numerical method in engineering trustworthy.

Here is the theme, and it connects to the last chapter. In Chapter 2 you saw that **continuity** forces structure: a continuous function can't get from one value to another without passing through everything in between. This

chapter is the sequel: **differentiability** forces structure too, and more of it. A function smooth enough to have a derivative everywhere is far more constrained than it looks — it cannot wander freely, and the Mean Value Theorem is the leash.

*(One dependency, named honestly: this chapter leans on the **Extreme Value Theorem** — a continuous function on a closed interval $[a, b]$ actually attains a highest and a lowest value somewhere on it. That belongs to the family of “continuity forces structure” results from Chapter 2; here we’ll use it as a tool.)*

6.1 First, the easy case: Rolle’s theorem

Start with a picture so simple it feels like it can’t be saying much. Suppose a smooth curve starts and ends at the **same height** — $f(a) = f(b)$. Draw any such curve in the air with your finger. To come back down to where you started, you had to stop climbing at some point and turn around: there is a peak, or a valley, somewhere strictly between the ends. And at that turning point, the curve is momentarily **flat** — its slope is zero.

That is the entire content of Rolle’s theorem.

Rolle’s Theorem. If f is continuous on $[a, b]$, differentiable on (a, b) , and $f(a) = f(b)$, then there is at least one point c in (a, b) where $f'(c) = 0$.

The proof rests on one small, sharp fact about what the derivative must do at a peak — a fact worth isolating, because we will reuse it when we hunt for maxima in Chapter 8.

◆ **Fermat's lemma: at an interior peak, the slope is zero**

Claim. If f has a local maximum at an interior point c (and $f'(c)$ exists), then $f'(c) = 0$. (Same for a local minimum.)

Why. Look at the difference quotient $\frac{f(c+h) - f(c)}{h}$ near the peak. Because c is a *maximum*, the numerator $f(c+h) - f(c)$ is ≤ 0 for every small h — you can't go higher than the top. Now watch the sign of the whole quotient from each side: - From the **right** ($h > 0$): numerator ≤ 0 , denominator > 0 , so the quotient is ≤ 0 . Taking $h \rightarrow 0^+$ gives $f'(c) \leq 0$. - From the **left** ($h < 0$): numerator ≤ 0 , denominator < 0 , so the quotient is ≥ 0 . Taking $h \rightarrow 0^-$ gives $f'(c) \geq 0$.

The derivative exists, so both one-sided limits equal $f'(c)$. The only number that is both ≤ 0 and ≥ 0 is 0. So $f'(c) = 0$. ■

With that in hand, Rolle falls out cleanly:

◆ **Proof of Rolle's theorem**

By the Extreme Value Theorem, f attains a maximum and a minimum somewhere on $[a, b]$. Two cases. - If **both** the max and the min happen at the endpoints, then since $f(a) = f(b)$, the highest and lowest values of f are equal — so f is *constant*, and $f'(c) = 0$ at every interior c . - Otherwise, at least one of them occurs at an **interior** point c . By Fermat's lemma, $f'(c) = 0$ there.

Either way, some interior c has $f'(c) = 0$. ■

(Figure 6.1: a curve with $f(a) = f(b)$, its interior peak marked, and a flat tangent line drawn at that peak.)

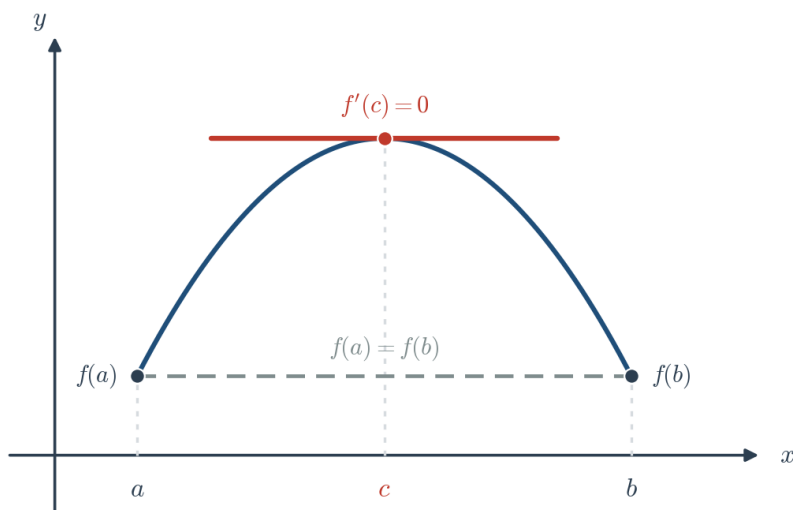


Figure 6.1

6.2 Now tilt it: the Mean Value Theorem

Rolle handled the flat case, where the endpoints are level. The Mean Value Theorem handles *every* case by a trick so good it's almost cheating: **tilt the whole picture until the endpoints are level, apply Rolle, then tilt back.**

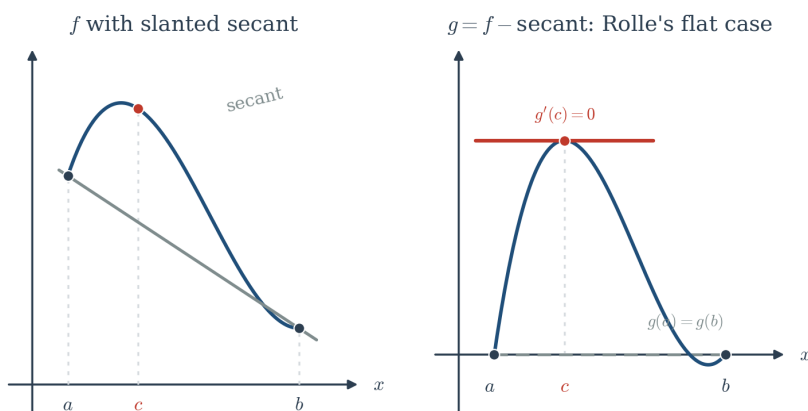


Figure 6.3

Mean Value Theorem. If f is continuous on $[a, b]$ and differentiable on (a, b) , then there is at least one point c in (a, b) where

$$f'(c) = \frac{f(b) - f(a)}{b - a}.$$

The right side is the **average rate of change** over the whole interval — the slope of the straight “secant” line connecting the two endpoints of the graph. The theorem says some interior tangent line is **parallel** to that secant: at the instant c , your instantaneous rate equals your trip average. (Figure 6.2: the secant line through the endpoints, and a parallel tangent line touching the curve at c .)

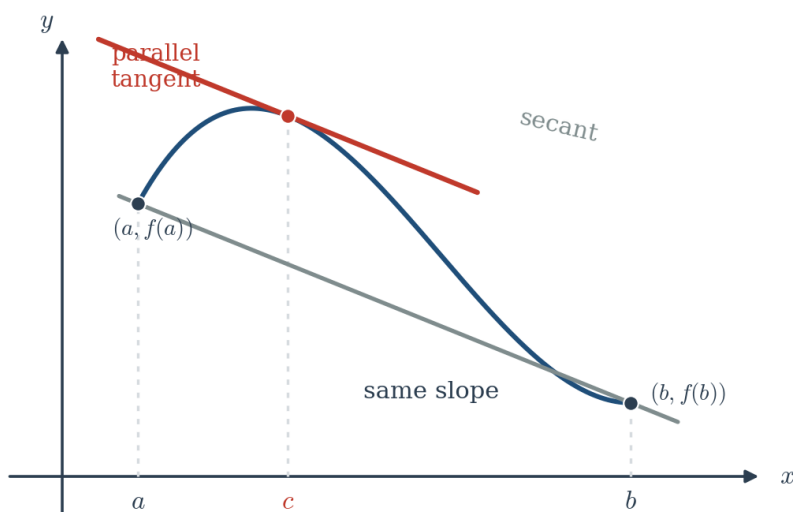


Figure 6.2

◆ Proof of the MVT

Subtract off the secant line to flatten the endpoints. Define

$$g(x) = f(x) - \left[f(a) + \frac{f(b) - f(a)}{b - a} (x - a) \right].$$

The bracket is exactly the secant line, so g measures how far f sits above or below it. Check the endpoints: $g(a) = f(a) - f(a) = 0$, and $g(b) = f(b) - f(b) = 0$. So $g(a) = g(b)$, and g inherits continuity and differentiability from f . **Rolle applies to g** : there's a c with $g'(c) = 0$. But

$$g'(x) = f'(x) - \frac{f(b) - f(a)}{b - a},$$

so $g'(c) = 0$ means precisely $f'(c) = \frac{f(b) - f(a)}{b - a}$. ■

Notice what just happened: the *general* theorem cost us almost nothing once we had the *special* one, because subtracting a straight line changes slopes by a constant and converts “average slope” into “zero.” That move — straighten the problem, solve the easy version, un-straighten — is one you’ll use for the rest of your mathematical life.

6.3 Why this chapter exists: three consequences

The MVT is a tool, and here is what it builds. Each of these is short to prove now and impossible to prove without it.

Consequence 1 — Zero slope means constant (the Chapter 5 debt, repaid)

If $f'(x) = 0$ at every point of an interval, then f is constant on that interval.

Proof. Take any two points $x < y$ in the interval. By the MVT, $f(y) - f(x) = f'(c)(y - x)$ for some c between them. But $f'(c) = 0$, so $f(y) - f(x) = 0$, i.e. $f(y) = f(x)$. Every two points have the same value: f is constant. ■

This looks almost too obvious to need proving — *of course* a function that never changes is constant. But “never has slope” and “never changes value” are genuinely different statements, and the only bridge between them is the MVT. Back in Chapter 5 we wanted to show Ce^{kx} was *the* solution to $\frac{dN}{dt} = kN$, not merely *a* solution, and the whole argument hinged on “ $g'(x) = 0$ everywhere, therefore g is constant.” We used it on faith. **That loan is now paid in full.**

A twin fact comes for free and will run all of integration: **if $f'(x) = g'(x)$ everywhere, then f and g differ by a constant** (apply Consequence 1 to $f - g$). When you reach Chapter 10 and wonder where the “+C” in an antiderivative comes from — this is its birth certificate.

Consequence 2 — The sign of the derivative controls rising and falling

If $f'(x) > 0$ on an interval, then f is increasing there. If $f'(x) < 0$, then f is decreasing.

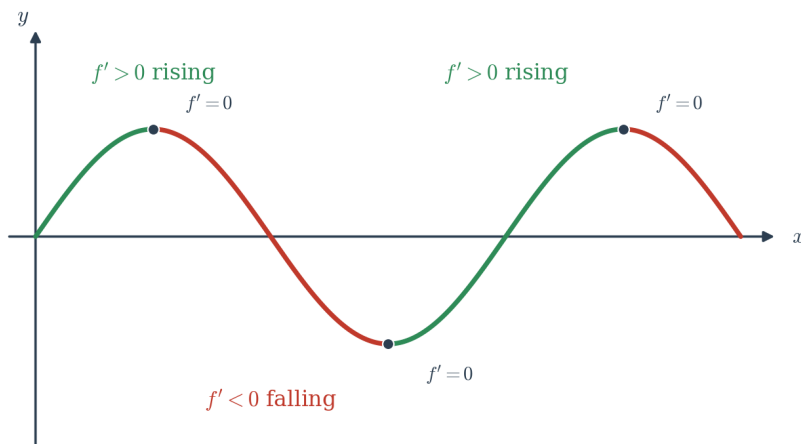


Figure 6.4

Proof. Take $x < y$. The MVT gives $f(y) - f(x) = f'(c)(y - x)$ with $y - x > 0$. If $f'(c) > 0$ the right side is positive, so $f(y) > f(x)$: increasing. If $f'(c) < 0$, it's negative, so $f(y) < f(x)$: decreasing. ■

This is the one that earns its keep in Chapter 8. The entire method of finding maxima and minima — locate where f' changes sign, because that's where f turns around — is just this consequence, read carefully. Without it, “set the derivative to zero” would be a ritual with no justification. With it, it's a theorem.

Consequence 3 — A function can't outrun its steepest slope

If $|f'(x)| \leq M$ everywhere on $[a, b]$, then $|f(b) - f(a)| \leq M(b - a)$.

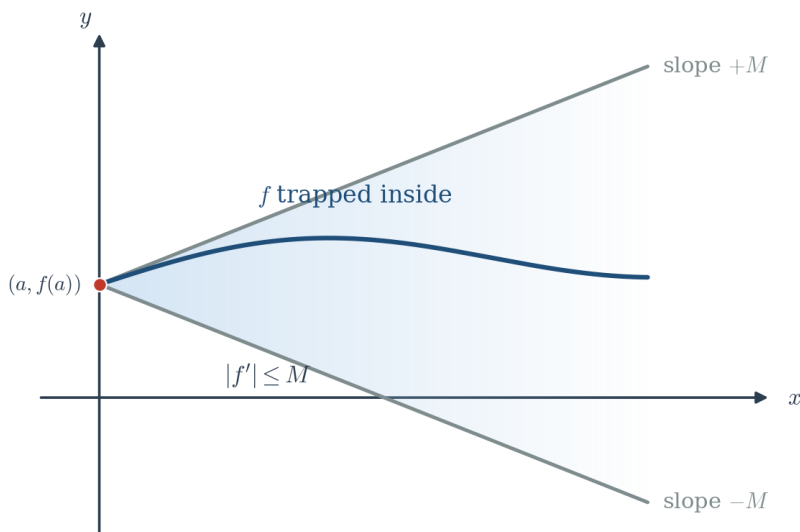


Figure 6.5

Proof. By the MVT, $f(b) - f(a) = f'(c)(b - a)$, so $|f(b) - f(a)| = |f'(c)|(b - a) \leq M(b - a)$. ■

This is the rigorous version of the robustness idea from Chapter 2: a function whose slope is capped simply *cannot* change too much over a short interval. It is the reason numerical methods come with error guarantees — bound the derivative, and you've bounded how wrong you can be. Engineers live on inequalities like this one.

◆ Layer 2 — The generalized MVT, and a glimpse of Taylor’s engine

There’s a two-function version of the MVT, due to Cauchy, that looks technical but is the secret machinery behind Chapter 7’s Taylor remainder (and behind L’Hôpital’s rule, if you meet it).

Cauchy’s MVT. If f and g are continuous on $[a, b]$ and differentiable on (a, b) , there is a c in (a, b) with

$$[f(b) - f(a)] g'(c) = [g(b) - g(a)] f'(c).$$

Proof (same tilt trick, cleverer line). Apply Rolle to $h(x) = [f(b) - f(a)] g(x) - [g(b) - g(a)] f(x)$. A quick check gives $h(a) = h(b)$ (the cross terms cancel), so some c has $h'(c) = 0$, which is exactly the claim. ■

Why care now? The ordinary MVT says $f(b) = f(a) + f'(c)(b - a)$ — it pins down f at b using *one* derivative and an unknown midpoint c . Taylor’s theorem will say $f(b) = f(a) + f'(a)(b - a) + \frac{1}{2} f''(c)(b - a)^2$ — the *same shape*, one derivative deeper, and Cauchy’s MVT is what proves that error term has exactly that form. The MVT isn’t just a theorem in this book; it’s the first rung of a ladder.

6.4 Where this is heading

You now hold the proof engine for the back half of Book 2. **Chapter 7 (Taylor)** uses the MVT — in its Cauchy form — to control the error when you approximate a curve by a polynomial, turning “the tangent line is close”

into “and here is *exactly* how close.” **Chapter 8 (maxima, minima, convexity)** runs almost entirely on Consequence 2: you’ll find a function’s highest and lowest points by tracking where its derivative changes sign, you’ll read its curvature off the *second* derivative, and you’ll meet **convex** functions — the ones that stay above all their tangent lines — which are precisely the functions on which the gradient descent of Chapter 9 is guaranteed to find the true bottom. Every one of those results is a consequence of the theorem you proved today.

Continuity forced structure. Differentiability forces more. That is the whole secret of calculus, and the Mean Value Theorem is where you first see it with full force.

6.5 Problems

Six tiers. *Fluency and Practice* are doing. *Concept & counterexample* and ♦ *Diamond* are thinking* — and in this chapter the Diamond tier is the heart of it, because the MVT’s value is in its proofs. **Teach it** is *teaching*. **Looking ahead** reaches into Chapters 7 and 8. Do all of Fluency; the Concept and Diamond tiers are where this chapter is actually learned.*

Fluency — *do these fast*

1. Verify Rolle’s hypotheses hold, then find every c it guarantees: (a) $f(x) = x^2 - 4x$ on $[0, 4]$; (b) $f(x) = x^3 - x$ on $[-1, 1]$.
2. Find every c the MVT guarantees: (a) $f(x) = x^2$ on $[1, 3]$; (b) $f(x) = x^3$ on $[0, 3]$.
3. For each, name which single hypothesis fails, so the theorems don’t apply: (a) $f(x) = |x|$ on $[-1, 1]$; (b) $f(x) = \frac{1}{x}$ on $[-1, 1]$; (c) $f(x) = \lfloor x \rfloor$ (the step function) on $[0, 2]$.
4. Given $f'(x) = (x-1)(x-3)$, state the intervals where f is increasing and where it is decreasing.

5. A car's position is $s(t) = t^2$ miles after t hours, on $[0, 3]$. Find its average speed over the trip, and the exact instant its speedometer matched that average.
6. If $|f'(x)| \leq 3$ for all x and $f(0) = 5$, what is the largest $f(4)$ could possibly be? The smallest?

Practice — *multi-step*

7. Find all c guaranteed by the MVT for $f(x) = x^3 - x$ on $[0, 2]$.
8. Show $f(x) = x^5 + 2x + 1$ has **exactly one** real root. (Existence: use the Intermediate Value Theorem on a sign change. Uniqueness: compute f' and use Consequence 2.)
9. Two functions satisfy $f'(x) = g'(x) = 2x$ for all x , with $f(0) = 3$ and $g(0) = 5$. Find $f(x) - g(x)$, then $f(x)$ and $g(x)$ themselves, citing the consequence you use.
10. Prove $|e^{-a} - e^{-b}| \leq |a - b|$ for all $a, b \geq 0$. (Apply the MVT to $f(x) = e^{-x}$ and bound $|f'|$.)
11. Prove the inequality $x > \ln(1 + x)$ for all $x > 0$. (Let $h(x) = x - \ln(1 + x)$, show $h(0) = 0$ and $h'(x) > 0$ for $x > 0$, then use Consequence 2.)
12. A differentiable f has $f(1) = 3$ and $f'(x) \geq 2$ for all x in $[1, 5]$. Prove $f(5) \geq 11$. (A one-sided version of Consequence 3.)

Concept & counterexample — *think; true/false with justification, or build the example*

13. (a) If f is continuous on $[a, b]$ and differentiable on (a, b) , some interior c has $f'(c)$ equal to the average slope. (b) If $f'(c) = 0$ at some interior c , then $f(a) = f(b)$. (c) If $f' > 0$ on (a, b) , then f is increasing there. (d) The MVT tells you how to *find* the point c .
14. **Find the error.** A student applies Rolle to $f(x) = |x|$ on $[-1, 1]$: “since $f(-1) = f(1) = 1$, there must be a c with $f'(c) = 0$.” But $|x|$

has slope ± 1 everywhere it has a slope at all. What went wrong?

15. **Construct** a function on $[0, 1]$ that is continuous with $f(0) = f(1)$ but has $f'(c) = 0$ at *no* interior point. (You'll have to break one hypothesis — which one, and why is that allowed to ruin the conclusion?)
16. **Construct** a function that is increasing on an interval yet has $f' = 0$ at one point inside it. (This shows “ f increasing” does *not* require “ $f' > 0$ everywhere” — the converse of Consequence 2 is false.)
17. **Is it possible?** A differentiable function with $f(a) = f(b)$ but $f'(x) \neq 0$ at *every* point of (a, b) . Argue from Rolle.
18. Explain in two or three sentences why “ $f' = g'$ everywhere” does **not** mean “ $f = g$,” and state exactly what it *does* mean, with a one-line example.

◆ Diamond — *the proofs are the chapter*

19. ◆ Prove **Fermat's lemma**: if f has an interior local maximum at c and $f'(c)$ exists, then $f'(c) = 0$. (Use one-sided difference quotients; reproduce the text's sign argument.)
20. ◆ Prove **Rolle's theorem** from the Extreme Value Theorem and Fermat's lemma, handling the case where both extremes sit at the endpoints.
21. ◆ Prove the **Mean Value Theorem** from Rolle by the tilt trick: construct the auxiliary g , verify it satisfies Rolle's hypotheses, and finish.
22. ◆ Prove **Consequence 1** ($f' = 0 \Rightarrow f$ constant) and its corollary ($f' = g' \Rightarrow f - g$ constant). State clearly where you invoke the MVT.
23. ◆ Prove **Consequence 2** ($f' > 0$ on an interval $\Rightarrow f$ strictly increasing).
24. ◆◆ Prove **Cauchy's generalized MVT** by applying Rolle to $h(x) = [f(b) - f(a)]g(x) - [g(b) - g(a)]f(x)$. Verify $h(a) = h(b)$ first.
25. ◆◆ Use the MVT (or a monotonicity argument) to prove $e^x \geq 1 + x$

for **all** real x , with equality only at $x = 0$. (Let $h(x) = e^x - 1 - x$; analyze the sign of h' on each side of 0.) Note for later: $1 + x$ is the tangent line to e^x at 0, and you've just shown e^x never dips below its tangent.

Teach it — *explain it, write it up, or fix it*

26. In **three or four sentences with the driving/speedometer story**, teach a friend what the MVT guarantees — and be careful to say what it does *not* give you (it doesn't tell you *where* c is).
27. **Debug your classmate.** They say: “ $f(x) = x^4 - 2x^2$ has $f(-\sqrt{2}) = f(\sqrt{2}) = 0$, so by Rolle there is *exactly one* point in between with zero slope.” Pinpoint the precise misreading and state what Rolle actually promises. (How many such points does this f really have?)
28. Write a complete, textbook-quality **worked solution** proving “ $x^5 + 2x + 1$ has exactly one real root,” narrating the two halves (IVT for existence, monotonicity for uniqueness) so a peer learns the reusable pattern.
29. **Design a problem.** Invent a cubic f and an interval $[a, b]$ on which Rolle applies, then write the solution key that finds every guaranteed c .
30. In two or three sentences, **teach the difference between the IVT and the MVT** — one is a promise about the *values* a continuous function must take, the other about the *slopes* a differentiable function must achieve.

Looking ahead — *Chapters 7 and 8 in your hands early*

31. The MVT says $f(b) = f(a) + f'(c)(b-a)$ for some midpoint c . Chapter 7 sharpens this to $f(b) = f(a) + f'(a)(b-a) + \frac{1}{2}f''(c)(b-a)^2$ — same shape, one derivative deeper. Guess the next version, with one more derivative still. (What power of $(b-a)$ and what factorial would you expect?)

32. Consequence 2 said the sign of f' controls rising and falling. In Chapter 8 you'll find maxima and minima by locating where f' *changes sign*. Predict: at a point where f switches from increasing to decreasing, what is f' doing there, and what kind of point have you found?
33. In problem 25 you proved e^x stays above its tangent line $1 + x$ everywhere. A function that lies above *all* of its tangent lines is called **convex** — the star of Chapter 8, and the exact condition under which gradient descent (Chapter 9) is guaranteed to reach the true minimum. Predict what the *second* derivative f'' (the slope of the slope) must be doing, in sign, for a convex function.
-

Chapter 7

Building a Function From Its Derivatives

Or: a curve is a line, then a parabola, then a polynomial — and you choose how close

Companion notebook. Play with this chapter's ideas — sliders, animations, and drills — in notebooks/ch7_taylor.ipynb.

By the end of this chapter you will be able to take almost any smooth function and replace it, near a point, with a **polynomial** that matches it as closely as you like — first matching its value, then its slope, then its curvature, then bending after bending — and you will know *exactly how wrong* the replacement is.

You will derive the one formula that runs the whole machine: the k -th coefficient is $\frac{f^{(k)}(a)}{k!}$. Not memorize it — *derive* it, in two lines, and see why the factorial is forced to be there.

You will build the **Taylor series of e^x** and watch it fall out as inevitable — $1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots$ — precisely *because* e^x is its own derivative, the property you fought for in Chapter 5. You'll do the same for $\frac{1}{1-x}$ (and meet an old friend, the geometric series) and for $\ln(1+x)$.

And you will meet the **remainder** — the honest accounting of the error — which turns out to be the Mean Value Theorem of Chapter 6, taken one derivative deeper. That is not a coincidence; it is the whole reason we proved the MVT first.

Here is the theme, and it is the third floor of a building we've been climbing. Continuity (Chapter 2) forced a function's *values* to behave. Differentiability and the MVT (Chapter 6) forced its *first derivative* to behave. Now: a function smooth enough to be differentiated again and again is pinned down, near a point, by its **whole tower of derivatives** at that one point. A function is — almost — nothing more than its derivatives, reassembled. “Almost” is the most interesting word in that sentence, and by the end you'll know exactly what it's hiding.

7.1 The tangent line was only a first draft

Back in Chapter 3 you learned that near a point a , a function is well approximated by its tangent line:

$$f(x) \approx f(a) + f'(a)(x - a).$$

Look at *why* this line, of all lines, is the right one. It agrees with f in two ways at a : same height ($f(a)$) and same slope ($f'(a)$). Among every line you could draw through the point $(a, f(a))$, the tangent is the one whose slope matches — and that makes it the one that hugs the curve most tightly right there. This is worth saying in plain words, because it is one of the central ideas of all of calculus and the bridge to everything you'll do later:

The derivative is the best linear approximation. To know f near a to first order is to know its tangent line.

But a line is straight, and almost nothing in the world is. The tangent is excellent in a tiny window around a and then peels away, because it cannot **bend** and f does. So the obvious move is: let our approximation bend too. (Figure 7.1: a curve with its tangent line, the two hugging near a and visibly separating as you move away.)

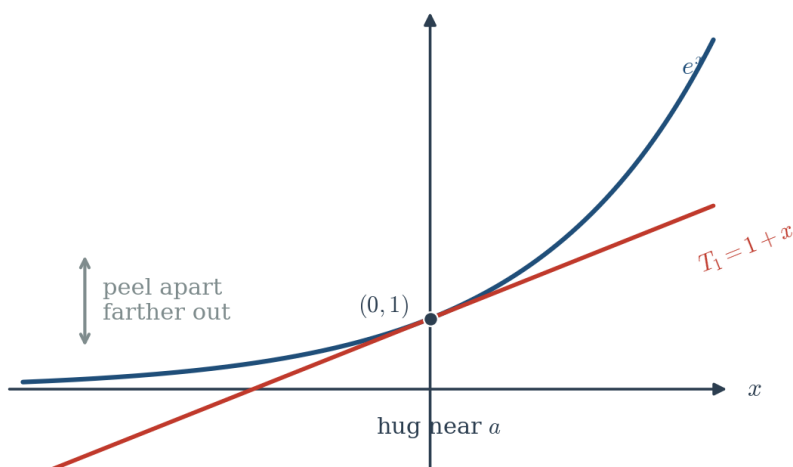


Figure 7.1

7.2 Match the curvature: the quadratic approximation

Bending is what the *second* derivative measures. So let's look for a parabola that matches f not in two ways but in three — value, slope, **and** curvature at a . Write the unknown parabola in powers of $(x - a)$, which makes everything easy to evaluate at a :

$$P(x) = c_0 + c_1(x - a) + c_2(x - a)^2.$$

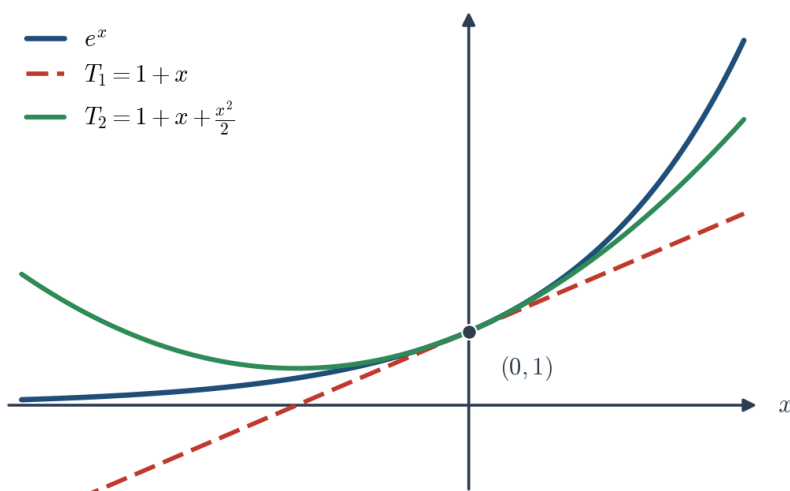
Now force the three matches, computing each derivative and plugging in $x = a$ (where every $(x - a)$ becomes 0):

- **Value:** $P(a) = c_0$, so matching $f(a)$ forces $c_0 = f(a)$.
- **Slope:** $P'(x) = c_1 + 2c_2(x - a)$, so $P'(a) = c_1$, forcing $c_1 = f'(a)$.
- **Curvature:** $P''(x) = 2c_2$, so $P''(a) = 2c_2$, forcing $c_2 = \frac{f''(a)}{2}$.

So the best quadratic is

$$P(x) = f(a) + f'(a)(x - a) + \frac{f''(a)}{2}(x - a)^2.$$

It contains the tangent line (the first two terms) and adds one bending term. It hugs the curve *better*, and over a *wider* window. (Figure 7.2: the same curve with its tangent and its matching parabola; the parabola tracks the curve far past where the line gave up.)



The pattern is already visible — keep matching higher derivatives, keep adding terms — so let's find the general rule, and the factorial it hides.

7.3 The general pattern, and why the factorial is forced

Suppose we want a degree- n polynomial matching f and its first n derivatives at a :

$$P(x) = \sum_{k=0}^n c_k (x - a)^k.$$

To find c_k , differentiate P exactly k times and evaluate at a . The key is what happens to a single power $(x - a)^j$ when you hit it with k derivatives:

- If $j < k$: you differentiate it down to a constant and then past it — it becomes 0.
- If $j > k$: after k derivatives it still carries a factor of $(x - a)$, which is 0 at $x = a$.
- If $j = k$: differentiating $(x - a)^k$ exactly k times peels off $k, k - 1, \dots, 1$ in turn, leaving the constant $k! = k \cdot (k - 1) \cdots 2 \cdot 1$.

So when you compute $P^{(k)}(a)$, **every term dies except the k -th**, which leaves $c_k \cdot k!$. Setting that equal to $f^{(k)}(a)$:

$$P^{(k)}(a) = c_k k! = f^{(k)}(a) \quad \implies \quad c_k = \frac{f^{(k)}(a)}{k!}$$

There's your factorial, and now you know *why* it's there: it's the $k!$ that falls out of differentiating $(x - a)^k$ down to the ground. We give the result a name.

The Taylor polynomial of degree n for f centered at a is

$$T_n(x) = \sum_{k=0}^n \frac{f^{(k)}(a)}{k!} (x-a)^k = f(a) + f'(a)(x-a) + \frac{f''(a)}{2!} (x-a)^2 + \cdots + \frac{f^{(n)}(a)}{n!} (x-a)^n.$$

Centered at $a = 0$ it's often called a *Maclaurin* polynomial. Let $n \rightarrow \infty$ and (where it's valid — see the remainder, below) you get the **Taylor series**.

Toolkit: Building a Taylor polynomial

1. Compute $f(a), f'(a), f''(a), \dots, f^{(n)}(a)$ — the derivatives **at the center**.

2. Divide the k -th one by $k!$.
 3. Assemble: $\sum_k \frac{f^{(k)}(a)}{k!}(x-a)^k$. Look for a pattern in the coefficients — the best functions have one.
-

7.4 The flagship: e^x rebuilds itself

Now the example the whole chapter was waiting for. Center at $a = 0$. We need all the derivatives of e^x at 0 — and here is where Chapter 5 pays an enormous dividend. Since e^x is **its own derivative**, every derivative of e^x is again e^x , and at $x = 0$ every one of them equals $e^0 = 1$. So *every* coefficient is $\frac{1}{k!}$, and

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \dots$$

Look at how inevitable that was. We didn't grind through a table of messy derivatives; the self-similarity of e^x handed us every coefficient at once. The most beautiful series in mathematics exists *because* its function refuses to change under differentiation. Test it: $e^{0.1} \approx 1 + 0.1 + \frac{0.01}{2} + \frac{0.001}{6} = 1.10517\dots$, and a calculator gives $e^{0.1} = 1.10517\dots$ — agreement to five places from four terms, because the factorials in the denominators crush the later terms fast. (Figure 7.3: e^x with T_1, T_2, T_3, T_4 layered on top, each clinging to the curve over a visibly wider stretch.)

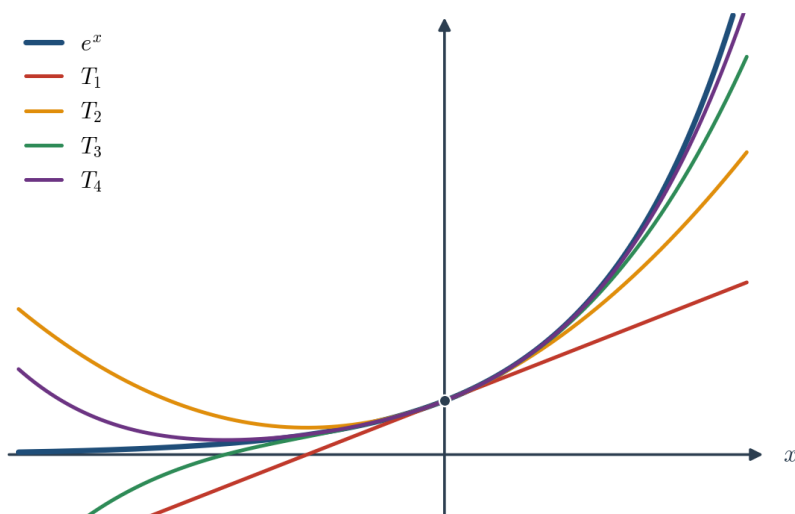


Figure 7.3

7.5 Two more, and an old friend returns

$\frac{1}{1-x}$ at 0. Its derivatives are $f^{(k)}(x) = \frac{k!}{(1-x)^{k+1}}$, so $f^{(k)}(0) = k!$, and the coefficient is $\frac{k!}{k!} = 1$. Every coefficient is 1:

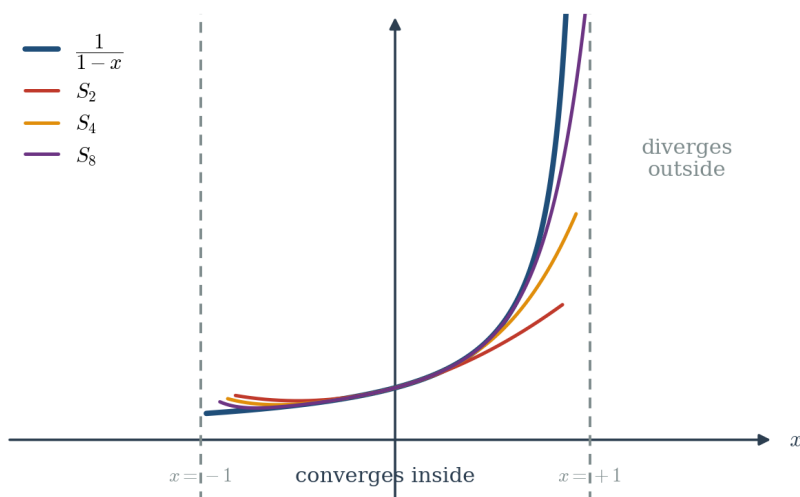


Figure 7.4

$$\frac{1}{1-x} = 1 + x + x^2 + x^3 + \dots$$

That is the **geometric series** from Chapter 1, now revealed as a Taylor series. And it carries a warning you must respect: this equality only holds for $|x| < 1$. Plug in $x = 2$ and the left side is $\frac{1}{1-2} = -1$ while the right side is $1 + 2 + 4 + \dots = +\infty$. A Taylor series is a *local* promise; it is true only where it converges. We'll make that precise with the remainder in a moment.

$\ln(1+x)$ at 0. Here $f'(x) = \frac{1}{1+x}$, and the derivatives give $f^{(k)}(0) = (-1)^{k-1}(k-1)!$, so the coefficient is $\frac{(-1)^{k-1}(k-1)!}{k!} = \frac{(-1)^{k-1}}{k}$:

$$\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \dots$$

Notice the leading term is just x . That is the exact reason behind a limit we *borrowed* back in Chapter 1: $\frac{\ln(1+t)}{t} \rightarrow 1$ as $t \rightarrow 0$, the fact that made $(1 + \frac{x}{n})^n \rightarrow e^x$ work. The “loan” was that $\ln(1+t) \approx t$ near 0 – and there it is, the first Taylor term, finally explained.

7.6 The other flagships: sine and cosine

Two functions belong beside e^x in any honest account of Taylor series, and now that you can differentiate them (Chapter 4), they fall out just as cleanly. Center both at 0.

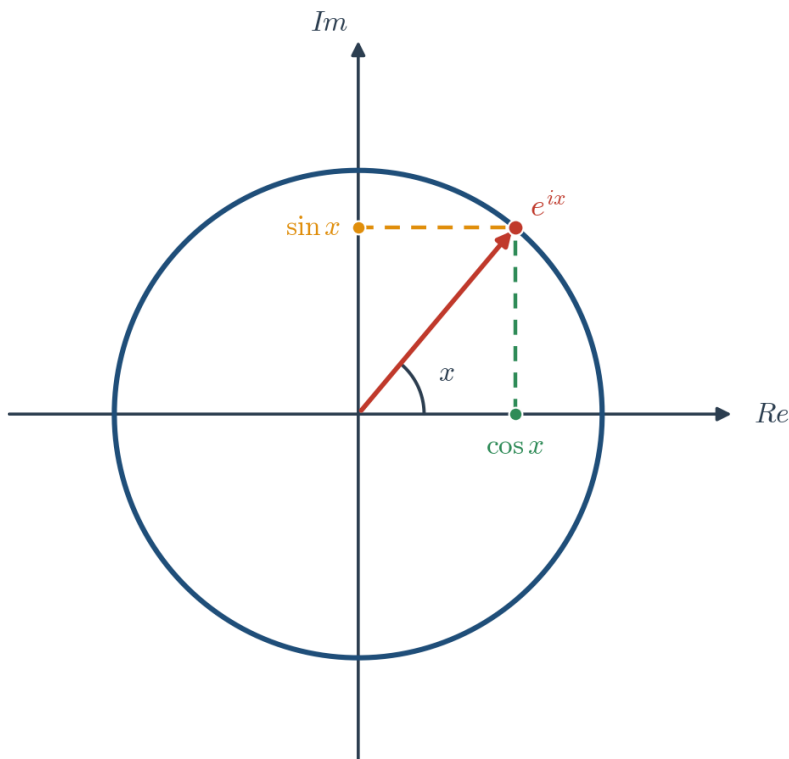


Figure 7.7

For $\sin x$, the derivatives **cycle**: $\sin x \rightarrow \cos x \rightarrow -\sin x \rightarrow -\cos x \rightarrow \sin x \rightarrow \dots$, repeating every four. Evaluated at 0 they read $0, 1, 0, -1, 0, 1, 0, -1, \dots$ — every even derivative is 0, every odd one is ± 1 . So only odd powers survive, with alternating signs:

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \dots$$

For $\cos x$, the same cycle evaluated at 0 reads $1, 0, -1, 0, \dots$ — only **even** powers survive:

$$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \dots$$

Two things to notice. First, $\sin$ keeps only odd powers and $\cos$ only even ones — which is exactly the statement that $\sin$ is an *odd* function (symmetric through the origin) and $\cos$ is *even* (mirror-symmetric across the y -axis). The symmetry of the function and the parity of its surviving powers are the same fact. Second, the leading term of $\sin x$ is just x — so $\frac{\sin x}{x} = 1 - \frac{x^2}{6} + \dots \rightarrow 1$, which is the famous limit from Chapter 2, now visible as nothing more than “the first Taylor term is x .” (Figure 7.6: $\sin x$ overlaid with its 1-, 3-, 5-term Taylor polynomials, each tracking the wave over more humps before peeling off.)

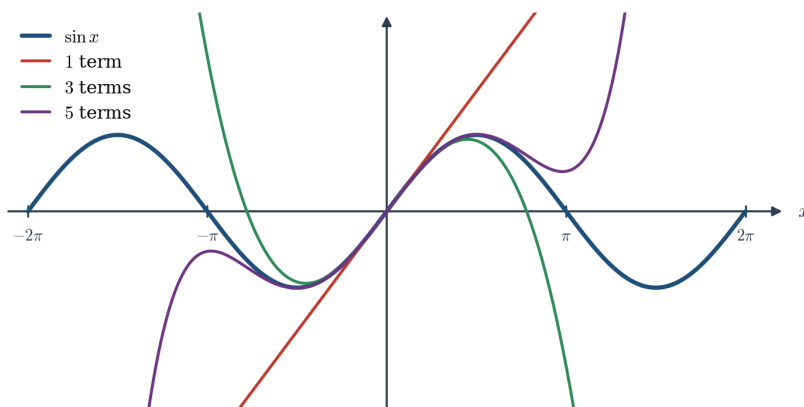


Figure 7.6

◆ **Layer 2 — Euler’s formula: where exponential meets trigonometry (and Book 1 pays off)**

Lay the three series side by side and something extraordinary appears. We have

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \dots$$

Now feed in an **imaginary** input — replace x with ix , where $i^2 = -1$ (the number you built in Book 1). The powers of i cycle $i, -1, -i, 1, i, \dots$, so the terms split into a real group and an imaginary group:

$$e^{ix} = \underbrace{\left(1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \dots\right)}_{\cos x} + i \underbrace{\left(x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots\right)}_{\sin x}.$$

The real part is *exactly* the cosine series; the imaginary part is *exactly* the sine series. So

$$e^{ix} = \cos x + i \sin x$$

This is **Euler’s formula**, and it is one of the deepest sentences in mathematics: the exponential that governs growth and decay, the circular functions that govern waves and rotation, and the imaginary unit from Book 1 are *the same object seen from three sides*. Set $x = \pi$ and, since $\cos \pi = -1$ and $\sin \pi = 0$, it collapses to **Euler’s identity**, $e^{i\pi} = -1$ — the single equation tying together e , π , i , 1, and 0. The complex numbers you met in Book 1, which had no payoff in single-variable real calculus, were waiting for exactly this moment.

A door left ajar. A Taylor series builds a function out of powers of $x - 1, x, x^2, x^3, \dots$ — and is at its best *near a single point*. But there is a second great expansion, just as

important, that builds a *repeating* function out of **waves** — $\sin x, \cos x, \sin 2x, \cos 2x, \dots$ — and is at its best *across a whole interval*. It is called a **Fourier series**, and through Euler’s formula those waves are exactly the complex exponentials e^{inx} you just met. Where Taylor asks “what polynomial matches this curve right here?”, Fourier asks “what mixture of pure tones makes up this signal?” — which is why it underlies music, images, data compression, and the analysis of anything that vibrates or repeats. The tools to build it (and to make sense of *infinitely many* waves added together) belong to Book 3; for now, just notice that powers are not the only basis a function can be written in, and file the word away.

7.7 The honest part: how wrong is the approximation?

A Taylor polynomial is an *approximation*, and an approximation you can’t bound is a guess. So we account for the gap exactly. Write

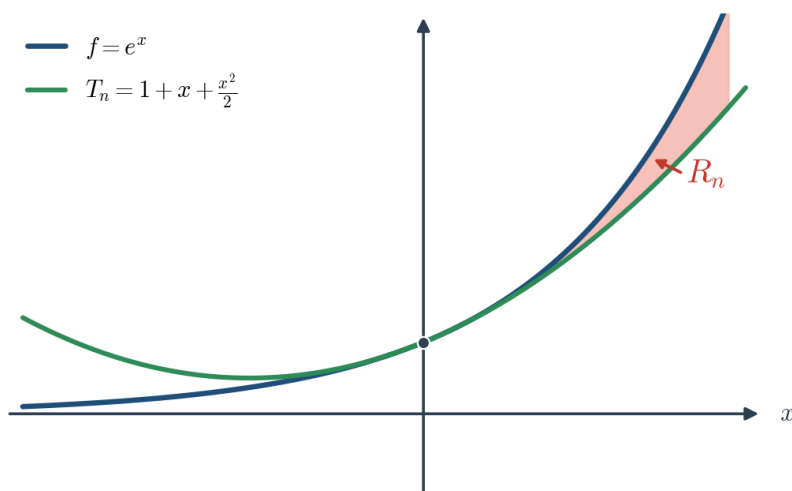


Figure 7.5

$$f(x) = T_n(x) + R_n(x),$$

where $R_n(x)$ is the **remainder** — everything the polynomial missed. Taylor's theorem gives it a stunningly clean form:

Taylor's Theorem (Lagrange remainder). For some point c between a and x ,

$$R_n(x) = \frac{f^{(n+1)}(c)}{(n+1)!}(x-a)^{n+1}.$$

Stare at that. It is *exactly the next term of the series* — the $(n+1)$ -th — except the derivative is evaluated not at the center a but at some unknown point c in between. And now look at the smallest case, $n = 0$:

$$f(x) = f(a) + f'(c)(x-a).$$

That is the Mean Value Theorem. The $n = 1$ case is $f(x) = f(a) + f'(a)(x-a) + \frac{1}{2}f''(c)(x-a)^2$ — the “second-order MVT” we previewed in Chapter 6. Taylor's remainder is the Mean Value Theorem, escalated one

derivative at a time. This is precisely why we proved the MVT — and its Cauchy cousin — before we got here: they are the engine that produces this error term.

The unknown c might look like it ruins everything — how can a formula with a mystery point be useful? Because you don't need to *find* c ; you only need to **bound** $f^{(n+1)}$ over the interval. If $|f^{(n+1)}| \leq M$ between a and x , then

$$|R_n(x)| \leq \frac{M |x - a|^{n+1}}{(n+1)!}.$$

Worked example — $e^{0.1}$ with a guarantee. Take $f = e^x$, $a = 0$, $n = 2$, so $T_2(0.1) = 1 + 0.1 + \frac{0.01}{2} = 1.105$. The remainder needs $f''' = e^x$, and on $[0, 0.1]$ we have $e^x \leq e^{0.1} < 1.2$. So

$$|R_2(0.1)| \leq \frac{1.2 \cdot (0.1)^3}{3!} = \frac{1.2 \times 0.001}{6} = 0.0002.$$

We have *proven*, without a calculator, that 1.105 is within 0.0002 of the true $e^{0.1}$. (It is: the true value is 1.10517, off by 0.00017.) That is the difference between Taylor as a suggestion and Taylor as mathematics: the remainder turns “approximately” into a number you can stand behind.

◆ Layer 2 — Proof of Taylor's theorem with Lagrange remainder

Fix the target x . Choose the number M that makes the equation exact: $f(x) = T_n(x) + M(x-a)^{n+1}$. We'll show $M = \frac{f^{(n+1)}(c)}{(n+1)!}$ for some c . Define, as a function of a *moving center* t ,

$$g(t) = f(x) - \sum_{k=0}^n \frac{f^{(k)}(t)}{k!} (x-t)^k - M(x-t)^{n+1}.$$

Two endpoint checks: $g(x) = f(x) - f(x) - 0 = 0$ (only the $k = 0$ term of the sum survives at $t = x$, giving $f(x)$), and $g(a) = 0$ by our choice of M . So $g(a) = g(x)$, and **Rolle's theorem** (Chapter 6) gives a c with $g'(c) = 0$. Now differentiate g . The sum **telescopes** — each term's product-rule pieces cancel against the next term's — leaving only

$$\frac{d}{dt} \sum_{k=0}^n \frac{f^{(k)}(t)}{k!} (x-t)^k = \frac{f^{(n+1)}(t)}{n!} (x-t)^n.$$

So $g'(t) = -\frac{f^{(n+1)}(t)}{n!} (x-t)^n + M(n+1)(x-t)^n$. Setting $g'(c) = 0$ and canceling $(x-c)^n$:

$$M(n+1) = \frac{f^{(n+1)}(c)}{n!} \implies M = \frac{f^{(n+1)}(c)}{(n+1)!}. \quad \blacksquare$$

One auxiliary function, one application of Rolle — the same tilt-and-flatten move that gave us the MVT itself, run at higher order.

◆ Layer 2 — Why e^x *actually equals* its series everywhere

Having a series isn't the same as the series being *true*. For e^x it is, for every x , and the remainder proves it. With $a = 0$, $|R_n(x)| = \frac{e^c |x|^{n+1}}{(n+1)!}$ for some c between 0 and x , so $|R_n(x)| \leq \frac{e^{|x|} |x|^{n+1}}{(n+1)!}$. Fix x and let $n \rightarrow \infty$: the factorial in the denominator overwhelms the power in the numerator (you saw in Chapter 1 how violently $n!$ outgrows any fixed base raised to n), so $R_n(x) \rightarrow 0$. The polynomials converge to e^x itself. *That* is what licenses writing “=” instead of “ $\approx$.”

7.8 Where this is heading

Three doors open from here, and you can already see through each.

The second-derivative test (Chapter 8). At a critical point, where $f'(a) = 0$, the quadratic approximation collapses to $f(x) \approx f(a) + \frac{1}{2}f''(a)(x - a)^2$ — a pure parabola. If $f''(a) > 0$ it opens upward and a is a *minimum*; if $f''(a) < 0$ it opens downward and a is a *maximum*. The whole second-derivative test is just *reading the leading Taylor term at a critical point*.

Convexity (Chapter 8). The error of the tangent-line approximation is $R_1 = \frac{1}{2}f''(c)(x - a)^2$. When $f'' \geq 0$, that remainder is never negative, so f never dips below its tangent line — it stays above all of them. A function that stays above all its tangents is **convex**, and convex is exactly the condition under which gradient descent is guaranteed to find the true bottom. The inequality $e^x \geq 1 + x$ you proved in Chapter 6 was this story's first instance.

Newton's quadratic convergence (Chapter 9). Newton's method replaces f with its tangent line T_1 and solves. The error it leaves behind is governed by the T_1 remainder, $\frac{1}{2}f''(c)(x - a)^2$ — proportional to the **square** of how far off you were. That square is the reason Newton's correct digits double each step.

A curve is a line, then a parabola, then a polynomial. Once you can choose how close, the rest of calculus is deciding how close you need.

7.9 Problems

Six tiers. **Fluency** and **Practice** are doing* — compute Taylor polynomials until the machine is automatic, then apply and estimate. **Concept & coun-**

terexample and **◆ Diamond** are *thinking* — the convergence subtleties and the proofs (the remainder is the real content). **Teach it** is *teaching*. **Looking ahead** opens Chapters 8 and 9. Do all of Fluency; the Concept tier here guards against the most common Taylor mistakes.*

Fluency — *do these fast, until the machine is automatic*

- Find the degree-2 Taylor polynomial at 0 for: (a) e^x ; (b) e^{2x} ; (c) e^{-x} ; (d) $e^{x/3}$.
- Write the first three nonzero terms at 0 for: (a) $\sin x$; (b) $\cos x$; (c) $\sin(2x)$; (d) $\cos \frac{x}{2}$.
- Find the degree-3 Taylor polynomial at 0 for: (a) $\frac{1}{1-x}$; (b) $\frac{1}{1+x}$; (c) $\frac{1}{1-2x}$. State the coefficient pattern in each.
- Write the degree-2 polynomial that matches each at 0 (compute $f(0), f'(0), f''(0)$ first): (a) $\sqrt{1+x}$; (b) $(1+x)^{1/3}$; (c) $\ln(1+x)$; (d) $\ln(1-x)$.
- By substituting or multiplying *known* series, write the first three nonzero terms at 0 for: (a) $\frac{1}{1+x^2}$; (b) e^{x^2} ; (c) $x e^x$.
- Compute $f^{(k)}(0)$ for $k = 0, 1, 2, 3$ and write the coefficient c_k for: (a) e^x ; (b) $\cos x$.
- Estimate with the first few series terms, then compare to a calculator: (a) $e^{0.2}$; (b) $\cos(0.1)$; (c) $\sin(0.3)$; (d) $\sqrt{1.1}$ via $(1+x)^{1/2}$.
- Taylor at a center *other than* 0, so $(x-a)$ powers matter: find the degree-2 Taylor polynomial of (a) e^x at $a = 1$; (b) $\sqrt{x}$ at $a = 4$.

Practice — *apply, estimate, and bound the error*

- Find the general-term series of e^{-x} at 0 and check it against e^x 's with the signs flipped.
- Derive the geometric series as the Taylor series of $\frac{1}{1-x}$ by computing $f^{(k)}(0)$, and state where the equality is valid.
- Find the first three nonzero terms of $\ln(1+x)$, and explain in one line how the leading term explains the Chapter 1 fact $\frac{\ln(1+t)}{t} \rightarrow 1$.

12. Estimate $e^{0.1}$ with T_2 , then **bound the error** using $|f'''| \leq 1.2$ on $[0, 0.1]$. Confirm the true value falls inside your bound.
13. Estimate $\cos(0.2)$ with $1 - \frac{x^2}{2}$. Then bound the error using the *next* nonzero term $\frac{x^4}{24}$, and explain why — for an even function like $\cos$ — the error jumps by **two** powers, not one.
14. The arctangent series. Given $\frac{d}{dx} \arctan x = \frac{1}{1+x^2} = 1 - x^2 + x^4 - \dots$, find the first three nonzero terms of $\arctan x$ at 0 (each term is an antiderivative of one above; match $\arctan 0 = 0$). Set $x = 1$ to glimpse $\frac{\pi}{4} = 1 - \frac{1}{3} + \frac{1}{5} - \dots$.
15. Find the first three terms of the binomial series for $(1+x)^{1/2}$ and $(1+x)^{-1}$, and check the second against the geometric series.
16. **Multiply series:** find the first three nonzero terms of $e^x \cos x$ by multiplying the two series. (Notice which power drops out.)
17. **The Gaussian.** By substituting $-x^2$ into the series for e^u , find the first three nonzero terms of e^{-x^2} — the bell curve at the heart of probability and machine learning. Is it an even or odd function, and where is it largest?
18. (*Notebook*) Plot $\sin x$ with its 1-, 3-, 5-, and 7-term Taylor polynomials on $[-2\pi, 2\pi]$, and describe how many humps each captures before peeling off.

Concept & counterexample — *think; true/false with justification, or build the example*

19. (a) T_n matches f and its first n derivatives at the center. (b) A larger n always gives a better approximation at *every* x . (c) The Taylor series of a polynomial is itself. (d) If two functions share the same Taylor polynomial of every degree at a , they're equal. (*Part (d) is subtle — see problem 31.*)
20. **Find the error.** A student writes: " $\frac{1}{1-x} = 1 + x + x^2 + \dots$, so at $x = 2$, $-1 = 1 + 2 + 4 + \dots$." Identify the precise mistake.
21. Explain why $\sin x$'s series has **only odd powers** and $\cos x$'s has **only**

- even powers**, by connecting each to the function's symmetry.
22. **Construct** a function whose degree-1 Taylor approximation at 0 is exactly $1 + 3x$, then a *different* one with the same approximation. What do all such functions share, and where do they differ?
 23. In one or two sentences, explain **why the factorial appears**, by reference to what k derivatives do to $(x - a)^k$.
 24. **Is it possible?** A function whose tangent-line error shrinks like $(x - a)^2$, and another whose error shrinks only like $|x - a|$. What property separates them?
 25. The series for $\ln(1 + x)$ is $x - \frac{x^2}{2} + \dots$. Without computing the radius of convergence formally, **predict** the value of x at which it must start to fail, and why. (Where does $\ln(1 + x)$ itself misbehave?)

◆ **Diamond — the remainder, and the deep identities, are the real content**

26. ◆ Derive the coefficient formula: for $P(x) = \sum c_k(x - a)^k$, show $P^{(k)}(a) = k! c_k$, hence $c_k = \frac{f^{(k)}(a)}{k!}$. (The “everything but one term dies” argument.)
27. ◆ Prove the quadratic Taylor polynomial is the **unique** parabola matching f 's value, slope, and curvature at a .
28. ◆ Show the $n = 0$ case of Taylor's remainder is exactly the Mean Value Theorem, and the $n = 1$ case is the second-order MVT from Chapter 6.
29. ◆ Prove $\cos x$ is an **even** function and $\sin x$ an **odd** function directly from their series (every power even, resp. odd).
30. ◆◆ **Derive Euler's formula** $e^{ix} = \cos x + i \sin x$ by substituting $x \rightarrow ix$ into e^x 's series, using $i^2 = -1$ to sort the terms into real and imaginary groups. Then set $x = \pi$ to obtain $e^{i\pi} = -1$.
31. ◆◆ Prove **Taylor's theorem with Lagrange remainder** by the auxiliary-function method: fix x , choose M with $f(x) = T_n(x) + M(x - a)^{n+1}$, define $g(t) = f(x) - \sum_{k=0}^n \frac{f^{(k)}(t)}{k!}(x - t)^k - M(x - t)^{n+1}$,

show $g(a) = g(x) = 0$, apply Rolle, and finish via the telescoping derivative.

32. ♦♦ Prove $\sin x$'s Taylor series converges to $\sin x$ for **every** x , using $|f^{(n+1)}| \leq 1$ to bound $|R_n(x)| \leq \frac{|x|^{n+1}}{(n+1)!} \rightarrow 0$. Note why this is *cleaner* than the e^x case.
33. ♦♦ The famous warning. Let $f(x) = e^{-1/x^2}$ for $x \neq 0$, $f(0) = 0$; accept that $f^{(k)}(0) = 0$ for all k . Write its Taylor series at 0, note it converges everywhere but **not to** f , and explain what this proves about 19(d).

Teach it — *explain it, write it up, or fix it*

34. In **three or four sentences to a friend**, teach “the derivative is the best linear approximation,” and how Taylor extends it to the best quadratic, cubic, and beyond.
35. **Debug your classmate.** They write: “ $e^x = 1 + x + x^2 + x^3 + \dots$, since every derivative of e^x is e^x .” Pinpoint the missing ingredient and give the correct series, with a one-line reason for the $\frac{1}{k!}$.
36. Write a complete **worked solution** estimating $\ln(1.1)$ from a Taylor polynomial of $\ln(1+x)$, *with* an error bound, narrated so a peer could redo it for $\ln(1.2)$.
37. **Design a problem.** Choose a function (try a trig one) and a center a , and write a problem asking for its degree-2 (or first three nonzero) Taylor terms, with a clean key.
38. In a few sentences, **teach Euler’s formula** to someone who knows the three series but has never seen them combined — what single substitution does all the work, and why is the result surprising?
39. In a few sentences, **teach the difference** between a Taylor *polynomial* (finite, carrying a remainder) and a Taylor *series* (infinite, equal to the function only where it converges and the remainder vanishes).

Looking ahead — Chapters 8 and 9 in your hands early

40. At a critical point ($f'(a) = 0$), the quadratic approximation is $f(x) \approx f(a) + \frac{1}{2}f''(a)(x-a)^2$. Predict: $f''(a) > 0$ — min or max? $f''(a) < 0$? (You've derived the second-derivative test.)
 41. The tangent-line error is $R_1 = \frac{1}{2}f''(c)(x-a)^2$. Predict the sign condition on f'' that makes $R_1 \geq 0$ always — i.e., makes f stay above all its tangents. (That's *convexity*, Chapter 8.)
 42. Newton's method (Chapter 9) replaces f by its tangent line and solves. Using the T_1 remainder, predict why the error *after* a Newton step is proportional to the **square** of the error before.
 43. In problem 17 you found the series of the Gaussian e^{-x^2} . Its cousin $e^{-x^2/2}$ is the bell curve of probability (Book 4). From the series, predict its symmetry and its single maximum — and notice you now understand the most important curve in statistics as nothing but a reshuffled exponential.
-

Chapter 8

Maxima, Minima, and the Shape of a Function

Or: where a curve peaks, where it bottoms out, and why some bowls have no false floors

Companion notebook. Play with this chapter's ideas — sliders, animations, and drills — in `notebooks/ch8_convexity.ipynb`.

By the end of this chapter you will be able to find the highest and lowest points of a function — the single most useful thing calculus does in the real world — and you will know *why* the methods work, because you built their foundations in the last two chapters.

You will locate the candidates with **Fermat's theorem** (extrema hide where the slope is zero), sort them with the **first-** and **second-derivative tests**, and read a function's curving from its **second derivative**. Every one of these is a consequence of the Mean Value Theorem (Chapter 6) or of Taylor's quadratic (Chapter 7) — you are not learning new magic, you are spending what you already own.

Then you will meet **convex functions** — the ones shaped like a single clean bowl — and prove the property that makes them the most important functions in optimization: *a convex function has no false floors*. Every place where its slope is zero is the true global minimum; there is nowhere to get stuck. That one fact is the dividing line between optimization problems that are guaranteed to work and the ones — like training a neural network

— where you cross your fingers. The two-valley landscape from Chapter 9’s gradient-descent lab, where the ball settled into the wrong valley, was exactly a *non-convex* function. This chapter names the condition that rules that out.

Here is the through-line. Finding a maximum or minimum is asking “where does this function turn around?” — and turning around is a statement about *slope*. So the whole subject is the derivative, read carefully. That’s why we needed the theorems first.

8.1 The problem, and why it is the whole game

A **maximum** is a high point of a function; a **minimum** is a low point. We split each into two kinds, and the difference matters enormously:

- A **local** maximum is a point higher than everything *nearby* — a hill-top, even if taller hills exist elsewhere.
- A **global** (or absolute) maximum is the single highest point *anywhere* on the function’s domain — the tallest hill of all.

(Same for minima, swapping high for low.) The reason this is *the* application of calculus: almost every practical question is secretly “make this as big or as small as possible.” An engineer minimizes the material in a can. A pharmacologist finds the dose at which a drug’s effect peaks. And — the thread we’ve followed all book — a machine-learning system minimizes a **loss**, a single number measuring how wrong its predictions are. Learning *is* minimizing. So learning to find minima is learning, in miniature, how a model is trained. (*Figure 8.1: a wavy curve with two local maxima of different heights, one global maximum, two local minima, and the global minimum marked — so the local/global distinction is unmistakable.*)

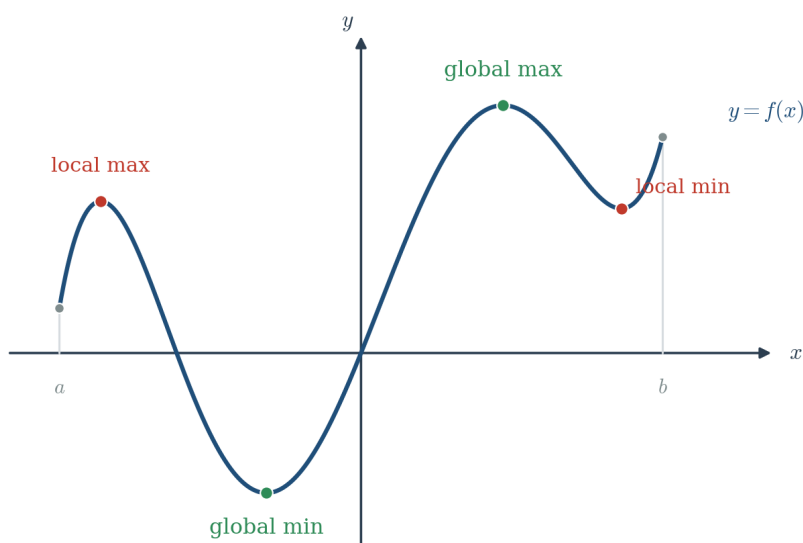


Figure 8.1

8.2 Where extrema hide: critical points

You cannot search every point of a curve. The first job is to narrow the hunt, and Chapter 6 already did the hard part. Recall **Fermat's lemma**: at an interior peak or valley where the derivative exists, the derivative is zero — the curve is momentarily flat at the top of a hill or the bottom of a dip. Promoted to a search strategy, this is:

Fermat's Theorem. If f has a local maximum or minimum at an interior point c , then either $f'(c) = 0$ or $f'(c)$ does not exist.

A point where $f'(c) = 0$ or f' fails to exist is called a **critical point**. Fermat's theorem says: *every interior extremum is a critical point*. So the complete list of places an extremum can hide is short — the critical points, plus (on a closed domain) the endpoints. Everything else is ruled out before

you start. That is the entire reason “set the derivative to zero” is the first move in every optimization problem: it’s not a ritual, it’s Fermat’s theorem narrowing an infinite search to a finite list.

A warning the theorem makes precise: $f'(c) = 0$ is **necessary** but not **sufficient**. The function $f(x) = x^3$ has $f'(0) = 0$, yet 0 is neither a max nor a min — the curve flattens for an instant and keeps climbing. So a critical point is a *candidate*, not a verdict. The next two tests deliver the verdict.

8.3 The first-derivative test: watch the slope change sign

At a critical point, ask what the slope is doing on either side — and Chapter 6’s Consequence 2 (the sign of f' controls rising and falling) tells you what that means for the shape:

- If f' goes from **positive to negative** as x passes through c — rising, then falling — then c is a **local maximum**. You climbed to a peak and came down.
- If f' goes from **negative to positive** — falling, then rising — then c is a **local minimum**. You descended to a valley floor and climbed out.
- If f' **does not change sign**, c is neither (like x^3 at 0: positive slope on both sides, just briefly zero).

That’s the whole test, and it rests entirely on monotonicity — which rests on the MVT. (*Figure 8.2: a curve annotated with + and – signs for f' on each interval, with peaks where $+ \rightarrow -$ and valleys where $- \rightarrow +$.*)

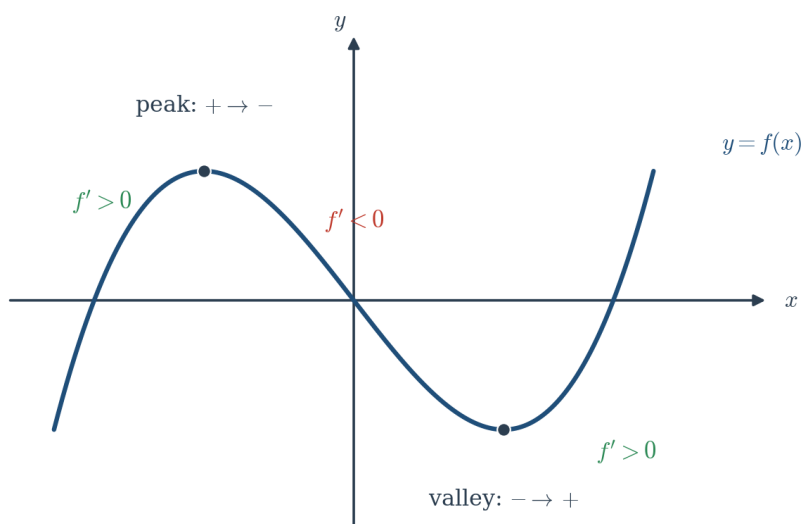


Figure 8.2

8.4 The second derivative: how a function curves

The derivative of the derivative, f'' , is the rate of change of the *slope* — and that measures **curving**. Watch what its sign means:

- $f'' > 0$: the slope is **increasing** (getting more positive or less negative as you move right), so the curve bends **upward** — like a cup, or a valley. We say f is **concave up**.
- $f'' < 0$: the slope is **decreasing**, so the curve bends **downward** — like a cap, or a hilltop. We say f is **concave down**.

A point where the concavity *switches* — where f'' changes sign — is an **inflection point**, the spot where a curve stops bending one way and starts bending the other (the steepest point of an S-shaped growth curve,

for instance). (Figure 8.3: a single curve colored by the sign of f'' — concave-up stretches as cups, concave-down as caps, with the inflection point where they meet.)

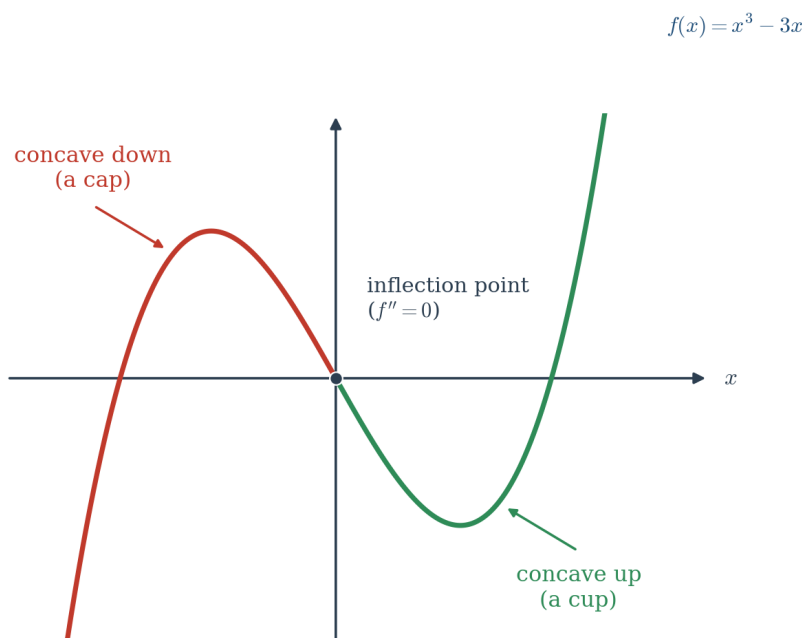


Figure 8.3

This gives a second, often faster, way to classify critical points.

8.5 The second-derivative test (and where it came from)

At a critical point a — where $f'(a) = 0$ — the concavity settles the question immediately:

Second-Derivative Test. Suppose $f'(a) = 0$. - If $f''(a) > 0$, then a is a **local minimum** (the curve cups upward there). -

- If $f''(a) < 0$, then a is a **local maximum** (it caps downward).
 - If $f''(a) = 0$, the test is **inconclusive** — a could be a max, a min, or neither.

Where does this come from? Straight from Chapter 7. Near a critical point, the Taylor quadratic loses its linear term (because $f'(a) = 0$) and becomes

$$f(x) \approx f(a) + \frac{1}{2}f''(a)(x - a)^2.$$

That is a parabola sitting at $(a, f(a))$. If $f''(a) > 0$ the parabola opens **upward** — a minimum; if $f''(a) < 0$ it opens **downward** — a maximum. The second-derivative test is nothing more than *reading the leading Taylor term at a critical point*. And the inconclusive case is honest: when $f''(a) = 0$, the parabola is flat and the *next* Taylor term takes over — for x^4 (a min), $-x^4$ (a max), and x^3 (neither), all have $f'(0) = f''(0) = 0$, and only the higher terms tell them apart. (Figure 8.4: the three functions x^4 , $-x^4$, x^3 near 0, all flat to second order, showing why $f'' = 0$ can't decide.)

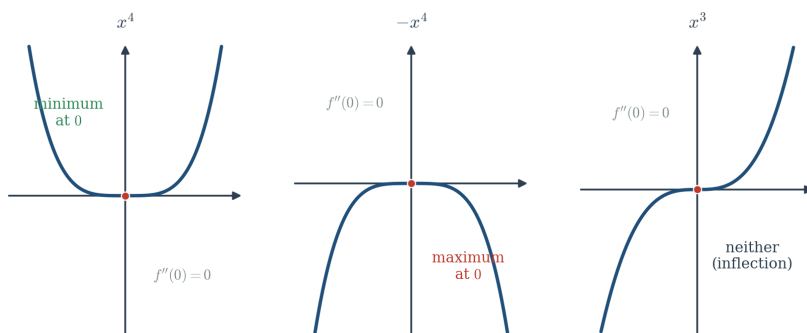


Figure 8.4

Toolkit: Classifying a critical point a (where $f'(a) = 0$)

1. **Second-derivative test** (fastest): compute $f''(a)$. Positive $\rightarrow$ min, negative $\rightarrow$ max, zero $\rightarrow$ fall back to step 2.
 2. **First-derivative test** (always works): check the sign of f' just left and just right of a . $+\rightarrow -\rightarrow$ max, $-\rightarrow +\rightarrow$ min, no change $\rightarrow$ neither.
-

8.6 Convex functions: one bowl, four faces

Now the centerpiece. A function is **convex** on an interval if it is shaped like a single upward bowl — and that shape can be described four different ways, all secretly the same. (A function is **concave** if $-f$ is convex — a single downward dome; everything below flips.)

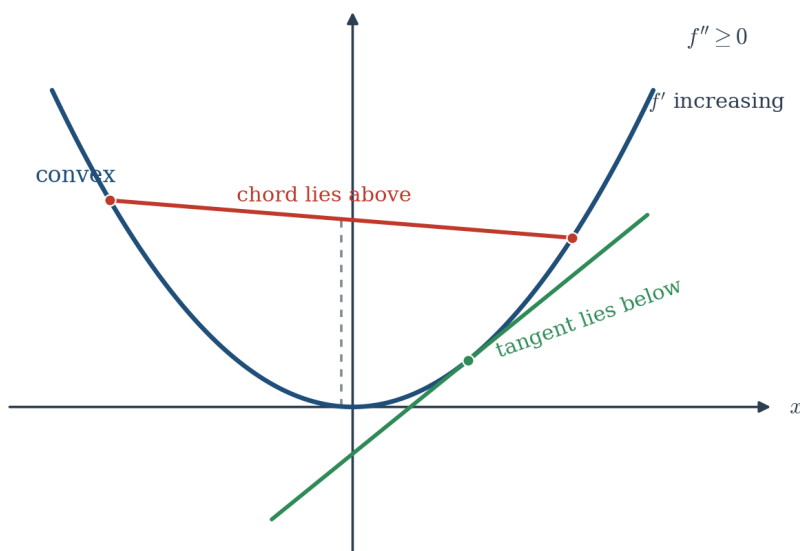


Figure 8.6

Four faces of convexity. For a (twice-differentiable) function f , these are equivalent: 1. **Chord above graph** — the straight line between any two points on the graph lies on or above the graph. (*This is the geometric definition.*) 2. **Tangent below graph** — the graph lies on or above every one of its tangent lines. 3. **Slopes increasing** — f' is an increasing function (the curve only ever steepens left to right). 4. **Nonnegative curvature** — $f''(x) \geq 0$ everywhere.

These look like four separate ideas and are one idea wearing four coats. The links are exactly the theorems of the last two chapters:

◆ Layer 2 — Why the four faces agree

Face 4 $\Leftrightarrow$ Face 3. “ $f'' \geq 0$ ” says the derivative of f' is nonnegative, which by Chapter 6’s Consequence 2 (applied to f') says f' is increasing. Same statement. **Face 4 $\Rightarrow$ Face 2.** By Taylor (Chapter 7), the gap between f and its tangent line at a is the remainder $f(x) - [f(a) + f'(a)(x - a)] = \frac{1}{2}f''(c)(x - a)^2$ for some c . If $f'' \geq 0$, this gap is ≥ 0 — the graph never dips below its tangent. (This is the inequality $e^x \geq 1 + x$ from Chapter 6, now understood as “ e^x is convex.”) **Face 2 $\Rightarrow$ Face 1** (chord above) is the geometric finish, and is left as a Diamond problem. The point: convexity isn’t a new mystery. It’s $f'' \geq 0$, dressed in geometry.

8.7 The theorem that matters: a convex bowl has no false floors

Here is why anyone cares. In a general function, the first-derivative test finds *local* minima — valley floors that might be nowhere near the lowest point overall. You could solve $f'(x) = 0$, find a dip, and be badly wrong about the true minimum. **Convexity destroys that danger completely.**

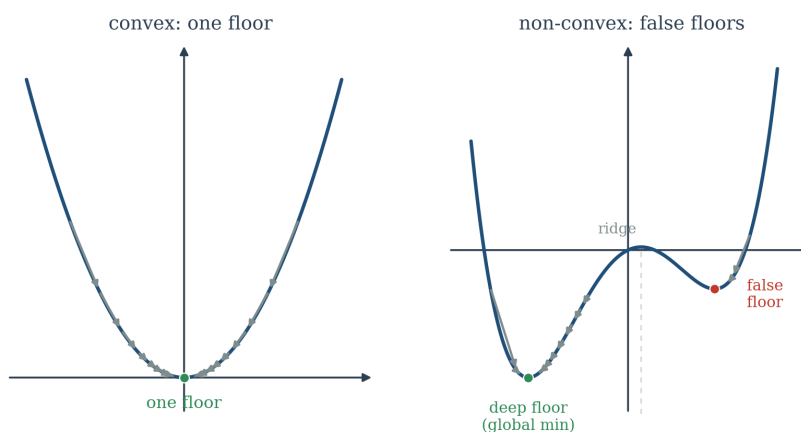


Figure 8.7

Theorem. If f is convex and $f'(a) = 0$, then a is the **global** minimum.

Proof. Since $f'(a) = 0$, the tangent line at a is *horizontal*: it is the flat line $y = f(a)$. By Face 2 of convexity, the graph lies on or above every tangent line — in particular, above this one. So $f(x) \geq f(a)$ for **every** x . That is exactly what it means for a to be the global minimum. ■

Read what just happened. For a convex function, you don't have to hunt for the best among many local minima, or worry that a better one lurks elsewhere. There *are* no others. Find one point where the slope is zero, and you have found the bottom of the world. A convex bowl has a single floor and no false ones.

Why this is the deepest idea in machine learning optimization

Recall the gradient-descent lab from Chapter 9 — the one where you ran the ball downhill on $f(x) = x^4 - 3x^2 + x$ and watched it settle into *different* valleys depending on where it started. That function is **non-convex**: it has two separate dips, and gradient descent, which only ever steps downhill, falls into whichever one is nearest and stops, with no way to know a deeper valley exists across the ridge. *That* is the entire reason the starting point mattered.

Now contrast a **convex** loss. By the theorem above, it has exactly one minimum, and it is global. Gradient descent cannot get stuck, because there is nothing to get stuck in — every downhill path leads to the same single bottom. **Convex problems are the ones where gradient descent is guaranteed to find the best answer.** This is why, for decades, “is your problem convex?” was the first question in optimization.

And the punchline of modern AI is that neural networks are *wildly* non-convex — their loss landscapes have astronomically many local minima — and we train them with gradient descent anyway. There is no guarantee it finds the global best. The astonishing empirical fact, still not fully understood, is that the local minima it does find are usually good enough. Every time you hear that training a model is part science and part art, this is what’s meant: we left the safe, convex world on purpose, because the non-convex models are the ones powerful enough to matter. You now know exactly what was traded away, and why.

8.8 Finding the global extremum on a closed interval

Often the domain is a closed interval $[a, b]$ — a quantity that can't go below a or above b . Here the **Extreme Value Theorem** from Chapter 2 makes a flat guarantee: a continuous function on a closed interval *does* attain a global maximum and a global minimum somewhere on it. They exist; we just have to find them. And the candidate list is finite:

Closed-interval method. To find the global max and min of a continuous f on $[a, b]$: 1. Find all critical points inside (a, b) . 2. Evaluate f at each critical point **and** at both endpoints a and b . 3. The largest value is the global max; the smallest is the global min.

The endpoints earn their place because Fermat's theorem only governs *interior* extrema — on a closed interval, the highest point can sit at an edge where the slope never had to be zero. (*Figure 8.5: a curve on $[a, b]$ whose global maximum is at an interior critical point but whose global minimum is at an endpoint, showing why both must be checked.*)

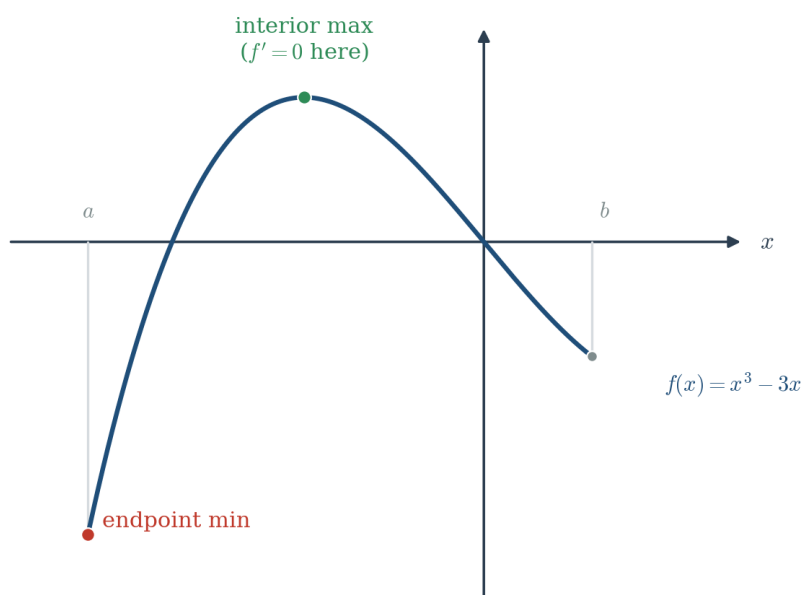


Figure 8.5

Worked example. Find the global extremes of $f(x) = x^3 - 3x$ on $[0, 2]$. Critical points: $f'(x) = 3x^2 - 3 = 0 \Rightarrow x = \pm 1$, of which only $x = 1$ is in $(0, 2)$. Evaluate at the candidates $\{0, 1, 2\}$: $f(0) = 0$, $f(1) = -2$, $f(2) = 2$. So the global minimum is -2 at $x = 1$, and the global maximum is 2 at the endpoint $x = 2$. (Notice the max hid at an endpoint — exactly the case the method is built to catch.)

8.9 Optimization in the wild: a worked design problem

The real payoff is turning a word problem into a function and minimizing it. Here is the genre's classic, an honest engineering question.

Problem. A cylindrical can must hold a fixed volume V . What dimensions use the *least metal* (least surface area)?

Set it up. A cylinder of radius r and height h has surface area $S = 2\pi r^2 + 2\pi rh$ (two circular ends plus the wall) and volume $V = \pi r^2 h$. We want to minimize S , but it has two variables — so use the volume constraint to eliminate one. From $V = \pi r^2 h$, $h = \frac{V}{\pi r^2}$, and substituting:

$$S(r) = 2\pi r^2 + 2\pi r \cdot \frac{V}{\pi r^2} = 2\pi r^2 + \frac{2V}{r}.$$

Optimize. Set $S'(r) = 0$: $S'(r) = 4\pi r - \frac{2V}{r^2} = 0 \Rightarrow 4\pi r = \frac{2V}{r^2} \Rightarrow r^3 = \frac{V}{2\pi}$. **Confirm it's a minimum** (not a max): $S''(r) = 4\pi + \frac{4V}{r^3} > 0$ for all $r > 0$ — so S is *convex*, and by our theorem this lone critical point is the **global** minimum. No further checking needed; convexity guarantees it.

Read the answer. With $r^3 = \frac{V}{2\pi}$, the height works out to $h = \frac{V}{\pi r^2} = 2r$. The most efficient can is exactly as **tall as it is wide** ($h = 2r$, height equals diameter). That's a clean, memorable design law — and the reason real cans aren't shaped that way is that they optimize for shelf appearance and grip, not metal, which is its own lesson about what “optimal” means once you choose the right thing to minimize.

8.10 Where this is heading

In Chapter 9, gradient descent **chases** the condition this chapter is built on — it steps downhill precisely to drive the slope f' toward zero, the critical-point equation you've been solving by hand. And the convexity you proved here is exactly the guarantee that its descent can't get lost. The two chapters are one idea from two sides: this one finds minima with algebra, the next one finds them by iteration when algebra can't.

And the whole story reopens, larger, in Book 3. There, x becomes a vector, the single slope f' becomes the **gradient**, “ $f' = 0$ ” becomes “the gradient vanishes,” the second derivative becomes a *matrix* of curvatures, and convexity becomes a statement about that matrix. The shape of a func-

tion in many dimensions is where optimization, and machine learning, actually live — and every word of it is built on the single-variable shapes you just learned to read.

8.11 Problems

Six tiers. **Fluency and Practice** are doing* — classify critical points and solve real optimization problems until it's automatic. **Concept & counterexample** and **◆ Diamond** are *thinking* — the convexity ideas and the proofs. **Teach it** is *teaching*. **Looking ahead** opens Chapter 9. Do all of Fluency; the Concept tier here is where the convexity ideas actually take hold.*

Fluency — *do these fast*

- Find all critical points of: (a) $f(x) = x^2 - 6x + 1$; (b) $f(x) = x^3 - 3x$; (c) $f(x) = x^4 - 2x^2$.
- Use the **second-derivative test** to classify each critical point in problem 1.
- State where each is concave up and concave down, and find any inflection points: (a) $f(x) = x^3$; (b) $f(x) = x^3 - 3x^2$.
- Find the critical points of $f(x) = \sin x + \cos x$ on $[0, 2\pi]$ and classify them. (Recall $f' = \cos x - \sin x$.)
- For each, decide whether it's convex, concave, or neither on all of $\mathbb{R}$, using the sign of f'' : (a) x^2 ; (b) e^x ; (c) $\ln x$ (on $x > 0$); (d) $-x^2$; (e) x^3 .
- Find the global max and min of $f(x) = x^2 - 4x + 3$ on $[0, 3]$ using the closed-interval method.

Practice — *apply and optimize*

- Find the global max and min of $f(x) = x^3 - 3x + 1$ on $[-2, 2]$.

8. A farmer has 200 m of fencing and wants to enclose the largest possible rectangular area against a straight river (no fence needed on the river side). Find the dimensions. (Set up area as a function of one side, then optimize.)
9. Find the point on the line $y = 2x + 3$ closest to the origin. (Minimize the squared distance — it has the same minimizer and easier algebra.)
10. A box with a square base and *open top* must hold 32 cm^3 . Find the dimensions using the least material. (Mirror the can example: express surface area in one variable, optimize, confirm with f'' .)
11. Find the maximum of $f(x) = x e^{-x}$ for $x \geq 0$, and confirm it's a maximum two ways (first- and second-derivative tests).
12. On $[0, \pi]$, find where $f(x) = \sin x (1 + \cos x)$ is largest. (A trig optimization — find f' , set it to zero.)

Concept & counterexample — *think; true/false with justification, or build the example*

13. (a) Every critical point is a local maximum or minimum. (b) If $f''(a) = 0$, then a is an inflection point. (c) A convex function can have two distinct local minima. (d) If $f'(a) = 0$ and $f''(a) = 0$, the point is definitely not an extremum.
14. **Find the error.** A student says: “ $f(x) = x^4$ has $f'(0) = 0$ and $f''(0) = 0$, so by the second-derivative test $x = 0$ is neither a max nor a min.” What’s wrong, and what is true at $x = 0$?
15. **Construct** a function with a local minimum that is *not* its global minimum. (This is exactly the “false floor” convexity forbids.)
16. **Construct** a critical point where $f'(a) = 0$ but a is neither a max nor a min, and a *different* one where $f''(a) = 0$ yet a is a minimum. (These show why each “inconclusive” case really is inconclusive.)
17. **Is it possible?** A convex function with a local maximum somewhere in the interior of its domain. Argue from the four faces why or why not.

18. Explain in two or three sentences why a convex loss function makes gradient descent *guaranteed* to find the best answer, while a non-convex one does not — referring to the two-valley picture.

◆ **Diamond — *the proofs are the foundation***

19. ◆ Prove **Fermat's theorem** (cite Fermat's lemma from Chapter 6) and explain why the conclusion is only " $f'(c) = 0$ **or** $f'(c)$ doesn't exist," not just the first.
20. ◆ Prove the **first-derivative test** for a local maximum: if $f' > 0$ just left of c and $f' < 0$ just right, then c is a local max. (Use Chapter 6's monotonicity consequence on each side.)
21. ◆ Prove the **second-derivative test**: if $f'(a) = 0$ and $f''(a) > 0$, then a is a local minimum. (Use the Chapter 7 quadratic $f(x) \approx f(a) + \frac{1}{2}f''(a)(x-a)^2$, and argue the quadratic term dominates for x near a .)
22. ◆ Prove that $f'' \geq 0$ implies the graph lies above each of its tangent lines (Face 4 $\Rightarrow$ Face 2), using the Taylor remainder. Then specialize to re-prove $e^x \geq 1 + x$.
23. ◆◆ Prove the **keystone theorem**: if f is convex and $f'(a) = 0$, then a is the global minimum. (Use the tangent-below face with the horizontal tangent at a .)
24. ◆◆ Prove **Face 2 $\Rightarrow$ Face 1** (tangent below $\Rightarrow$ chord above): take two points, and show the chord between them stays on or above the graph. (Hint: a clean route is to first prove $f'' \geq 0 \Rightarrow$ for any x between p and q , $f(x)$ lies below the chord, using that f' is increasing.)

Teach it — *explain it, write it up, or fix it*

25. In **three or four sentences with the bowl-and-floor picture**, teach a friend what convexity is and why it means "no false floors."
26. **Debug your classmate.** They claim: "I found a point where $f'(x) = 0$, so it's the minimum of f ." Explain in two sentences what they've

actually found and the two things they still have to check before calling it a global minimum.

27. Write a complete **worked solution** to “find the rectangle of maximum area with perimeter 40,” narrating each step — setup, the single-variable function, optimize, confirm — so a peer could redo it for any perimeter.
28. **Design a problem.** Invent an applied optimization problem (a box, a fence, a can, or a dose) whose answer is a clean number, and write the full solution key including the f'' check.
29. In a few sentences, **teach the difference** between a *local* and a *global* extremum, and explain why convexity is exactly the condition that makes the two coincide.

Looking ahead — *Chapter 9 and Book 3 in your hands early*

30. Gradient descent (Chapter 9) finds minima by stepping downhill until the slope is zero — i.e., it *chases* the equation $f'(x) = 0$ you’ve been solving by hand. For the convex function $f(x) = x^2$, predict: starting anywhere, will gradient descent find the global minimum? Why does convexity guarantee it?
 31. For the non-convex $f(x) = x^4 - 3x^2 + x$ (the two-valley function), predict what determines *which* minimum gradient descent lands in, and connect this to why “where you start” mattered in the Chapter 9 lab.
 32. In Book 3, x becomes a vector and “ $f' = 0$ ” becomes “the gradient is the zero vector.” Predict: what should replace the *second derivative* when there are many input directions, and what might “convex” mean for a bowl in many dimensions? (You’re guessing the shape of multivariable optimization — the real home of machine learning.)
-

Chapter 9

When You Can't Solve It, Chase It

Or: How to find an answer by taking smarter and smarter guesses

Companion notebook. Play with this chapter's ideas — sliders, animations, and drills — in notebooks/`ch9_newton_gradient_descent.ipynb`

By the end of this chapter you will be able to do three things, and the third one is, more or less, how every modern AI system is trained.

You will be able to find where a function crosses zero even when no formula on Earth can do it for you — by starting with a guess and improving it. This is **Newton's method**, and you will *derive* it yourself out of the one idea from Chapter 6 that we said would keep paying off: near any point, a curve looks like its tangent line.

You will be able to find where a function is *smallest* by repeatedly stepping downhill. This is **gradient descent**, and you will understand exactly why it sometimes glides smoothly to the bottom, sometimes bounces around, and sometimes flies off the edge of the world — all controlled by a single dial called the learning rate.

And you will see that these two methods, which look like they are doing different jobs, are secretly the same idea pointed at two different targets. Understanding *how* they differ is the difference between knowing what a neural network does when it learns and just believing that it does.

Here is the mindset for the whole chapter. Up to now, “solving” a problem has meant finding a formula for the exact answer. But most equations

worth caring about have no such formula. The honest move, the move that runs the real world, is to stop demanding the exact answer in one shot and instead build a *guess that gets better every time you repeat a step*. That repeated step is called **iteration**, and it is one of the most powerful ideas you will learn in this book.

9.1 The wall: equations you can't solve

In Chapter 7 we learned that to minimize a function f , we look for where its slope is zero — we solve $f'(x) = 0$. That is the right idea. The trouble is that “solve” usually fails.

Try to find where $f(x) = x - \cos x$ equals zero. Set $x = \cos x$ and... there is no algebra that isolates x . The answer exists — it is about 0.739 — but no rearranging, factoring, or formula will hand it to you. Equations mixing polynomials with $\cos$, or with e^x , or with $\ln$, are the *normal* case, not the exotic one. The clean solvable equations you met in algebra are the rare, lucky exceptions.

So we need a method that finds the answer without a formula for it. Watch how the tangent line — the star of Chapter 6 — turns out to be exactly the tool.

9.2 Newton's method: replace the curve with its tangent, and repeat

We want a **root** of f — a place x_* where $f(x_*) = 0$. Suppose we have a guess x_n that is in the neighborhood but not exact. From Chapter 6 we know that near x_n , the function is almost equal to its tangent line:

$$f(x) \approx f(x_n) + f'(x_n)(x - x_n).$$

The curve is hard. The tangent line is easy — it is just a straight line, and a straight line crosses zero in exactly one obvious place. So here is the whole idea: *instead of asking where the hard curve hits zero, ask where its easy tangent line hits zero, and use that as our next guess.* Set the right-hand side to 0 and call the solution x_{n+1} :

$$0 = f(x_n) + f'(x_n)(x_{n+1} - x_n).$$

Solve for x_{n+1} — just rearranging:

$$x_{n+1} - x_n = -\frac{f(x_n)}{f'(x_n)} \quad \Longrightarrow \quad \boxed{x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}}$$

That is Newton's method. You did not memorize it; you built it. Geometrically: stand on the curve above your guess, slide down the tangent line until it hits the x -axis, and step over to that spot. The new spot is almost always closer to the true root. Then do it again. (*Figure 9.1: the curve, the tangent at x_0 sliding to the axis at x_1 , the tangent at x_1 sliding to x_2 , the guesses marching in toward the root.*)

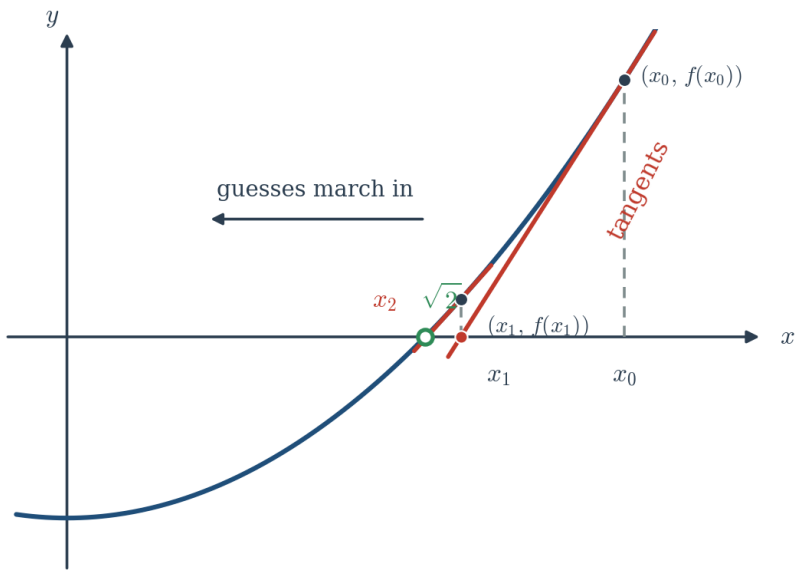


Figure 9.1

Watching it work: computing $\sqrt{2}$ by hand

In Book 1, the very first irrational number we met was $\sqrt{2}$ — a number with no exact fraction and no terminating decimal. We could prove it existed but we could not write it down. Now we can *chase* it.

Finding $\sqrt{2}$ means finding the root of $f(x) = x^2 - 2$ (since $x^2 - 2 = 0$ means $x = \sqrt{2}$). Its derivative is $f'(x) = 2x$. Drop these into Newton’s formula:

$$x_{n+1} = x_n - \frac{x_n^2 - 2}{2x_n} = \frac{x_n}{2} + \frac{1}{x_n} = \frac{1}{2} \left(x_n + \frac{2}{x_n} \right).$$

Start with a rough guess, $x_0 = 1.5$, and turn the crank:

step	value	correct digits
x_0	1.5	—
x_1	1.416 666 ...	≈ 2
x_2	1.414 215 7 ...	≈ 5

step	value	correct digits
x_3	1.414 213 562 3 ...	≈ 11

The true value is $\sqrt{2} = 1.414\,213\,562\,37\dots$. Look at the last column: the number of correct digits roughly **doubles** every step — two, then five, then eleven. After three steps from a sloppy start we have eleven correct digits. This blistering speed is called **quadratic convergence**, and it is Newton’s signature. *(That formula $x_{n+1} = \frac{1}{2}(x_n + 2/x_n)$, by the way, is the square-root method the Babylonians used four thousand years ago. They had no calculus; they found the rule by cleverness. You just rediscovered it from a tangent line.)*

Toolkit: Newton’s method (finding a root)

To solve $f(x) = 0$ when no formula will: 1. Pick a starting guess x_0 somewhere near the root. 2. Repeat: $x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$. 3. Stop when the guesses stop changing (to as many digits as you need).

What it’s really doing: approximating the curve by its tangent line and jumping to where the *line* hits zero. Fast — digits roughly double each step — *when it works*. The next section is about when it doesn’t.

9.3 When Newton breaks

A method you do not understand the failures of is a method you cannot trust. Newton has a clear weak spot, and it is written right into the formula:

we **divide** by $f'(x_n)$. If the slope at our guess is close to zero — if the tangent line is nearly flat — then a nearly-horizontal line crosses the axis *very far away*, and the next guess hurls itself off into the distance instead of homing in. (Figure 9.2: a near-flat tangent throwing x_{n+1} far from the root.)

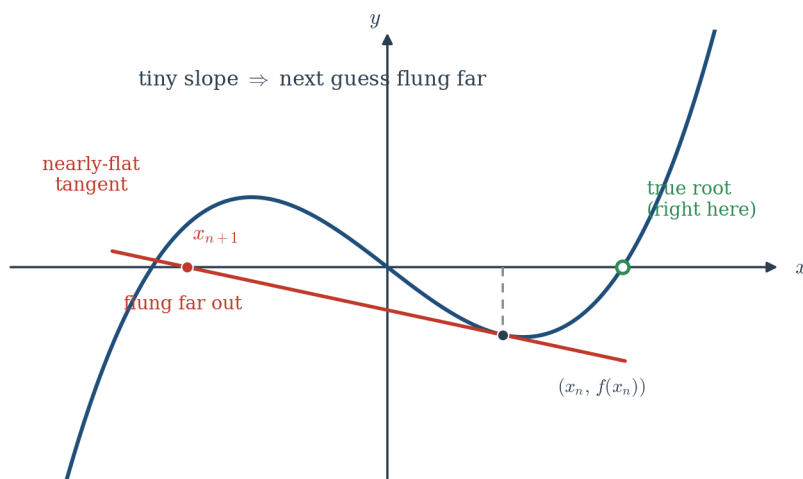


Figure 9.2

The cleanest disaster: try to find the root of $f(x) = x^{1/3}$ (the cube root), whose only root is at $x = 0$. Its derivative is $f'(x) = \frac{1}{3}x^{-2/3}$. Run Newton's formula and watch (this is a diamond problem below, but here is the punchline): every step gives

$$x_{n+1} = -2x_n.$$

The guess flips sign and *doubles in size* every single step: 1, -2, 4, -8, 16, ... It runs screaming away from the very root it was supposed to find. Same elegant method, same formula, complete failure — because the function's slope misbehaves at the root.

The lesson is not “Newton is bad.” The lesson is that fast and reliable are different promises. Newton is the fastest method in this chapter when you start near a well-behaved root, and it is the most dangerous when you don't. Hold that thought, because the slower method we meet next is

slower *precisely because it is built not to blow up*.

9.4 Gradient descent: just walk downhill

Now the target changes. We do not want a place where f is *zero*; we want the place where f is *smallest*. (Remember the reframe from Chapter 7: a “loss” — a number measuring how wrong a model is — is just a function, and learning means making it as small as possible. Finding minima is the whole game.)

Here is the most intuitive idea in the book, and a 12-year-old and a professional ML engineer use the exact same picture. Imagine standing on the graph of f in the dark. You cannot see the bottom of the valley. But you *can* feel which way the ground slopes under your feet — that is exactly what the derivative $f'(x)$ tells you. So take a step **downhill**: if the ground slopes up to the right ($f' > 0$), step left; if it slopes up to the left ($f' < 0$), step right. Either way, step *against* the slope. And take a bigger step where the slope is steeper. All of that is captured in one line:

$$x_{n+1} = x_n - \eta f'(x_n)$$

The minus sign is “go against the slope, i.e., downhill.” The number η (the Greek letter *eta*) is how big a step you take per unit of slope — it is called the **learning rate**, and choosing it is the entire art. At the bottom of the valley the slope is zero, so the update stops moving: you have arrived. (Figure 9.3: a ball on a U-shaped curve taking downhill steps that shrink as it nears the bottom.)

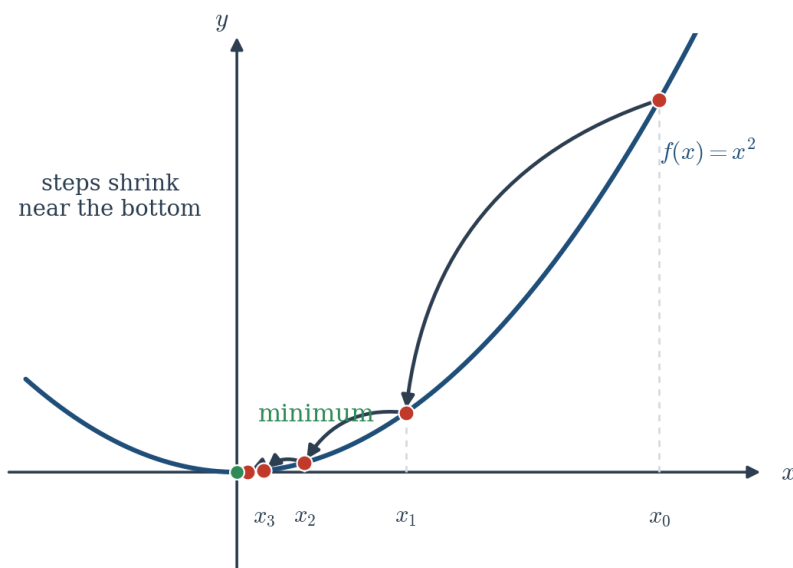


Figure 9.3

The learning rate is everything — and we can prove it with algebra

Let us minimize the simplest possible valley, $f(x) = x^2$, whose bottom is obviously at $x = 0$. Its derivative is $f'(x) = 2x$, so the update rule becomes:

$$x_{n+1} = x_n - \eta(2x_n) = (1 - 2\eta) x_n.$$

Every step just *multiplies the current position by the fixed number* $(1 - 2\eta)$. But multiplying by the same number over and over is exactly a **geometric sequence** — the thing we studied all the way back in Chapter 1. So after n steps, starting from x_0 :

$$x_n = (1 - 2\eta)^n x_0.$$

And we already know, from Chapter 1, *exactly* when a geometric sequence shrinks to zero: when the thing being raised to the n th power has size less than 1. So whether gradient descent works is decided entirely by

the single number $|1 - 2\eta|$. Let us start at $x_0 = 1$ and turn the same crank for four different learning rates:

learning rate η	factor $1 - 2\eta$	the sequence	what happens
		$1, (1-2\eta), (1-2\eta)^2, \dots$	
0.1	0.8	1, 0.8, 0.64, 0.512, ...	slides smoothly to 0 ✓
0.9	-0.8	1, -0.8, 0.64, -0.512, ...	overshoots, but still closes in ✓
1.0	-1	1, -1, 1, -1, ...	bounces forever, never arrives ✗
1.5	-2	1, -2, 4, -8, ...	explodes off to infinity ✗

Read that table slowly, because it is the whole story of training a model in miniature. A *small* learning rate ($\eta = 0.1$) creeps down patiently and lands. A *medium* one ($\eta = 0.9$) overshoots the bottom each time but the overshoots shrink, so it spirals in. At the *critical* value ($\eta = 1$) it bounces between the same two points forever, getting nowhere. And a *too-large* rate ($\eta = 1.5$) overshoots so hard that each step lands farther out than the last, and the whole thing detonates. The condition for success is just $|1 - 2\eta| < 1$, which works out to $0 < \eta < 1$. (Figure 9.4: the four trajectories on the parabola — the smooth glide, the shrinking spiral, the stuck bounce, the explosion.)

gradient descent on $y = x^2$ from $x_0 = 1$: $x_n = (1 - 2\eta)^n$

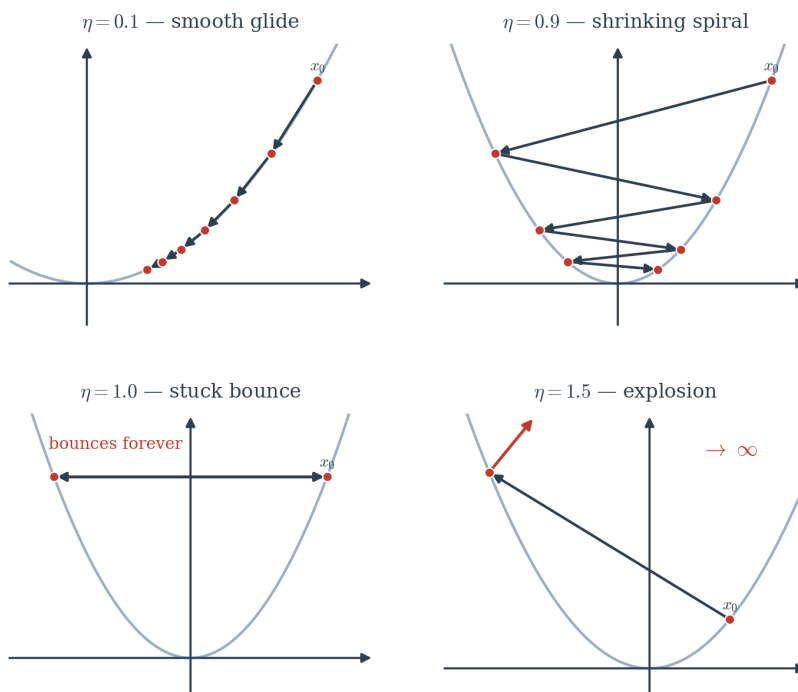


Figure 9.4

There is one more magic value worth seeing: $\eta = \frac{1}{2}$ makes the factor $1 - 2\eta = 0$, so $x_1 = 0$ — gradient descent reaches the exact minimum in a *single step*. Tuck that away. It is not a coincidence, and the next section explains it.

Toolkit: Gradient descent (finding a minimum)

To minimize f : 1. Pick a starting guess x_0 and a learning rate $\eta > 0$. 2. Repeat: $x_{n+1} = x_n - \eta f'(x_n)$. 3. Stop when the steps become tiny (the slope is near zero — you're at the bottom).

The learning rate is the dial that decides everything: too small and it crawls; too large and it overshoots and can explode. The sweet spot is the whole craft. Unlike Newton, gradient descent never divides by anything, so it can't blow up from a flat spot — its failures come only from a step size you chose.

9.5 The two methods were the same idea all along

We learned Newton's method to find where a function is **zero**, and gradient descent to find where a function is **smallest**. Now watch them collapse into one.

A minimum of f is a place where $f'(x) = 0$. But “find where f' is zero” is a *root-finding problem* — it is exactly the job Newton's method does, applied to the function f' instead of f . So take Newton's formula and replace f with f' (which means replacing f' with f'' , the derivative of the derivative — the *curvature* from Chapter 7):

$$x_{n+1} = x_n - \frac{f'(x_n)}{f''(x_n)}.$$

Now line up the two ways to hunt for a minimum:

$$\text{gradient descent: } x_{n+1} = x_n - \eta f'(x_n) \qquad \text{Newton's: } x_{n+1} = x_n - \frac{1}{f''(x_n)} f'(x_n).$$

They are *the same shape*. Both step against the slope $f'(x_n)$. The only difference is the step size in front. Gradient descent uses a fixed number η that **you** pick and hope is good. Newton's method uses $1/f''(x_n)$ — it reads the curvature off the function and **sets its own step size automatically**, taking small steps where the valley is sharply curved and big steps where

it is gently curved. That is why Newton is so fast: it is not guessing the step size, it is computing the right one.

This even explains the magic $\eta = \frac{1}{2}$ from a moment ago. For $f(x) = x^2$ the curvature is $f'' = 2$, so Newton's automatic step size is $1/f'' = 1/2$ — exactly the η that jumped to the answer in one step. Gradient descent with the perfect hand-tuned rate *is* Newton's method, for that function.

So why does AI use the slow one?

Here is the honest payoff, and it is one of the central facts about how modern AI is actually built. If Newton's method is so much faster — picks its own step size, converges in a handful of iterations — why does essentially every neural network on the planet get trained with plain gradient descent and its fixed, fiddly learning rate?

Because of what f'' becomes when x is not a single number but a list of millions or billions of numbers — which is exactly the situation for a neural network, whose “position” is the gigantic list of all its adjustable parameters. With one variable, f'' is one number. With N variables, the “curvature” is no longer a number; it becomes a square table with N rows and N columns — N^2 entries. For a model with a billion parameters, that table has a *billion squared* — a billion billion — entries. You cannot store it. You certainly cannot do the division (which becomes something far more expensive than ordinary division). Newton's method, the fast one, becomes physically impossible at the scale that matters.

Gradient descent only ever needs the slope f' — which, for N parameters, is just N numbers, one downhill direction per parameter. That you *can* compute, even for a billion parameters. So the field made a trade it did not exactly choose: give up Newton's speed and self-tuning, accept a learning rate you have to babysit, in exchange for a method that is cheap enough to actually run. **That trade — fast-but-impossible second-order versus slow-but-affordable first-order — is why “tuning the learning rate” is a phrase every person who trains models knows in their bones.**

You now know why it exists, from a parabola you analyzed with Chapter 1's geometric sequences.

9.6 Where this is heading

Everything in this chapter happened on a number line — x was a single number, f' a single slope, “downhill” a choice between left and right. In Book 3, x becomes a **vector** — a point in a space of many dimensions — and the derivative f' becomes the **gradient**, a vector that points in the steepest uphill direction. “Step against the slope” becomes “step against the gradient,” and the update rule keeps the exact same shape: $x_{n+1} = x_n - \eta(\text{gradient at } x_n)$. The picture you built here — a ball rolling to the bottom of a valley, a learning rate that must be neither too timid nor too bold — is the *whole* picture. Book 3 just gives the ball more directions it is allowed to roll.

When you eventually watch a neural network train and see its “loss” tick downward step by step, you will not be watching magic. You will be watching this chapter, run a few million times.

9.7 Problems

*This chapter is about **iteration**, so its “doing” is mostly computational — you learn gradient descent by running it and watching it converge or blow up. So the tiers are shaped differently here. The **Computational Lab** is the heart of the practice: structured experiments in the companion notebook. **By hand** is the short list genuinely worth doing on paper. **Concept** tests judgment. ♦ **Diamond** is where this chapter's real paper depth lives — the proofs are the practice. **Teach it** and **Looking ahead** close it out.*

Computational Lab — *the real practice; run these in the notebook*

1. **Learning-rate sweep.** On $f(x) = x^2$, run gradient descent from $x_0 = 1$ for $\eta = 0.1, 0.3, 0.5, 0.7, 0.9, 1.0, 1.1, 1.5$. Plot x_n versus n for each. Sort them into four behaviors (smooth glide, shrinking spiral, stuck bounce, explosion) and check the boundary against the prediction $|1 - 2\eta| < 1$.
2. **Hunt the fastest rate.** Still on x^2 : which η reaches within 10^{-6} of the minimum in the *fewest* steps? Compare your experimental winner with the predicted optimum $\eta = \frac{1}{2}$.
3. **Newton, fast and broken.** Implement $x_{n+1} = \frac{1}{2}(x_n + \frac{2}{x_n})$ for $\sqrt{2}$; print x_n and the error $|x_n - \sqrt{2}|$ each step and watch the correct digits roughly double. Then point Newton at $f(x) = x^{1/3}$ from $x_0 = 1$ and watch it fly apart.
4. **A landscape with two valleys.** Run gradient descent on $f(x) = x^4 - 3x^2 + x$ from several starting points. Show that different starts land in different minima. (You've just discovered, by experiment, the central headache of training: gradient descent finds *a* local minimum, not necessarily the *best* one — and which one depends on where you begin.)
5. **The race.** On a single quadratic, count the steps gradient descent (good η) and Newton's method each need to reach a fixed tolerance. Watch Newton win by a landslide — then find a setup where Newton fails and gradient descent doesn't.
6. **(Stretch) A first real optimizer.** Add “momentum”: $x_{n+1} = x_n - \eta f'(x_n) + \beta(x_n - x_{n-1})$ with $\beta = 0.9$. On a stretched bowl $f(x) = 10x^2$, compare plain gradient descent against the momentum version. (This is a baby version of the optimizers that actually train neural networks.)

By hand — worth doing on paper

7. For $f(x) = x^2 - 5$: write f' , write Newton's rule, simplify it, then do two steps from $x_0 = 2$ toward $\sqrt{5}$.
8. Gradient descent on $f(x) = x^2$ with $\eta = 0.1$, $x_0 = 1$: compute x_1, x_2, x_3 .
9. For $f(x) = x^2 - 4x + 7$: find f' , find the exact minimizer from $f' = 0$, then take two gradient-descent steps from $x_0 = 0$ with $\eta = 0.1$ and check you're heading the right way.
10. Classify the behavior for each factor $1 - 2\eta$: $\eta = 0.2, 0.5, 0.8, 1.2$ (monotone glide / one-step / oscillating glide / explosion).

Concept & counterexample — think; true/false with justification, or build the case

11. (a) A smaller learning rate is always safer, so always better. (b) Gradient descent can overshoot the minimum. (c) Newton's method always converges faster than gradient descent. (d) If gradient descent's steps keep shrinking, it must be approaching the minimum.
12. **Predict, then check.** For $f(x) = x^2$ with $\eta = 0.5$, where is x_1 — without computing step by step? For $\eta = 1$, describe the long-run behavior in one sentence.
13. **When it breaks.** Give (a) a function/start where Newton diverges, and (b) a function/start where gradient descent diverges, naming the cause of each. Which failure is fixable by changing only one number, and which isn't?
14. **The two-valley trap (on paper).** Sketch a function with a shallow valley and a deeper one. Mark a start that gradient descent sends to the *shallow* (worse) minimum and one that sends it to the deeper one. In one sentence, what does this say about “gradient descent finds the best answer”?

15. **An asymmetry worth seeing.** For $f(x) = x^2$, argue that *too small* an η only makes you slow (you still converge), while *too large* an η makes you fail outright. Why are the two kinds of “wrong learning rate” not symmetric?

◆ Diamond — *the proofs are the practice*

16. ◆ Derive Newton’s method from the tangent-line approximation $f(x) \approx f(x_n) + f'(x_n)(x - x_n)$: set the line to zero and solve for x_{n+1} .
17. ◆ Show Newton on $f(x) = x^2 - A$ gives the Babylonian rule $x_{n+1} = \frac{1}{2}(x_n + \frac{A}{x_n})$, and use it to compute $\sqrt{7}$ to five decimals from $x_0 = 3$.
18. ◆ The cube-root disaster: for $f(x) = x^{1/3}$, show Newton’s rule simplifies to $x_{n+1} = -2x_n$; conclude divergence from every nonzero start and explain geometrically (what is the tangent near $x = 0$ doing?).
19. ◆ **The learning-rate range, in general.** For $f(x) = \frac{a}{2}(x - m)^2$ (a bowl of curvature $a > 0$, minimum at m), show gradient descent gives $x_{n+1} - m = (1 - \eta a)(x_n - m)$, hence $x_n - m = (1 - \eta a)^n(x_0 - m)$. Find the exact convergence range and the optimal η . Then state the single most important practical lesson: *what must happen to the safe learning rate as the curvature a grows?*
20. ◆◆ **Why Newton’s error squares.** For $x_{n+1} = \frac{1}{2}(x_n + \frac{A}{x_n})$, let $e_n = x_n - \sqrt{A}$. Prove the exact identity $e_{n+1} = \frac{e_n^2}{2x_n}$. Conclude that near the root the error roughly squares each step — which is exactly why the correct digits double (quadratic convergence).
21. ◆◆ **Newton reaches a quadratic’s minimum in one step.** Using $x_{n+1} = x_n - \frac{f'(x_n)}{f''(x_n)}$, show that for any $f(x) = ax^2 + bx + c$ ($a > 0$) the first step lands on the exact minimizer, regardless of x_0 . Explain via Chapter 6 why this had to happen (how well does a quadratic match its own second-order Taylor model?).
22. ◆◆ **The fixed-point view that unifies everything.** Both methods have the form $x_{n+1} = g(x_n)$, and a limit L must satisfy $g(L) = L$ —

a *fixed point*, exactly like Chapter 1's recursive sequences. A general fact (which you may use): the iteration converges to L when $|g'(L)| < 1$, and the smaller $|g'(L)|$, the faster. (a) For gradient descent, $g(x) = x - \eta f'(x)$; compute $g'(L)$ at a minimum L and recover the convergence range from problem 19. (b) For Newton, $g(x) = x - \frac{f(x)}{f'(x)}$; show $g'(L) = 0$ at a simple root. Explain how this *one* criterion accounts for both the learning-rate range *and* Newton's blazing speed.

Teach it — *explain it, write it up, or fix it*

23. In **three or four sentences with the ball-in-a-valley picture**, teach a friend what the learning rate is and what goes wrong at both extremes — too big and too small.
24. **Debug your classmate.** They say: "Newton's method is faster, so real AI should obviously use it instead of gradient descent." Give the one true sentence that explains why neural networks are trained with gradient descent anyway.
25. Write a **worked solution** for "use Newton's method to find $\sqrt{10}$ to four decimals," narrating the setup, the rule, and each iteration so a peer could reuse it for any square root.
26. **Design a problem.** Cook up a specific quadratic f and a learning rate η for which gradient descent reaches the minimum in *exactly one step*. Show why your η works (relate it to f''), and write the solution key.
27. In a few sentences, **teach the connection** between "finding where $f = 0$ " (Newton's first job) and "finding where f is smallest" (gradient descent's job) — how does the same machine do both?

Looking ahead — *Book 3, where x becomes a vector*

28. In Book 3, x becomes a vector and f' becomes the **gradient** — a vector pointing in the steepest uphill direction. Predict what the rule

$x_{n+1} = x_n - \eta f'(x_n)$ becomes, and what stays exactly the same.

29. Your safe range was $0 < \eta < \frac{2}{f''}$. In many dimensions there isn't one curvature f'' but many — different in different directions. Predict why a single number η might be *too big* in one direction and *too small* in another at the same time. (This is why advanced optimizers adapt the rate per direction.)
30. Newton needs f'' : one number here, an $N \times N$ table in N dimensions. For a model with one million parameters, how many entries is that table? Say in one sentence why that number ends Newton's method at real scale — and reconnect it to this chapter's punchline.
-

Chapter 10

Adding Up the Infinite

Or: how to measure the area under a curve — the second great engine of calculus

Companion notebook. Play with this chapter’s ideas — sliders, animations, and drills — in notebooks/`ch10_integral.ipynb`.

For nine chapters, calculus has meant the derivative: given a curve, find its slope; given a position, find its velocity; given a quantity, find its *rate of change*. This chapter runs the machine in reverse. We start with a rate and ask for the **total accumulated** — given a velocity, how far did you travel? Given a curve, what is the **area underneath it**? That question, which sounds like a separate subject from slopes, turns out to be the other half of calculus, and the two halves meet in the next chapter in the most important theorem you will learn.

By the end of this chapter you will be able to define the **definite integral** as a limit of sums (the same “approximate, then take the limit” move that built the derivative), compute one from scratch, and find **antiderivatives** — functions whose derivative is a given function — for the whole standard library, trigonometric functions included. You’ll learn **substitution**, which is the chain rule run backward. And you’ll meet the strange fact that some perfectly ordinary functions have no antiderivative you can write down at all.

Here is the shape of the thing. The integral has *two faces* that look completely unrelated. One is geometric — the area under a curve, built by

adding up infinitely many thin rectangles. The other is algebraic — the reverse of differentiation, found by undoing the rules of Chapter 4. This chapter develops both faces separately and leaves them staring at each other across a gap. Chapter 11 closes the gap, and that closing is the climax of the book.

10.1 The accumulation problem

The derivative answered: *given where you are at each moment, how fast are you going?* Now reverse it: *given how fast you're going at each moment, how far have you gone?* If you drive at a steady 60 mph for 2 hours, you've gone 120 miles — speed times time. But if your speed *varies*, you can't just multiply; you have to add up the contribution of each instant. That adding-up is integration.

And it has a beautiful geometric meaning. Plot velocity against time. Over a tiny slice of time Δt , your speed is nearly constant, so the distance covered is (speed) $\times \Delta t$ — which is the **area of a thin rectangle** under the velocity curve. Add all the slices, and the total distance is the **total area under the velocity curve**. This is the master idea: *the accumulated total of a rate is the area under its graph*. Distance is the area under velocity; total rainfall is the area under the rainfall-rate curve; total charge is the area under the current. To accumulate is to measure area. (Figure 10.1: *a varying velocity curve over $[a, b]$ with the region beneath it shaded, labeled “total distance = area.”*)

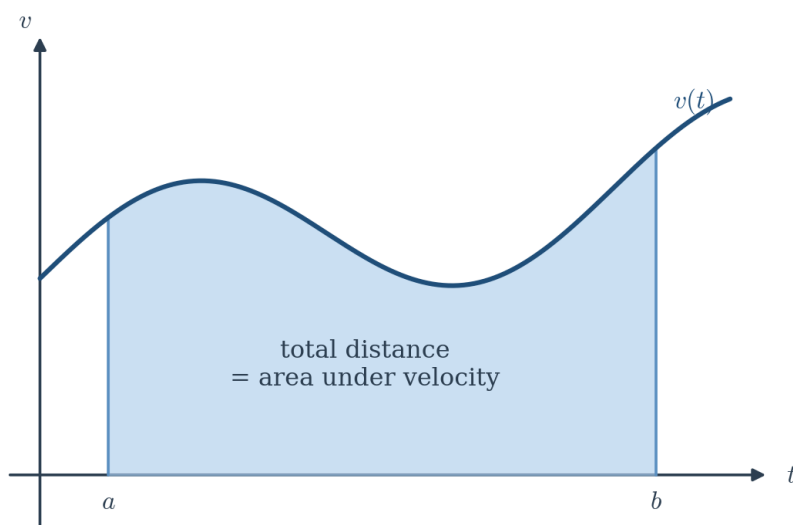


Figure 10.1

So the central technical problem is simply: **find the area under a curve**. For a straight line it's a triangle or trapezoid and you already know how. For a genuine curve, we need a new idea.

10.2 Riemann sums: approximate with rectangles, then take the limit

We cannot find the area under a curve directly, so we *approximate* it with shapes we can measure — rectangles — and then make the approximation exact with a limit. (This is the identical strategy from Chapter 3, where we approximated a slope with a secant line and took the limit. Calculus has essentially one idea, and this is its second great use.)

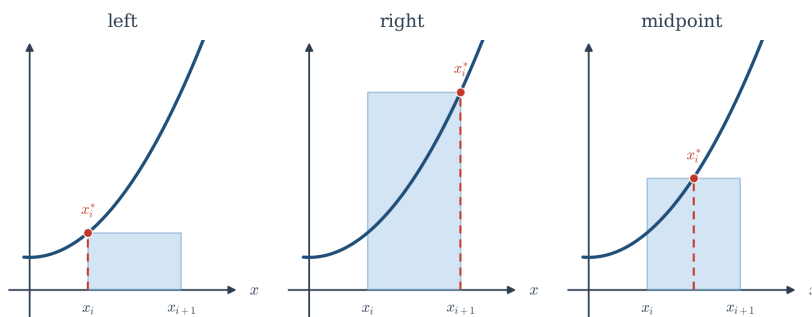


Figure 10.3

To estimate the area under f on $[a, b]$: chop the interval into n equal slices, each of width $\Delta x = \frac{b-a}{n}$. On each slice, build a rectangle whose height is the value of f at some sample point x_i in that slice. The rectangle areas are $f(x_i) \Delta x$, and their sum approximates the total area:

$$\text{Area} \approx \sum_{i=1}^n f(x_i) \Delta x.$$

This is a **Riemann sum**. Using the left edge of each slice as the sample point gives a *left sum*, the right edge a *right sum*, the middle a *midpoint sum* — and for a great many functions, all of them close in on the same number as the rectangles get thinner. (Figure 10.2: the region under a curve filled with rectangles, shown at $n = 4$, $n = 8$, $n = 16$ — the staircase hugging the curve more tightly each time.)

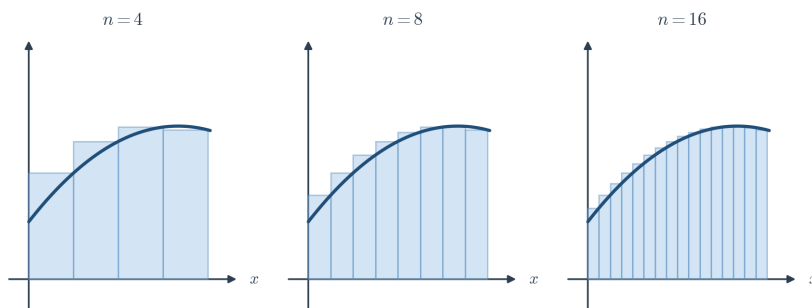


Figure 10.2

Now take the limit. As $n \rightarrow \infty$ the rectangles become infinitely thin, the staircase melts onto the curve, and the approximation becomes the exact area.

10.3 The definite integral

The **definite integral** of f from a to b is the limit of its Riemann sums:

$$\int_a^b f(x) dx = \lim_{n \rightarrow \infty} \sum_{i=1}^n f(x_i) \Delta x.$$

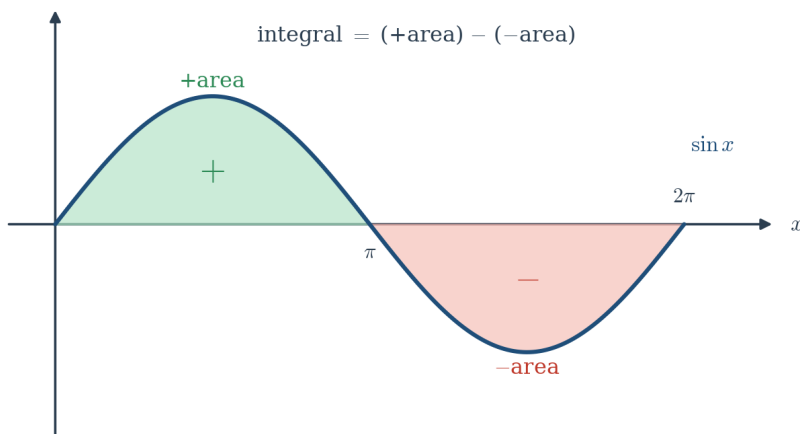


Figure 10.4

The notation is a story in itself. The sign $\int$ is an elongated **S**, for “sum”; it is the Σ of the Riemann sum, stretched out as the sum becomes continuous. The dx is the width Δx in the limit — the infinitely thin slice. Together, $\int_a^b f(x) dx$ reads “sum up $f(x)$ times a sliver of width dx , as x runs from a to b .” The result is a single **number**: the area.

One subtlety: it is *signed* area. Where f is below the axis, $f(x_i)$ is negative, so those rectangles contribute negatively. The definite integral counts area above the axis as positive and area below as negative. (For distance, this is exactly right — moving backward subtracts.)

Computing one from the definition. Let's find $\int_0^1 x^2 dx$ — the area under the parabola from 0 to 1 — with nothing but the definition. Slice $[0, 1]$ into n pieces of width $\frac{1}{n}$ and sample at the right edges $x_i = \frac{i}{n}$:

$$\sum_{i=1}^n \left(\frac{i}{n}\right)^2 \cdot \frac{1}{n} = \frac{1}{n^3} \sum_{i=1}^n i^2 = \frac{1}{n^3} \cdot \frac{n(n+1)(2n+1)}{6} = \frac{(n+1)(2n+1)}{6n^2}.$$

(We used the sum-of-squares formula $\sum i^2 = \frac{n(n+1)(2n+1)}{6}$ from Chapter 1.) As $n \rightarrow \infty$, the dominant terms give $\frac{2n^2}{6n^2} = \frac{1}{3}$. So $\int_0^1 x^2 dx = \frac{1}{3}$. We found the exact area under a parabola by adding up infinitely many rectangles — but it took real work, and we'd have to start over for every new function. Hold that frustration; the next chapter eliminates it entirely.

Properties you'll use constantly

A few facts follow straight from the definition (sums behave this way, so their limits do too):

- **Linearity:** $\int_a^b [f(x) + g(x)] dx = \int_a^b f + \int_a^b g$, and $\int_a^b c f(x) dx = c \int_a^b f$.
- **Additivity over intervals:** $\int_a^b f + \int_b^c f = \int_a^c f$ — areas of adjacent pieces add.
- **A point has no width:** $\int_a^a f = 0$.
- **Reversing orientation flips the sign:** $\int_b^a f = - \int_a^b f$.

10.4 The other face: the antiderivative

Now set area aside entirely and ask a question that sounds unrelated. Differentiation turns F into F' . Can we run it backward — given a function f , find a function F whose derivative is f ? Such an F is called an **antiderivative** of f .

For example, an antiderivative of $2x$ is x^2 , because $\frac{d}{dx}x^2 = 2x$. But so is $x^2 + 1$, and $x^2 - 7$ — adding any constant leaves the derivative unchanged. This is not sloppiness; it is the whole truth, and Chapter 6 already proved it: if two functions have the *same* derivative, they differ by a constant (because $F' = G'$ means $(F - G)' = 0$, and a function with zero derivative everywhere is constant — a consequence of the Mean Value Theorem). So antiderivatives come in families, all parallel, separated by a constant:

The indefinite integral. $\int f(x) dx = F(x) + C$, where $F' = f$ and C is an arbitrary constant — the whole family of antiderivatives at once.

(Notice the same symbol $\int$ now appears *without* limits a and b . That collision of notation is a deliberate hint: the area problem and the antiderivative problem are about to turn out to be the same problem. For now, treat them as separate — definite integral with limits is a *number*, indefinite integral without limits is a *family of functions*.)

The standard antiderivative library

Finding antiderivatives is just running the Chapter 4 rules backward. Every derivative you learned becomes an antiderivative read right to left:

- **Power rule, reversed:** $\int x^n dx = \frac{x^{n+1}}{n+1} + C$ (for $n \neq -1$). *Check by differentiating the answer.* (Raise the power by one and divide by the new power — the opposite of “drop the power and subtract one.”)

- **The exception, $n = -1$:** $\int \frac{1}{x} dx = \ln |x| + C$. The power rule would divide by zero here, so this one stands apart — and it is no accident, as Chapter 11's capstone will reveal.
- **Exponential:** $\int e^x dx = e^x + C$ (it is its own antiderivative, just as it's its own derivative).
- **Trigonometric** — now within reach, because Chapter 4 derived the matching derivatives: since $\frac{d}{dx} \sin x = \cos x$,

$$\int \cos x dx = \sin x + C, \quad \int \sin x dx = -\cos x + C, \quad \int \sec^2 x dx = \tan x + C.$$

(Mind the minus sign on $\int \sin$: the antiderivative of sine is *negative* cosine, because the derivative of cosine carries the minus.)

- **Linearity**, as for definite integrals: antiderivative of a sum is the sum of antiderivatives, and constants pull out.

So $\int (3x^2 - 4 \cos x + e^x) dx = x^3 - 4 \sin x + e^x + C$ — reverse each term, attach one C at the end. Differentiate it back to check, and you recover the integrand exactly.

10.5 Substitution: the chain rule, run backward

Most antiderivatives aren't in the table — but many can be *transformed* into ones that are, by the single most useful integration technique, and it is precisely the **chain rule reversed**. Recall the chain rule: $\frac{d}{dx} F(g(x)) = F'(g(x)) g'(x)$. Read that equation from right to left and it becomes an antiderivative rule:

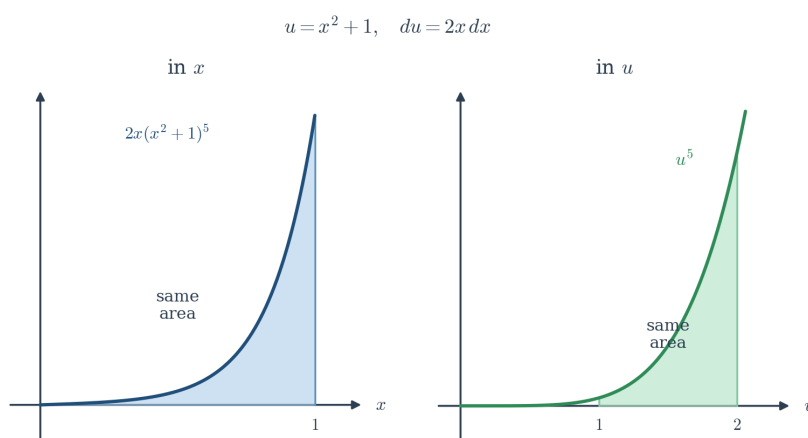


Figure 10.5

$$\int F'(g(x)) g'(x) \, dx = F(g(x)) + C.$$

In practice you spot an “inside function” $g(x)$ whose derivative $g'(x)$ is *also sitting in the integrand* (up to a constant factor), and you rename: let $u = g(x)$, so that $du = g'(x) \, dx$. The integral collapses into one in u that you can do from the table.

Example. $\int 2x(x^2 + 1)^5 \, dx$. The inside is $g(x) = x^2 + 1$, and its derivative $2x$ is right there. Let $u = x^2 + 1$, $du = 2x \, dx$:

$$\int 2x(x^2 + 1)^5 \, dx = \int u^5 \, du = \frac{u^6}{6} + C = \frac{(x^2 + 1)^6}{6} + C.$$

Differentiate the answer (chain rule!) and you get $2x(x^2 + 1)^5$ back — substitution and the chain rule are genuinely inverse moves.

Example. $\int x e^{x^2} \, dx$. Inside is x^2 , derivative $2x$ — present up to the factor of 2. Let $u = x^2$, $du = 2x \, dx$, so $x \, dx = \frac{1}{2} \, du$:

$$\int x e^{x^2} \, dx = \int e^u \cdot \frac{1}{2} \, du = \frac{1}{2} e^u + C = \frac{1}{2} e^{x^2} + C.$$

Example — the logarithm pattern. $\int \tan x \, dx = \int \frac{\sin x}{\cos x} \, dx$. The inside is $\cos x$, and its derivative $-\sin x$ is sitting on top (up to the minus sign). Let $u = \cos x$, $du = -\sin x \, dx$:

$$\int \frac{\sin x}{\cos x} \, dx = \int \frac{-1}{u} \, du = -\ln |u| + C = -\ln |\cos x| + C.$$

Whenever the top of a fraction is the derivative of the bottom, substitution turns the whole thing into a logarithm — a pattern worth knowing on sight:

$$\int \frac{g'(x)}{g(x)} \, dx = \ln |g(x)| + C.$$

This is the mirror image of Chapter 4. There, the product rule and chain rule were tools for *taking* derivatives; here, reversed, they become tools for *finding* antiderivatives. (The product rule's reverse — **integration by parts** — is the other great technique, and it arrives in Chapter 11 as a payoff of the Fundamental Theorem.) For now, substitution applies cleanly to indefinite integrals; using it on definite integrals, where the limits themselves transform, is a Chapter 11 refinement.

Toolkit: Substitution

1. Find an inside function $u = g(x)$ whose derivative $g'(x)$ appears in the integrand (up to a constant).
 2. Compute $du = g'(x) \, dx$ and rewrite the *entire* integral in terms of u — no x may remain.
 3. Integrate in u using the table, then substitute x back.
 4. **Always check by differentiating your answer.** Substitution is the chain rule reversed, so the chain rule is your proofreader.
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10.6 When there is no antiderivative — and a whisper about frequencies

Two honest remarks before the climax.

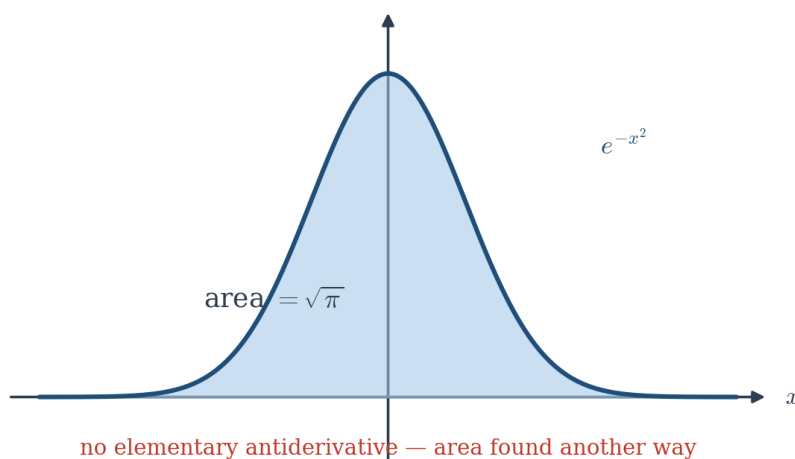


Figure 10.6

Some functions have no antiderivative you can write down. This is genuinely surprising. The function e^{-x^2} — the bell curve, whose Taylor series you found in Chapter 7 — is perfectly smooth and perfectly well-behaved, yet *no* combination of powers, roots, exponentials, logarithms, and trig functions has it as a derivative. Its antiderivative exists (it's a real, smooth function), but it simply isn't expressible in our elementary vocabulary; the same is true of $\sin(x^2)$ and many others. When this happens, the antiderivative method of the next chapter cannot even begin, and we fall back on what never fails — Riemann sums, computed numerically. The *area* is always there to be measured; we just sometimes can't name the antiderivative. And the most important such area, the total under the bell curve,

$$\int_{-\infty}^{\infty} e^{-x^2} dx = \sqrt{\pi},$$

is one of the loveliest numbers in mathematics — central to all of probability

(Book 4), and found by an ingenious trick rather than by any antiderivative.

Integrals can measure more than area — they can measure resonance. Here is a thread to pick up much later. If you integrate a signal $f(x)$ against a pure wave, say $\int f(x) \sin(nx) dx$ over one period, the resulting number tells you *how much of that frequency f contains* — how strongly the signal “rings” at that pitch. Those numbers are the **Fourier coefficients** glimpsed back in Chapter 7, and they are how a sound is decomposed into tones, an image into patterns, a data stream into its essential frequencies. The integral, in this guise, is a frequency detector. The machinery belongs to Book 3; file it away beside the word *Fourier*.

10.7 Where this is heading: the two faces meet

Step back and look at what this chapter built. We now have **two** things, both called integration, that appear to have nothing to do with each other:

- the **definite integral** $\int_a^b f dx$ — a *number*, the signed area under a curve, defined as a limit of sums and (as our $\int_0^1 x^2 dx$ showed) genuinely laborious to compute;
- the **antiderivative** $\int f dx$ — a *function*, the reverse of differentiation, found quickly by running the Chapter 4 rules backward.

One is about adding up infinitely many rectangles. The other is about undoing a derivative. There is no obvious reason they should be related at all — and yet they share a symbol, which I promised was a hint.

The hint is true. The **Fundamental Theorem of Calculus**, the summit of this entire book, says these two faces are one: *to find the area under a curve, you do not add up rectangles — you find an antiderivative and subtract its values at the endpoints*. The hard limit of sums and the easy

reverse-derivative compute the same number. The area under x^2 from 0 to 1, which cost us a page of algebra, will become a five-second calculation. Differentiation and integration, the two engines we built separately, are about to be revealed as the same engine running in opposite directions.

Turn the page.

10.8 Problems

Six tiers. **Fluency and Practice** are doing* — build Riemann sums and find antiderivatives until both are automatic (and like Chapter 4, there's a lot of antiderivative drill, because it's the same kind of mechanical skill). **Concept & counterexample** and **◆ Diamond** are *thinking* — signed area, the $+C$, and the proofs. **Teach it** is *teaching*. **Looking ahead** opens Chapter 11. Do all of Fluency; antiderivative fluency is what makes the Fundamental Theorem usable.*

Fluency — *do these fast*

- Antiderivatives, reverse power rule: (a) $\int x^4 dx$; (b) $\int x^7 dx$; (c) $\int x^{-2} dx$; (d) $\int \sqrt{x} dx$; (e) $\int 1 dx$.
- With constants and sums: (a) $\int (3x^2 - 2x + 5) dx$; (b) $\int (6x^5 + 4) dx$; (c) $\int \left(\frac{1}{x^2} + x\right) dx$.
- Trig and exponential antiderivatives: (a) $\int \cos x dx$; (b) $\int \sin x dx$; (c) $\int e^x dx$; (d) $\int \sec^2 x dx$; (e) $\int \frac{1}{x} dx$.
- Mixed library: (a) $\int (2 \cos x - 3 \sin x) dx$; (b) $\int (e^x + x^2) dx$; (c) $\int (4 \sec^2 x - \frac{5}{x}) dx$.
- Rewrite first, then integrate: (a) $\int \frac{x^2-1}{x} dx$ (split the fraction); (b) $\int x^2 \sqrt{x} dx$ (one power); (c) $\int (x+1)^2 dx$ (expand).
- Simple substitution (the inside's derivative is present): (a) $\int 2x(x^2 + 1)^5 dx$; (b) $\int 3x^2(x^3 + 1)^4 dx$; (c) $\int \cos x e^{\sin x} dx$; (d) $\int 4x^3(x^4 + 2)^6 dx$; (e) $\int \sin^4 x \cos x dx$.

7. Substitution needing a constant factor, or landing on a log: (a) $\int x e^{x^2} dx$; (b) $\int x(x^2 + 1)^3 dx$; (c) $\int \sin(3x) dx$; (d) $\int \cos(5x) dx$; (e) $\int \frac{x^2}{x^3 + 1} dx$ (the logarithm pattern).
8. Estimate $\int_0^4 f dx$ for $f(x) = x^2$ with a right Riemann sum using $n = 4$ rectangles (widths of 1). Is your estimate above or below the true area, and why?
9. Evaluate by recognizing the area as a simple shape: (a) $\int_0^3 2 dx$ (a rectangle); (b) $\int_0^2 x dx$ (a triangle); (c) $\int_{-1}^1 x dx$ (watch the sign).

Practice — *multi-step*

10. Compute $\int_0^1 x^3 dx$ from the definition, using $\sum_{i=1}^n i^3 = \left[\frac{n(n+1)}{2} \right]^2$ and a right Riemann sum. (Mirror the $\int_0^1 x^2$ computation in the chapter.)
11. Substitution: (a) $\int \frac{2x}{x^2 + 1} dx$; (b) $\int \frac{x}{\sqrt{x^2 + 9}} dx$; (c) $\int \sin x \cos x dx$ (let $u = \sin x$).
12. Substitution with rearrangement: (a) $\int \frac{\ln x}{x} dx$ (let $u = \ln x$); (b) $\int x\sqrt{x+1} dx$ (let $u = x+1$, then write $x = u-1$).
13. Find the function F with $F'(x) = 3x^2 - 2$ **and** $F(1) = 5$. (Antiderivative, then pin down C from the condition — your first “initial value problem.”)
14. A ball’s velocity is $v(t) = -10t + 20$ (m/s). Find its position $s(t)$ given $s(0) = 0$, by antidifferentiating. (You’re undoing the Chapter 3 derivative that produced v from s .)
15. Set up — but do not yet evaluate — the area under $f(x) = e^{-x^2}$ from 0 to 1 as a limit of Riemann sums, and explain why you *cannot* finish it by finding an antiderivative.
16. Sketch $f(x) = x^2 - 1$ on $[0, 2]$ and explain, using *signed* area, why $\int_0^2 (x^2 - 1) dx$ is smaller than the unsigned area between the curve and the axis.

Concept & counterexample — *think; true/false with justification, or build the example*

17. (a) $\int_a^b f \, dx$ is a number; $\int f \, dx$ is a function. (b) If $F' = G'$, then $F = G$. (c) $\int_a^b f \, dx$ can be negative even when f is a “real” function. (d) Every continuous function has an antiderivative that can be written with elementary functions.
18. **Why the +C?** Explain, citing the Mean Value Theorem, why two antiderivatives of the same function must differ by a constant — and why this means $\int f \, dx$ is a whole *family*.
19. **Find the error.** A student writes $\int \sin x \, dx = \cos x + C$. What’s wrong, and how would differentiating the answer have caught it?
20. **Construct** a function f on $[-1, 1]$ whose definite integral $\int_{-1}^1 f \, dx$ is 0 even though f is never zero. (Use signed area — what symmetry forces cancellation?)
21. Explain when substitution will and won’t help: why does it work for $\int 2x(x^2+1)^5 \, dx$ but not, as written, for $\int (x^2+1)^5 \, dx$? What’s missing in the second?
22. **Is it possible?** A definite integral $\int_a^b f \, dx$ that equals 0 with $a < b$ and $f > 0$ somewhere on $[a, b]$. (What must f do elsewhere?)

◆ **Diamond — *the definitions and the proofs***

23. ◆ Compute $\int_0^1 x^2 \, dx$ from the definition in full, deriving the limit of the right Riemann sum and using $\sum i^2 = \frac{n(n+1)(2n+1)}{6}$. (Reproduce the chapter’s computation, justifying each step.)
24. ◆ Prove the **linearity** of the definite integral, $\int_a^b (f+g) = \int_a^b f + \int_a^b g$, from the Riemann-sum definition. (A sum of sums is the sum of the sums; take the limit.)
25. ◆ **Derive substitution from the chain rule.** Starting from $\frac{d}{dx}F(g(x)) = F'(g(x))g'(x)$, integrate both sides to obtain the substitution rule $\int F'(g(x))g'(x) \, dx = F(g(x)) + C$, and explain why

it is exactly the chain rule read backward.

26. ♦ Prove that a function with $F'(x) = 0$ for all x on an interval is constant there (the fact behind the $+C$). (This is a one-line corollary of the Mean Value Theorem — state it.)
27. ♦♦ **The midpoint sum is sharp.** For $f(x) = x^2$ on $[0, 1]$, compute the *midpoint* Riemann sum with n rectangles and show it also converges to $\frac{1}{3}$ — but, by comparing to the right sum, argue that it approaches the true value *faster*. (You'll need $\sum(i - \frac{1}{2})^2$.)
28. ♦♦ Explain why $\int_a^b f dx$ being defined as a limit means it can *fail* to exist for badly behaved f (for instance, a function that jumps infinitely often). What property of f — met by every continuous function — guarantees the Riemann sums actually converge? (Name it; the proof is a real-analysis matter.)

Teach it — *explain it, write it up, or fix it*

29. In **three or four sentences with the rectangles picture**, teach a friend why the area under a curve is a limit of sums, and how thinner rectangles make the estimate exact.
30. **Debug your classmate.** They compute $\int 2x(x^2 + 1)^5 dx = \frac{(x^2+1)^6}{6} \cdot 2x + C$ — they “kept the $2x$.” Explain where their substitution went wrong and give the correct answer.
31. Write a complete **worked solution** to $\int x e^{x^2} dx$, narrating the choice of u , the handling of the stray constant, and the back-substitution, so a peer could redo it for $\int x \sin(x^2) dx$.
32. **Design a problem.** Invent an integral that *requires* a substitution, choose it so the answer is clean, and write the full solution key (including the differentiate-to-check step).
33. In a few sentences, **teach the difference** between the definite integral (a number, an area) and the indefinite integral (a family of functions), and why they nonetheless share a symbol.

Looking ahead — Chapter 11, the summit, in your hands early

34. Define $A(x) = \int_0^x t^2 dt$ — the area under t^2 from 0 out to a movable right edge x . Using your result $\int_0^1 t^2 dt = \frac{1}{3}$, guess a formula for $A(x)$, then compute $A'(x)$. **What do you notice about $A'(x)$ compared to the integrand t^2 ?** (You are about to discover the Fundamental Theorem yourself.)
35. If finding the area under f amounts to finding an antiderivative F and looking at its endpoint values, **predict** the formula for $\int_a^b f dx$ in terms of $F(a)$ and $F(b)$. (Chapter 11 proves your guess and turns every area problem into a subtraction.)
36. The one antiderivative that escaped the power rule was $\int \frac{1}{x} dx = \ln|x|$. **Predict** what the function $L(x) = \int_1^x \frac{1}{t} dt$ has to do with the logarithm — in particular, what is $L(1)$, and is L increasing or decreasing for $x > 1$? (Chapter 11 *defines* $\ln$ this way and repays every loan from Chapter 5.)
-

Chapter 11

The Fundamental Theorem of Calculus

Or: why finding an area is the same as undoing a derivative — the summit of the book

Companion notebook. Play with this chapter’s ideas — sliders, animations, and drills — in `notebooks/ch11_fundamental_theorem.ipynb`.

This is the chapter everything has been climbing toward. In Chapter 10 you built two things and left them estranged: the **definite integral**, a signed area defined as a punishing limit of rectangle-sums, and the **antiderivative**, found in seconds by running the Chapter 4 rules backward. They share a symbol and seem to share nothing else. This chapter proves they are the same — that *the area under a curve can be computed by undoing a derivative* — and that single fact is the most consequential theorem in all of calculus, perhaps in all of the mathematics that science uses.

By the end you will be able to evaluate a definite integral in one line instead of a page; you’ll learn **integration by parts**, the product rule run backward; you’ll see how *every* “total” — an area between curves, a volume, the length of a curve — is the same slicing-and-summing idea, so you never have to memorize a formula you can rederive; and, as the book’s final act, you will **define the logarithm as an integral** and watch every loan we took on about e and $\ln$ — taken on faith chapters ago — get paid in full.

The whole chapter rests on one picture: let the right edge of an area move, and watch how fast the area grows.

11.1 The bridge, discovered: a moving area

Take a continuous function f and define the **area-so-far function**

$$A(x) = \int_a^x f(t) dt,$$

the signed area under f from the fixed left edge a out to a movable right edge x . (You met this in Chapter 10's last problem, where $A(x) = \int_0^x t^2 dt$ quietly turned out to have derivative x^2 .) Now ask the central question: *how fast does this area grow as you slide x to the right?* — that is, what is $A'(x)$?

Push the right edge out by a tiny amount h . The area gains a thin vertical strip of width h and height about $f(x)$, so the extra area is approximately $f(x) \cdot h$:

$$A(x+h) - A(x) \approx f(x) \cdot h \quad \implies \quad \frac{A(x+h) - A(x)}{h} \approx f(x).$$

Let $h \rightarrow 0$ and the approximation becomes exact. The left side is the *definition of $A'(x)$* . So $A'(x) = f(x)$: **the rate at which area accumulates is exactly the height of the curve.** (Figure 11.1: the region under f from a to x , with the thin strip of width h and height $f(x)$ added on at the right edge — the strip whose area is $A(x+h) - A(x)$.)

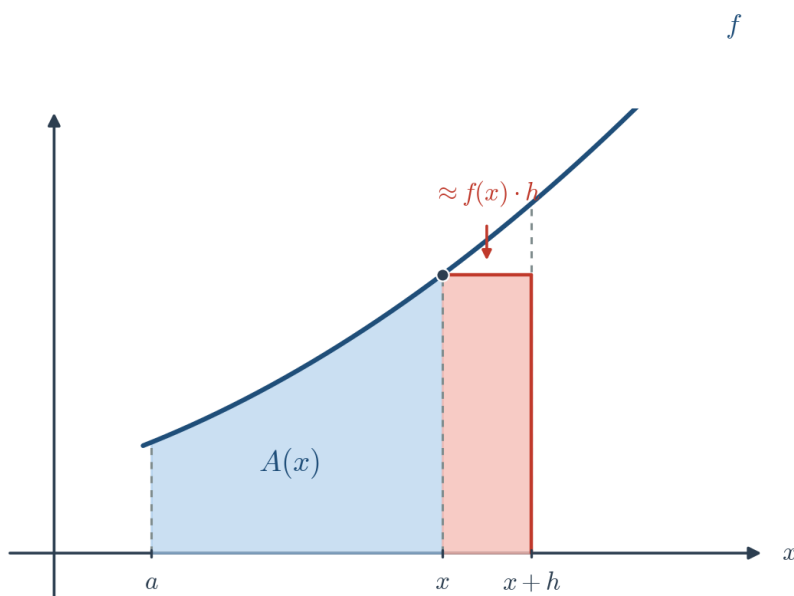


Figure 11.1

That is the entire secret. Differentiating the area gives back the curve. Read it the other way and it says the area function is an *antiderivative* of f — which is the bridge between the two faces.

◆ Layer 2 — The strip argument, made rigorous

“The strip has height about $f(x)$ ” needs tightening. On the little interval $[x, x+h]$, the continuous function f has a minimum value m_h and a maximum M_h (Extreme Value Theorem, Chapter 2). The strip’s area $A(x+h) - A(x)$ is squeezed between the areas of the short and tall rectangles:

$$m_h \cdot h \leq A(x+h) - A(x) \leq M_h \cdot h.$$

Divide by h : $m_h \leq \frac{A(x+h)-A(x)}{h} \leq M_h$. As $h \rightarrow 0$ the interval shrinks to the point x , and by continuity *both* m_h and M_h converge to $f(x)$. The middle is squeezed (Chapter 2's Squeeze Theorem) to $f(x)$. Hence $A'(x) = f(x)$, exactly. ■

This is half of the theorem.

Fundamental Theorem of Calculus, Part 1. If f is continuous and $A(x) = \int_a^x f(t) dt$, then A is differentiable and

$$A'(x) = f(x).$$

Every continuous function is the derivative of its own accumulated area.

11.2 The payoff: areas become subtractions

Part 1 says the area function is *an* antiderivative of f . But antiderivatives are easy to find — just reverse the Chapter 4 rules — and they all differ by a constant (Chapter 6's Mean Value Theorem, via Chapter 10's $+C$). Chain those two facts together and the laborious definite integral collapses into arithmetic.

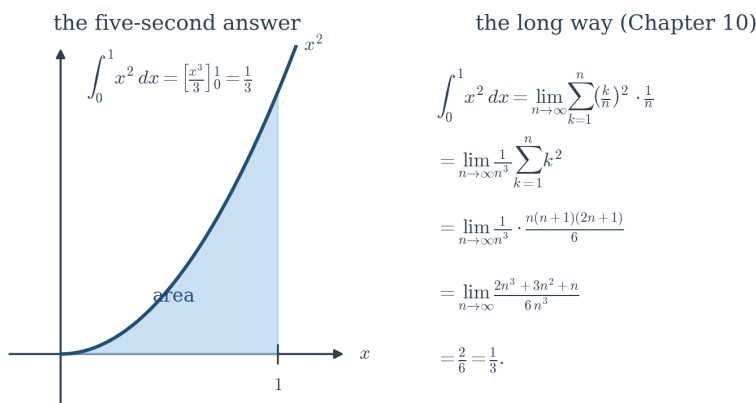


Figure 11.5

Fundamental Theorem of Calculus, Part 2. If f is continuous and F is *any* antiderivative of f (any function with $F' = f$), then

$$\int_a^b f(x) dx = F(b) - F(a).$$

Why it's true. The area function $A(x) = \int_a^x f$ is one antiderivative (Part 1). Your F is another, so they differ by a constant: $F(x) = A(x) + C$. Then

$$F(b) - F(a) = (A(b) + C) - (A(a) + C) = A(b) - A(a) = A(b) - 0 = \int_a^b f(x) dx,$$

since $A(a) = \int_a^a f = 0$. The constant cancels; *any* antiderivative works. ■

We write $F(b) - F(a)$ as $[F(x)]_a^b$. And now watch Chapter 10's hardest computation evaporate. There, finding $\int_0^1 x^2 dx$ took a page of Riemann sums and the sum-of-squares formula. Here:

$$\int_0^1 x^2 dx = \left[\frac{x^3}{3} \right]_0^1 = \frac{1^3}{3} - \frac{0^3}{3} = \frac{1}{3}.$$

Five seconds. Find an antiderivative, plug in the endpoints, subtract. Every area you can find an antiderivative for is now this easy: $\int_0^\pi \sin x dx =$

$$[-\cos x]_0^\pi = -\cos \pi + \cos 0 = 1 + 1 = 2; \int_1^2 \frac{1}{x} dx = [\ln |x|]_1^2 = \ln 2.$$

The miracle, stated plainly

Stand back and see what just happened. **Differentiation and integration are inverse operations** — the two engines we built in total isolation, one measuring slopes and one measuring areas, are a single engine running in opposite directions. Part 1 says: integrate, then differentiate, and you return to where you started ($\frac{d}{dx} \int_a^x f = f$). Part 2 says: to integrate, just undo a derivative. Slope and area, which have no visible relationship, turn out to be reciprocal questions. That two such different ideas are secretly the same is the reason calculus is not two subjects but one, and it is the engine under essentially all of physics, engineering, and quantitative science.

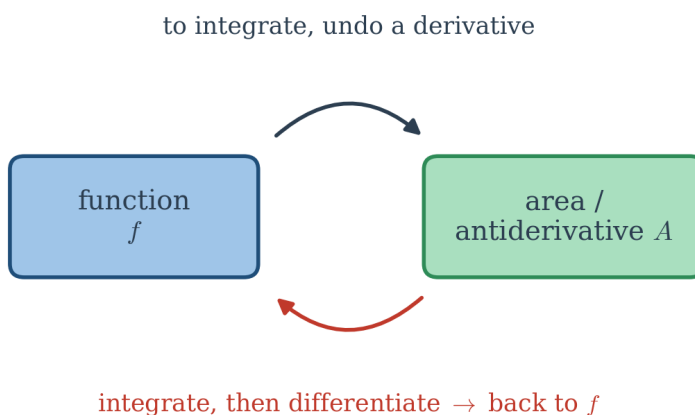


Figure 11.3

11.3 Integration by parts: the product rule, backward

Substitution (Chapter 10) was the chain rule reversed. The other great technique is the **product rule reversed**, and it's derived the same way. Start from the product rule, $(fg)' = f'g + fg'$, and integrate both sides. The left gives fg ; rearrange:

$$\int f g' dx = fg - \int f' g dx, \quad \text{often written} \quad \int u dv = uv - \int v du.$$

You use it when the integrand is a product, and differentiating one factor makes the integral *simpler*. Pick that factor to be u (so it gets differentiated) and the rest to be dv (so it gets integrated).

- $\int x e^x dx$: let $u = x$ (its derivative 1 simplifies things), $dv = e^x dx$. Then $\int x e^x dx = x e^x - \int e^x dx = x e^x - e^x + C$.
- $\int x \cos x dx$: let $u = x$, $dv = \cos x dx$, giving $x \sin x - \int \sin x dx = x \sin x + \cos x + C$.
- $\int \ln x dx$: the slick one. Take $u = \ln x$ and $dv = dx$ (yes, the whole rest is just dx). Then $v = x$, $du = \frac{1}{x} dx$, and $\int \ln x dx = x \ln x - \int x \cdot \frac{1}{x} dx = x \ln x - x + C$. We integrated a function that *isn't even a product* by inventing one.
- $\int x^2 e^x dx$: when one pass isn't enough, *repeat*. Let $u = x^2$, $dv = e^x dx$, giving $x^2 e^x - \int 2x e^x dx$ — and the leftover integral is the first example (times 2), so parts again finishes it: $x^2 e^x - 2(x e^x - e^x) + C = (x^2 - 2x + 2) e^x + C$. Each pass knocks the power down by one.

That's the principle — the product rule, read right to left. (Substitution and parts are the reverse-images of Chapter 4's two big rules; together they handle an enormous range of integrals, and when they fail, the inte-

gral often has no elementary antiderivative at all, and we fall back on the Riemann sums of Chapter 10.)

11.4 Every “total” is one idea: slice and sum

A standard calculus course now unloads a warehouse of formulas — areas between curves, volumes by disks, volumes by shells, arc length, surfaces of revolution, work, fluid pressure — each with its own boxed equation to memorize. **Don’t.** They are all the *same* idea, the idea the integral was built from: to measure a total, **cut it into infinitesimal pieces simple enough to measure, then add the pieces up (integrate).** Learn that move and you can derive any of the formulas on demand; memorizing them is a confession that you didn’t see they were one thing. Three quick illustrations, then we move on:

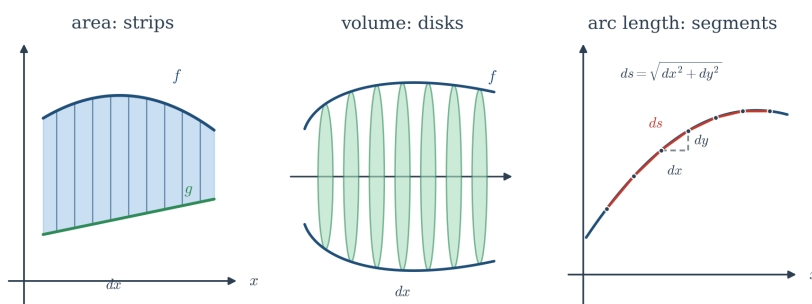


Figure 11.4

- **Area between two curves.** Slice the region into thin vertical strips. Each strip has width dx and height (top curve – bottom curve), so its area is (top – bottom) dx . Add them: $\int_a^b (\text{top} - \text{bottom}) dx$.
- **Volume of a solid of revolution.** Spin a curve $y = f(x)$ around the x -axis and slice the solid into thin coins. Each coin is a disk of

radius $f(x)$ and thickness dx , with volume $\pi f(x)^2 dx$. Add them:

$$\int_a^b \pi f(x)^2 dx.$$

- **Length of a curve.** Slice the curve into tiny nearly-straight segments. By the Pythagorean theorem each segment has length

$$\sqrt{dx^2 + dy^2} = \sqrt{1 + (f'(x))^2} dx. \text{ Add them: } \int_a^b \sqrt{1 + (f'(x))^2} dx.$$

In every case the recipe was identical: *what does one infinitesimal piece contribute, and what do they sum to?* Notice, too, the honesty in the last one — that arc-length integrand is a square root of a polynomial, and for most curves (even a humble parabola) it has *no elementary antiderivative*. So you set up the integral by the principle and then, as often as not, hand it to a Riemann sum. This is exactly why memorizing the formula is pointless and understanding the slice is everything.

11.5 The capstone: defining the logarithm, and settling every debt

We end by paying our debts. All through this book we *borrowed*: in Chapter 1 we asserted the number e exists; in Chapter 5 we used $\ln' x = \frac{1}{x}$ and the laws of logarithms; each time we flagged it as a loan against a later, rigorous account. This is that account. And the instrument that settles everything is the one antiderivative the power rule could never produce — the missing $n = -1$ case, $\int \frac{1}{x} dx$.

So we turn it around and make it a **definition**:

$$\ln x := \int_1^x \frac{1}{t} dt \quad (x > 0).$$

We *define* the natural logarithm to be the area under $\frac{1}{t}$ from 1 to x . From this one line, with nothing assumed, every property of $\ln$ we ever

used falls out as a theorem:

- $\ln 1 = 0$: it's $\int_1^1 \frac{1}{t} dt$, an integral over no width. ✓ (The loan that $\ln 1 = 0$ — repaid by definition.)
- $\ln' x = \frac{1}{x}$: immediate from the Fundamental Theorem, Part 1 — the derivative of $\int_1^x \frac{1}{t} dt$ is the integrand $\frac{1}{x}$. The fact we *assumed* in Chapter 5 is now a one-line consequence. ✓
- **$\ln$ is increasing for $x > 0$** : its derivative $\frac{1}{x}$ is positive. And it grows without bound as $x \rightarrow \infty$ (the area under $\frac{1}{t}$ never stops accumulating). ✓
- **The product law** $\ln(ab) = \ln a + \ln b$: split the defining integral and substitute. We have $\ln(ab) = \int_1^{ab} \frac{1}{t} dt = \int_1^a \frac{1}{t} dt + \int_a^{ab} \frac{1}{t} dt$. In the second piece let $u = t/a$ (so $dt = a du$, and $t : a \rightarrow ab$ becomes $u : 1 \rightarrow b$): it turns into $\int_1^b \frac{1}{u} du = \ln b$. Hence $\ln(ab) = \ln a + \ln b$ — the signature property of logarithms, emerging from integral additivity and a substitution. ✓
- **The number e exists**: define e as the unique number with $\ln e = 1$ — i.e. the right edge at which the area under $\frac{1}{t}$ first equals 1. It *exists* because $\ln$ is continuous with $\ln 1 = 0 < 1$ and grows past 1, so by the **Intermediate Value Theorem** (Chapter 2) it hits 1 somewhere. This is the very loan from Chapter 1 — *that e exists at all* — now discharged completely. ✓
- **e^x and its derivative**: the exponential is simply the inverse function of $\ln$, and $\frac{d}{dx} e^x = e^x$ follows from the inverse-function rule (Chapter 4). ✓

Every promissory note the book ever wrote is now paid. The whole edifice — e , $\ln$, e^x , their derivatives and laws — which we used freely and honestly flagged as borrowed, rests in the end on a single integral and the Fundamental Theorem. The account balances. (Figure 11.2: the curve $y = \frac{1}{t}$ with the area from 1 to x shaded and labeled $\ln x$; a second marker at $t = e$ where the accumulated area first reaches exactly 1.)

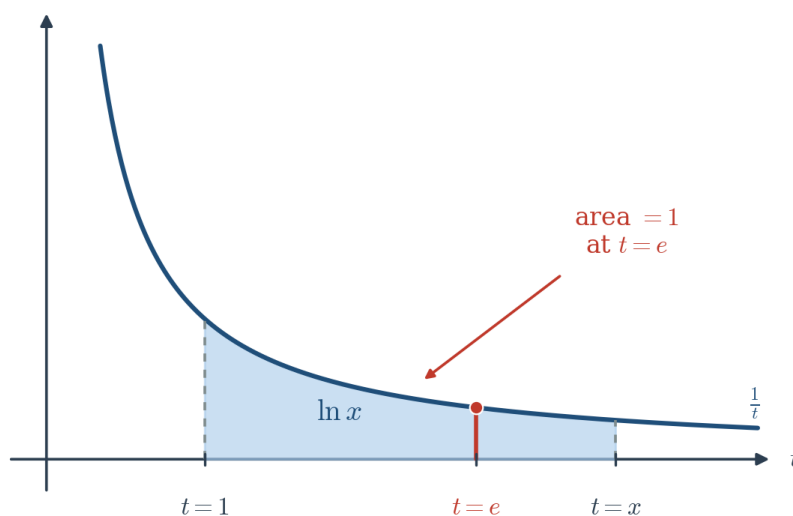


Figure 11.2

11.6 Where this is heading

You have now built single-variable calculus from the ground up — limits, derivatives, the rules, the transcendental functions, the great theorems, the integral, and the Fundamental Theorem that binds the two halves into one — with every result derived and every loan repaid. Two doors stand open.

Chapter 12 stays in this book and finishes a thread from Chapter 1. We have added up infinitely many *infinitesimal* pieces (integrals); what about adding up infinitely many *ordinary* numbers — an **infinite series**? When does $1 + \frac{1}{2} + \frac{1}{4} + \dots$ settle to a value and $1 + \frac{1}{2} + \frac{1}{3} + \dots$ run off to infinity? The tool that decides it is the **integral test**, which weighs a sum against an area using the very picture this chapter is made of — and it will let us finally say, rigorously, *where* the Taylor series of Chapter 7 are allowed to live.

Book 3 opens the door to many dimensions: the derivative becomes the gradient, the single integral becomes a double and triple integral

(volume measured exactly the way we measured area), optimization becomes the real landscape that machine learning descends, and the waves of Fourier finally get the orthogonal-basis home they were waiting for. The slope of a single point started all of this; it doesn't stop here.

11.7 Problems

Six tiers. **Fluency and Practice** are doing* — evaluate definite integrals with the FTC and integrate by parts until both are automatic (plenty of drill, as in Chapter 4). **Concept & counterexample** and **◆ Diamond** are *thinking* — the theorem itself and the proofs. **Teach it** is *teaching*. **Looking ahead** opens Chapter 12. The applications problems below ask you to **set up** the integral from the slicing principle — not to recall a formula. Do all of Fluency; FTC fluency is the working core of all of integral calculus.*

Fluency — *evaluate with the FTC, fast*

1. $\int_0^2 x^3 dx$. 2. $\int_1^4 \sqrt{x} dx$. 3. $\int_0^\pi \sin x dx$. 4. $\int_0^{\pi/2} \cos x dx$. 5. $\int_1^e \frac{1}{x} dx$.
2. $\int_0^1 e^x dx$. 7. $\int_{-1}^1 (x^2 - 1) dx$ (and say what the sign means). 8. $\int_1^2 (3x^2 - 2x) dx$. 9. $\int_0^{\pi/4} \sec^2 x dx$. 10. $\int_1^3 \left(x + \frac{1}{x}\right) dx$.
3. Use FTC Part 1 to differentiate without integrating: (a) $\frac{d}{dx} \int_0^x \sin(t^2) dt$; (b) $\frac{d}{dx} \int_2^x e^{-t^2} dt$; (c) $\frac{d}{dx} \int_1^x \frac{1}{1+t^4} dt$.
4. Integration by parts: (a) $\int xe^x dx$; (b) $\int x \cos x dx$; (c) $\int x \sin x dx$; (d) $\int xe^{-x} dx$; (e) $\int (x + 1) \sin x dx$.
5. Integration by parts, harder: (a) $\int \ln x dx$; (b) $\int xe^{2x} dx$; (c) $\int x^2 e^x dx$ (parts twice); (d) $\int x^2 \cos x dx$ (parts twice); (e) $\int x \ln x dx$ (take $u = \ln x$).

Practice — multi-step, and set up the slice

14. Evaluate $\int_0^1 xe^{x^2} dx$ two ways — by substitution, and by recognizing an antiderivative — and confirm they agree.
15. Find the area **between** $y = x$ and $y = x^2$ from $x = 0$ to $x = 1$. (Set up top — bottom, then integrate.)
16. A region under $y = \frac{1}{x}$ from $x = 1$ to $x = 4$ is spun around the x -axis. **Set up** the volume by slicing into disks, then evaluate.
17. **Set up** (do not evaluate) the length of the parabola $y = x^2$ from $x = 0$ to $x = 1$ using the slicing principle, and explain why you should expect *not* to get a clean closed form.
18. Find the function F with $F'(x) = \cos x$ and $F(0) = 3$. Then use it to compute $\int_0^\pi \cos x dx$ two ways (antiderivative-and-subtract, and directly), confirming they match.
19. Using $\ln x = \int_1^x \frac{1}{t} dt$, show directly from the integral that $\ln(2 \cdot 3) = \ln 2 + \ln 3$, by performing the substitution in the chapter on $\int_2^6 \frac{1}{t} dt$.

Concept & counterexample — think; true/false with justification, or build the example

20. (a) $\frac{d}{dx} \int_a^x f(t) dt = f(x)$. (b) $\int_a^b f'(x) dx = f(b) - f(a)$. (c) If F and G are both antiderivatives of f , then $\int_a^b f = F(b) - F(a) = G(b) - G(a)$. (d) Every continuous function has an antiderivative.
21. Explain, in your own words and citing FTC Part 1, why $\int_a^x f(t) dt$ is *guaranteed* to be an antiderivative of f even when no elementary antiderivative of f exists (as for e^{-t^2}).
22. **Find the error.** A student computes $\int_{-1}^1 \frac{1}{x^2} dx = \left[-\frac{1}{x} \right]_{-1}^1 = -1 - 1 = -2$, a negative number for a positive function. What went wrong? (Where does $\frac{1}{x^2}$ fail the FTC's hypotheses on $[-1, 1]$?)
23. The two parts of the FTC say differentiation and integration undo each other. State precisely what “undo” means in each part — which operation is done first, and what you get back.

24. **Is it possible?** Two different functions $f \neq g$ with $\int_a^x f(t) dt = \int_a^x g(t) dt$ for all x . (Differentiate both sides — what must be true?)

◆ **Diamond — *the theorem and the capstone***

25. ◆ Prove **FTC Part 1** rigorously using the strip argument: bound $A(x+h) - A(x)$ between $m_h \cdot h$ and $M_h \cdot h$, divide by h , and squeeze as $h \rightarrow 0$ (citing the Extreme Value and Squeeze Theorems of Chapter 2).
26. ◆ Prove **FTC Part 2** from Part 1, using the fact (Chapter 6) that two antiderivatives differ by a constant. Identify the exact step where $A(a) = 0$ is used.
27. ◆ Derive **integration by parts** from the product rule, and use it to evaluate $\int \ln x dx$, explaining the unusual choice $dv = dx$.
28. ◆◆ From the definition $\ln x = \int_1^x \frac{1}{t} dt$, prove the **power law** $\ln(x^r) = r \ln x$ for the case r a positive integer. (Hint: use the product law repeatedly, or substitute.)
29. ◆◆ Prove that e , defined as the number with $\ln e = 1$, **exists and is unique**, citing the Intermediate Value Theorem (existence) and the fact that $\ln$ is strictly increasing (uniqueness). Explain how this discharges the loan first taken in Chapter 1.
30. ◆◆ Define $E(x)$ as the inverse function of $\ln$. Using the inverse-function rule (Chapter 4) and $\ln' x = \frac{1}{x}$, prove $E'(x) = E(x)$ — i.e., the inverse of the log-integral is its own derivative, which is why it deserves the name e^x .

Teach it — *explain it, write it up, or fix it*

31. In **three or four sentences with the moving-edge picture**, teach a friend why differentiating an area gives back the curve (FTC Part 1).
32. **Debug your classmate.** They evaluate $\int_0^\pi \sin x dx$ as $[\cos x]_0^\pi = \cos \pi - \cos 0 = -2$. Find both errors (the antiderivative *and* the sign)

and give the correct value.

33. Write a complete **worked solution** to $\int x^2 e^x dx$ (integration by parts, applied twice), narrating why each step reduces the power, so a peer could do $\int x^3 e^x dx$.
34. In a few sentences, **teach the slicing principle** — how area-between-curves, volume, and arc length are all “what does one thin piece contribute, summed up” — and why that means you never need to memorize their formulas.
35. **Teach the capstone.** In a short paragraph, explain to someone who has finished Chapter 5 why *defining* $\ln x = \int_1^x \frac{1}{t} dt$ is more honest than just asserting $\ln' x = \frac{1}{x}$ — what does the definition give you that the assertion didn't?

Looking ahead — *Chapter 12 in your hands early*

36. The integral $\int_1^\infty \frac{1}{x} dx$ is the area under $\frac{1}{x}$ stretching out forever. Using $\int_1^b \frac{1}{x} dx = \ln b$, decide whether this area is **finite or infinite** as $b \rightarrow \infty$. (You've just decided whether the series $1 + \frac{1}{2} + \frac{1}{3} + \dots$ converges — Chapter 12 makes the link exact.)
 37. Compare $\int_1^\infty \frac{1}{x^2} dx$ (compute it, as $b \rightarrow \infty$) with the previous problem. One area is finite, the other infinite. **Predict** which of $\sum \frac{1}{n^2}$ and $\sum \frac{1}{n}$ converges, and articulate the rule you're sensing: a sum and the matching area live or die together. (That rule is Chapter 12's integral test, and it is made of this chapter's pictures.)
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Chapter 12

Adding Up Forever

Or: when does an infinite sum settle to a number — and where do Taylor's series really live?

Companion notebook. Play with this chapter's ideas — sliders, animations, and drills — in notebooks/`ch12_series.ipynb`.

This is the last chapter, and it finishes a story that started in the first. In Chapter 1 you learned what it means for a *sequence* of numbers to close in on a limit. In Chapter 7 you wrote functions as infinite sums of powers — Taylor series — and used them freely, while quietly stepping over a hard question: *what does it even mean to add infinitely many things, and when is the answer a finite number?* This chapter answers it, and in doing so ties the whole book into a single knot.

By the end you will know what an **infinite series** is (the limit of its partial sums — pure Chapter 1), which famous sums converge and which famously don't, and **two principles** for telling them apart: **comparison** (trap your sum between sums you understand) and the **integral test** (weigh your sum against an area — the rectangle-and-curve picture that has run through this entire book). And you will finally pin down the thing Chapter 7 left vague: the precise interval where each Taylor series is actually allowed to equal its function — its **radius of convergence**.

A word on spirit before we start. A typical course buries this topic under a dozen named tests to be memorized and matched to problems like keys to locks. We will not do that. There are really just two ideas here —

compare to something known, and *compare to an area* — and almost everything else is those two ideas wearing costumes. Learn the two, and you can rederive the rest.

12.1 What it means to add infinitely many numbers

You cannot actually perform infinitely many additions. So, exactly as with every infinite process in this book, we don't — we add *finitely* many and take a limit. Given a list $a_1, a_2, a_3, \dots$, form the **partial sums**

$$S_1 = a_1, \quad S_2 = a_1 + a_2, \quad S_3 = a_1 + a_2 + a_3, \quad \dots, \quad S_N = \sum_{n=1}^N a_n.$$

These partial sums form a *sequence* — and we already know, from Chapter 1, what it means for a sequence to converge. So:

An infinite series $\sum_{n=1}^{\infty} a_n$ **converges** to S if its sequence of partial sums S_N converges to S . If the partial sums have no limit, the series **diverges**.

That's the whole definition, and notice it contains nothing new: *series convergence is sequence convergence of partial sums*. The infinite sum $\frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \dots$ converges to 1 because its partial sums $\frac{1}{2}, \frac{3}{4}, \frac{7}{8}, \frac{15}{16}, \dots$ climb toward 1 — the staircase of Chapter 1, now built from a running total. (*Figure 12.1: the partial sums of $\sum 2^{-n}$ plotted as a sequence of dots climbing toward the line $y = 1$.*)

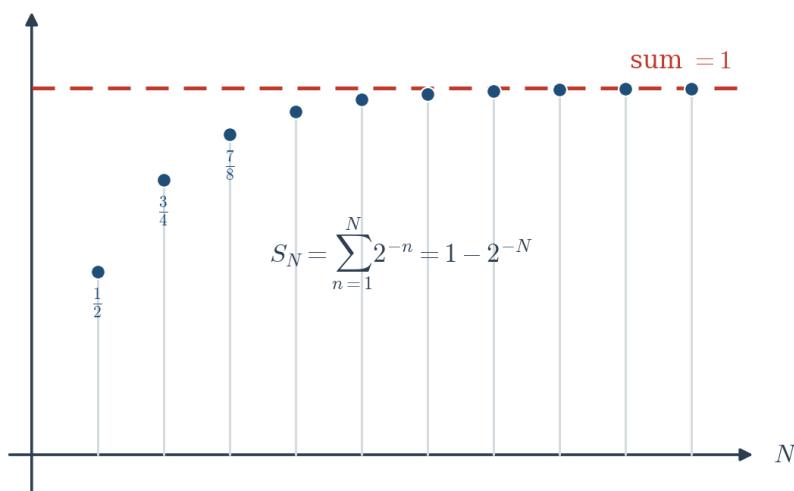


Figure 12.1

12.2 The one sum you can compute exactly: the geometric series

Almost no series can be summed in closed form. One family can, and it's the benchmark for everything else — and you've already met it, as the Taylor series of $\frac{1}{1-x}$ in Chapter 7. The **geometric series** adds powers of a fixed ratio r :

$$\sum_{n=0}^{\infty} r^n = 1 + r + r^2 + r^3 + \dots.$$

Its partial sum has a clean form (multiply $S_N = 1 + r + \dots + r^N$ by r and subtract): $S_N = \frac{1 - r^{N+1}}{1 - r}$. Now take $N \rightarrow \infty$. If $|r| < 1$, then $r^{N+1} \rightarrow 0$ and the sum converges:

$$\sum_{n=0}^{\infty} r^n = \frac{1}{1-r} \quad (|r| < 1).$$

If $|r| \geq 1$, the terms don't shrink and the partial sums run away — divergent. So $\frac{1}{2} + \frac{1}{4} + \dots = \frac{1}{1-1/2} = 1$ ✓, while $1 + 2 + 4 + \dots$ explodes. This is precisely Chapter 7's $\frac{1}{1-x} = 1 + x + x^2 + \dots$ with $x = r$, and the condition $|r| < 1$ is the same “wall at ± 1 ” you saw the series slam into there. The geometric series is the ruler we measure other series against.

12.3 A first sanity check: the terms must die

Here is the simplest possible test, and the one principle you can always apply first.

The term test. If $\sum a_n$ converges, then $a_n \rightarrow 0$.

The reason is one line: $a_n = S_n - S_{n-1}$, and if both partial sums converge to the same S , their difference goes to $S - S = 0$. The useful form is the *contrapositive*: **if the terms do not go to zero, the series diverges**. So $\sum \frac{n}{n+1}$ diverges instantly (its terms approach 1, not 0) — no further work needed.

But beware the trap that ensnares everyone at first: the term test is **necessary, not sufficient**. Terms going to zero does *not* guarantee convergence. The most important example in the subject proves it.

12.4 The harmonic series: every term vanishes, yet the sum is infinite

Consider $\sum_{n=1}^{\infty} \frac{1}{n} = 1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \dots$. The terms $\frac{1}{n}$ shrink to zero, so the term test is *passed* — and yet this series **diverges to infinity**. This is one of the genuine surprises of mathematics, and a medieval one: the argument is due to Nicole Oresme, around 1350, and it is pure cleverness.

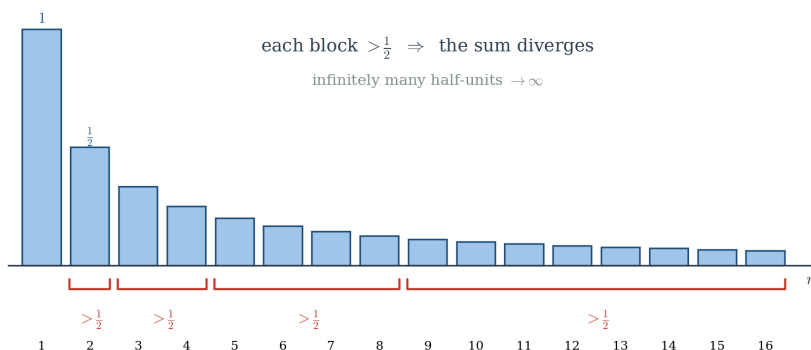


Figure 12.5

◆ Layer 2 — Oresme’s proof that the harmonic series diverges

Group the terms in blocks of doubling length and bound each block from below:

$$1 + \frac{1}{2} + \underbrace{\left(\frac{1}{3} + \frac{1}{4}\right)}_{> \frac{1}{4} + \frac{1}{4} = \frac{1}{2}} + \underbrace{\left(\frac{1}{5} + \frac{1}{6} + \frac{1}{7} + \frac{1}{8}\right)}_{> 4 \cdot \frac{1}{8} = \frac{1}{2}} + \underbrace{\left(\frac{1}{9} + \dots + \frac{1}{16}\right)}_{> 8 \cdot \frac{1}{16} = \frac{1}{2}} + \dots$$

Every block exceeds $\frac{1}{2}$ (each has terms no smaller than its last, and there are exactly enough of them). Since there are infinitely many blocks, each contributing more than $\frac{1}{2}$, the partial sums climb past $\frac{1}{2}, 1, \frac{3}{2}, 2, \dots$ without bound. The sum is infinite. ■

Keep the harmonic series close; it is the dividing line. “The terms go to zero” feels like it should be enough — and the harmonic series is the eternal reminder that it isn’t. To decide convergence, we need real tools. Here are the two.

12.5 Principle 1 — Comparison: trap it between sums you know

The first principle is exactly what you’d do by instinct: if your series is term-by-term *smaller* than one you know converges, it must converge too; if it’s *larger* than one you know diverges, it must diverge.

Comparison. Suppose $0 \leq a_n \leq b_n$ for all n . If $\sum b_n$ converges, then $\sum a_n$ converges. If $\sum a_n$ diverges, then $\sum b_n$ diverges.

The geometric series is usually the thing you compare against. For instance, $\sum \frac{1}{2^n + 1}$ has every term smaller than $\frac{1}{2^n}$, and $\sum \frac{1}{2^n}$ is a convergent geometric series — so by comparison, $\sum \frac{1}{2^n + 1}$ converges. The whole skill is recognizing what to compare to: bound your complicated series above by a simple convergent one (to prove convergence), or below by a simple divergent one (to prove divergence). (*Figure 12.2: two stacks of bars, the a_n bars sitting entirely under the b_n bars, with the b_n total marked finite — so the a_n total is trapped finite beneath it.*)

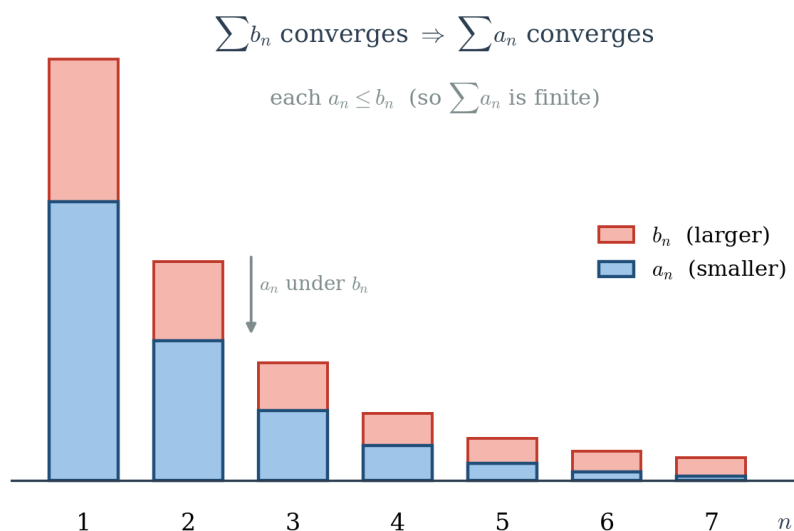


Figure 12.2

12.6 Principle 2 — The integral test: weigh the sum against an area

The second principle is the one that earns this chapter its place at the *end* of a calculus book, because it connects a sum to an **integral** — and it does so with the very rectangle-and-curve picture you built in Chapters 10 and 11.

Take a series $\sum f(n)$ whose terms come from a function f that is positive, decreasing, and continuous. Draw the terms as rectangles of width 1 and height $f(n)$, and lay them against the graph of f . Because f is decreasing, each rectangle is trapped between the area just under and just over the curve — and summing across all of them traps the *whole sum* between two copies of the same integral:

$$\sum_{n=2}^{\infty} f(n) \leq \int_1^{\infty} f(x) dx \leq \sum_{n=1}^{\infty} f(n).$$

The sum and the integral differ by at most the single term $f(1)$, so they are *finite together or infinite together*. That is the test:

The integral test. For positive, decreasing, continuous f , the series $\sum_{n=1}^{\infty} f(n)$ and the integral $\int_1^{\infty} f(x) dx$ either **both converge or both diverge**.

Look at what just happened: the integral you learned to compute with the Fundamental Theorem now *decides infinite sums*. The rectangles that defined the integral in Chapter 10 reappear here as the terms of a series, and the area they approximate settles whether the sum is finite. Adding up infinitely many numbers and measuring the area under a curve are, once again, the same question. (Figure 12.3: the curve $y = f(x)$ with the series' rectangles drawn against it — one set tucked under the curve, one rising above — visibly squeezing the sum to the area.)

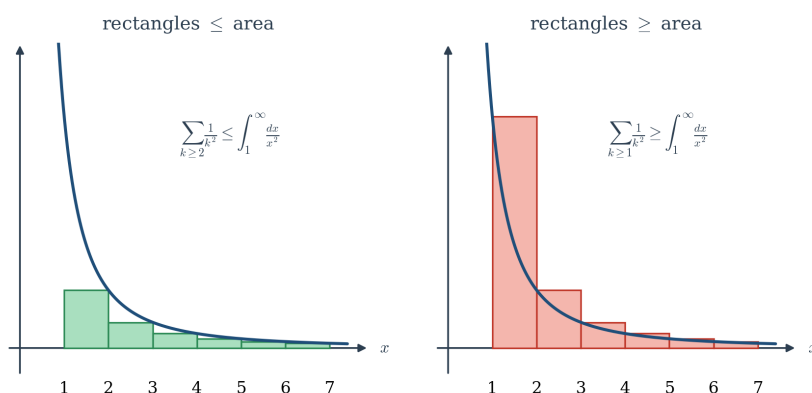


Figure 12.3

The p-series, settled at last — and a debt from Chapter 11 repaid

Apply the integral test to the family $\sum_{n=1}^{\infty} \frac{1}{n^p}$. Convergence hinges on $\int_1^{\infty} \frac{1}{x^p} dx$ — which we computed in Chapter 11: it is **finite exactly when** $p > 1$. Therefore:

$$\sum_{n=1}^{\infty} \frac{1}{n^p} \text{ converges if and only if } p > 1.$$

This resolves the cliffhanger Chapter 11 left you with. There, $\int_1^{\infty} \frac{1}{x} dx$ was infinite while $\int_1^{\infty} \frac{1}{x^2} dx$ was finite — and you were asked to predict which of $\sum \frac{1}{n}$ and $\sum \frac{1}{n^2}$ converges. Now you know, and you know *why* they match: the harmonic series ($p = 1$) diverges because its integral does; $\sum \frac{1}{n^2}$ ($p = 2$) converges because *its* integral does. (That sum, remarkably, equals $\frac{\pi^2}{6}$ — a fact whose proof waits for the Fourier methods of Book 3.) The sum and the area live or die together, exactly as Chapter 11 promised.

12.7 Power series: where Taylor's expansions actually live

Now the payoff Chapter 7 has been owed. A **power series** $\sum_{n=0}^{\infty} c_n x^n$ is a Taylor series viewed as a sum that may or may not converge depending on x . The fundamental fact is clean:

Radius of convergence. A power series $\sum c_n x^n$ converges for all $|x| < R$ and diverges for all $|x| > R$, for some number R (possibly 0 or ∞) called the **radius of convergence**.

So every Taylor series comes with an interval $(-R, R)$ — centered at its center — inside which it genuinely equals its function, and outside which it is meaningless. This is the rigor Chapter 7 promised and postponed. And the practical tool for *finding* R is the **ratio test**: form the ratio of consecutive terms and see what it approaches. (We introduce it here, where it actually matters — for power series — rather than as one more entry in a catalog.) If $\left| \frac{c_{n+1}}{c_n} \right| \rightarrow L$, then $R = \frac{1}{L}$, because the series behaves geometrically with ratio $\approx L|x|$, which converges precisely when $L|x| < 1$.

Turn it on Chapter 7's flagship series:

- $e^x = \sum \frac{x^n}{n!}$: the ratio of coefficients is $\frac{1/(n+1)!}{1/n!} = \frac{1}{n+1} \rightarrow 0$, so $L = 0$ and $R = \infty$. The series converges for *every* x — confirming, rigorously, the claim you proved by remainder in Chapter 7.
- $\frac{1}{1-x} = \sum x^n$: the coefficients are all 1, so $L = 1$ and $R = 1$. Converges for $|x| < 1$, diverges beyond — *that* is the wall at ± 1 in Chapter 7's convergence figure, now derived rather than drawn.
- $\ln(1+x)$ **and** $\arctan x$: both have $R = 1$, so each is valid only on $(-1, 1)$ — which is why setting $x = 1$ in the arctangent series to reach $\frac{\pi}{4}$ was living dangerously, right at the boundary (it works, but only barely, and for a subtler reason).

So Chapter 7's series were never just formal manipulations: each carries a precise domain of validity, and you can now compute it. (*Figure 12.4: a power series' interval of convergence drawn on the x -axis — solid inside $(-R, R)$, hatched outside — for e^x (the whole line) and $\frac{1}{1-x}$ (a segment ending at ± 1).)*

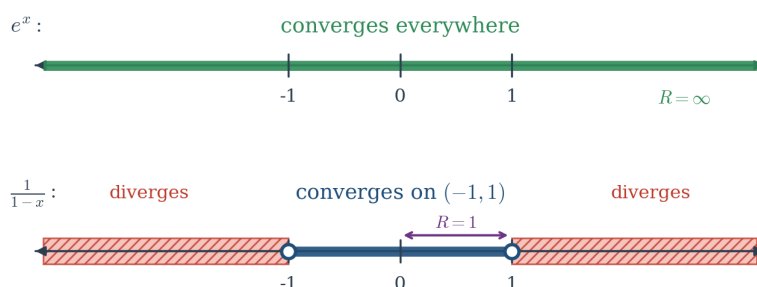


Figure 12.4

A last word: when alternating signs rescue a sum

One pattern deserves a closing mention, because Chapter 7's sine, cosine, and arctangent series all have it: the terms **alternate in sign**. Alternation helps enormously, because consecutive terms partly cancel. The principle: *if the terms alternate in sign and shrink steadily to zero, the series converges* — even when the same terms with all-positive signs would diverge. The cleanest example is the **alternating harmonic series** $1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \dots$, which converges (to $\ln 2$, as the Chapter 7 series for $\ln(1+x)$ at $x = 1$ suggests) — although, strip the signs and it becomes the harmonic series, which diverges. Cancellation is a real force. We won't drill it into a formal apparatus; the idea — *alternation tames a sum* — is what to carry forward.

12.8 Where this leaves us — and where it goes

Look back at the knot this chapter tied. An infinite series is a *limit of partial sums* — the sequences of Chapter 1. The benchmark series is the *geometric* one — the Taylor series of Chapter 7. The deepest tool for convergence is the *integral test* — the rectangles and areas of Chapters 10 and 11. And the whole apparatus finally pins down *where Taylor series live* — closing the one

question Chapter 7 had to leave open. Nothing here was new machinery; it was every idea in the book, brought together to make sense of the infinite. That is what a capstone should do.

And it opens the last door. We have now added up infinitely many *powers* of x . **Book 3** will add up infinitely many *waves* — the **Fourier series** glimpsed in Chapters 7 and 10, where a function is built not from $1, x, x^2, \dots$ but from $\sin x, \cos x, \sin 2x, \dots$. Deciding when those infinite sums of waves converge will use exactly the convergence ideas you just built — but *constructing* them, and seeing why waves are the natural building blocks, needs a genuinely new language: vectors, inner products, and orthogonal bases, the machinery of linear algebra. That is where the next book begins, and it is the mathematical ground that modern data analysis, signal processing, and machine learning actually stand on.

You started this book with the slope of a single point. You end it able to add up infinitely many things without fear — to take the infinite, hold it at arm's length through a limit, and make it give you a finite, exact answer. Every result was built, never received; every loan was repaid. That is calculus, and it is yours now.

12.9 Problems

*Six tiers. **Fluency and Practice** are doing** — decide convergence and find radii, using the handful of principles, not a catalog of tests. **Concept & counterexample** and **◆ Diamond** are *thinking* — the necessary-not-sufficient trap and the proofs. **Teach it** is *teaching*. **Looking ahead** opens Book 3. Throughout, name *which principle* you're using (term test, geometric, comparison, integral test, ratio for radius) — that habit matters more than volume here.*

Fluency — *decide it, fast*

1. Compute the sum of each convergent geometric series, or say it diverges: (a) $\sum_{n=0}^{\infty} \left(\frac{1}{3}\right)^n$; (b) $\sum_{n=0}^{\infty} \left(\frac{3}{2}\right)^n$; (c) $\sum_{n=1}^{\infty} \left(\frac{1}{5}\right)^n$; (d) $\sum_{n=0}^{\infty} \left(-\frac{1}{2}\right)^n$.
2. Apply the **term test**: which of these diverge because their terms don't go to 0? (a) $\sum \frac{n}{2n+1}$; (b) $\sum \frac{1}{n}$; (c) $\sum \frac{n^2}{n^2+1}$; (d) $\sum \frac{1}{\sqrt{n}}$. (For the ones the term test *can't* rule out, say so — it's inconclusive, not a pass.)
3. **p-series**: state convergence or divergence and the value of p : (a) $\sum \frac{1}{n^3}$; (b) $\sum \frac{1}{n^{1/2}}$; (c) $\sum \frac{1}{n}$; (d) $\sum \frac{1}{n^{1.01}}$.
4. **Comparison**: decide convergence by comparing to a geometric or p -series: (a) $\sum \frac{1}{2^{n+n}}$; (b) $\sum \frac{1}{n^2+1}$; (c) $\sum \frac{1}{n^2-n}$ (for $n \geq 2$).
5. **Radius of convergence** via the ratio test: (a) $\sum \frac{x^n}{n!}$; (b) $\sum x^n$; (c) $\sum \frac{x^n}{2^n}$; (d) $\sum n! x^n$.
6. Write the first four partial sums of $\sum_{n=1}^{\infty} \frac{1}{n(n+1)}$ and guess the limit. (The terms telescope: $\frac{1}{n(n+1)} = \frac{1}{n} - \frac{1}{n+1}$.)

Practice — *multi-step*

7. Use the **integral test** to decide $\sum \frac{1}{n(\ln n)}$ (for $n \geq 2$). (Set up $\int_2^{\infty} \frac{1}{x \ln x} dx$ and substitute $u = \ln x$.)
8. Use the **integral test** to confirm $\sum \frac{1}{n^2}$ converges, by computing $\int_1^{\infty} \frac{1}{x^2} dx$. Then use the integral to also confirm $\sum \frac{1}{n}$ diverges.
9. Find the radius of convergence and state the interval $(-R, R)$ for: (a) $\sum \frac{x^n}{n}$; (b) $\sum \frac{x^n}{n^2}$; (c) $\sum \frac{(x-2)^n}{3^n}$ (centered at 2 — find where it converges).
10. The geometric series gives $\frac{1}{1-x} = \sum x^n$ for $|x| < 1$. By substituting, write a series for $\frac{1}{1+x^2}$ and state where it converges — then connect it to the arctangent series of Chapter 7.
11. A bouncing ball rises to $\frac{3}{4}$ of its previous height on each bounce, starting from 2 meters. Find the **total vertical distance** it travels before coming to rest. (A geometric series — mind the up-and-down.)
12. Decide convergence and **name your principle** for each: (a) $\sum \frac{n}{n^3+1}$; (b) $\sum \frac{1}{2^n}$; (c) $\sum \frac{1}{\sqrt{n}}$; (d) $\sum \frac{n+1}{n}$.

Concept & counterexample — *think; true/false with justification, or build the example*

13. (a) If $a_n \rightarrow 0$, then $\sum a_n$ converges. (b) If $\sum a_n$ converges, then $a_n \rightarrow 0$. (c) The harmonic series diverges even though its terms go to 0. (d) A geometric series $\sum r^n$ converges for every r with $|r| < 1$.
14. **The fundamental trap.** Explain, using the harmonic series as your example, exactly why “the terms go to zero” is *necessary but not sufficient* for convergence.
15. **Construct** a divergent series whose terms go to zero (other than the harmonic series). (A p -series with the right p will do.)
16. **Find the error.** A student writes “ $\sum \frac{1}{n}$ converges because $\frac{1}{n} \rightarrow 0$.” Identify the precise misuse of the term test and give the correct conclusion with a reason.
17. Explain why the integral test requires f to be **positive and decreasing** — what could go wrong with the rectangle-vs-area comparison if f wandered up and down?
18. **Is it possible?** A power series with radius of convergence $R = 0$ (it converges *only* at its center). What would the coefficients have to do? (Look at $\sum n! x^n$.)

◆ Diamond — *the proofs behind the principles*

19. ◆ Derive the **geometric series sum** $\sum_{n=0}^{\infty} r^n = \frac{1}{1-r}$ for $|r| < 1$ by finding a closed form for the partial sum S_N and taking the limit. State exactly where $|r| < 1$ is used.
20. ◆ Prove the **term test**: if $\sum a_n$ converges, then $a_n \rightarrow 0$. (Write $a_n = S_n - S_{n-1}$.) Then explain why the converse is false in one sentence.
21. ◆ Prove that the **harmonic series diverges** using Oresme’s grouping argument (blocks of doubling length, each exceeding $\frac{1}{2}$). Write out the first three blocks explicitly.

22. ♦♦ Prove the **integral test** by establishing the rectangle inequalities $\sum_{n=2}^{\infty} f(n) \leq \int_1^{\infty} f \leq \sum_{n=1}^{\infty} f(n)$ for positive decreasing f , and conclude the sum and integral converge together. (Draw the two rectangle pictures — under and over the curve.)
23. ♦♦ Prove the **p -series result** ($\sum \frac{1}{n^p}$ converges iff $p > 1$) by applying the integral test and the value of $\int_1^{\infty} x^{-p} dx$ from Chapter 11. Handle $p = 1$ separately.
24. ♦♦ Use the ratio test to prove $e^x = \sum \frac{x^n}{n!}$ converges for **every** real x ($R = \infty$), and reconcile this with the remainder argument you gave in Chapter 7 — two different proofs of the same fact.

Teach it — *explain it, write it up, or fix it*

25. In **three or four sentences**, teach a friend what it means for an infinite series to converge — that it's the limit of partial sums, nothing more exotic — using $\frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \cdots = 1$.
26. **Debug your classmate.** They claim $1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \cdots$ converges “because each term is smaller than the last and they’re heading to zero.” Give them the one fact that demolishes this, in a sentence.
27. Write a complete **worked solution** deciding whether $\sum \frac{1}{n^2+1}$ converges, naming your comparison series and justifying the inequality, so a peer could do $\sum \frac{1}{n^3+5}$.
28. In a few sentences, **teach the integral test as a picture** — how rectangles drawn against a curve trap a sum between two copies of an area — and why that means a sum and an integral share the same fate.
29. **Design a problem.** Invent a series whose convergence is best decided by the integral test, and write the full solution key, including the substitution in the integral.

Looking ahead — Book 3, and the waves to come

30. You can sum powers of x into functions (Taylor) and decide when those sums converge (this chapter). Book 3 sums **waves** — $\sin x, \cos x, \sin 2x, \dots$ — into functions (Fourier). **Predict** one reason waves might be better building blocks than powers for a *repeating* signal like a sound or a heartbeat. (Hint: what do $\sin$ and $\cos$ do that x^n never does?)
31. The remarkable fact $\sum \frac{1}{n^2} = \frac{\pi^2}{6}$ is proved with Fourier series. Knowing only that the sum *converges* (this chapter) and that the answer involves π , **speculate** why a sum of pure number-fractions might secretly know about circles. (No proof expected — just notice that π appearing here is strange and wonderful, and Book 3 explains it.)
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